

From Agriculture to Services: Development without Industrialization

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Abstract Does industrialization leave a lasting legacy after manufacturing declines? I argue that it does. Large manufacturing firms create demand for managerial and technical capabilities that remain valuable as economies transition toward services. I study this mechanism using cross-country evidence, U.S. county data from 1850 to 2000, and a quantitative growth model in which firm scale creates demand for high-skill workers to handle coordination problems, generating organizational knowledge that diffuses across firms and sectors. In the data, countries with larger historical manufacturing peaks develop service sectors more intensive in high-skill producer services and grow faster even after manufacturing has declined. Across counties in the United States, a 10pp higher manufacturing peak employment share driven by national growth in industries where large-scale coordination was technologically important raises the share of high-skill producer services in total private-service employment by 2.8 percentage points in 2000, or 45% of the cross-county average. The calibrated model shows that this mechanism goes a long way toward explaining cross-country differences in service composition, skill accumulation, and growth, and identifies firm scale and the diffusion of organizational knowledge as central forces. Because manufacturing’s lasting effects operate through the organizational knowledge created by production at scale rather than manufacturing employment itself, subsidizing manufacturing leaves little persistent legacy. Lasting gains instead arise when skill formation is combined with reforms that allow firms to grow and improve how production is organized.

Keywords: structural transformation, industrialization, producer services, firm scale, organizational capital, service-led growth.

JEL Classification: O14, O41, O33, O15, J24.

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1 Introduction

For most of economic history, development meant industrialization. Labor moved out of agriculture, manufacturing expanded, and services became dominant only after a substantial industrial phase.¹ That sequence is no longer typical. Many developing economies now move directly from agriculture into services, reaching lower manufacturing-employment peaks and deindustrializing at lower income levels than earlier industrializers (Rodrik, 2016). This raises a natural question: does a service economy that emerges directly from agriculture have the same foundations for sustained development as one that follows industrialization?

I argue that the answer depends on whether the development path passes through production at sufficient scale to make investment in organization worthwhile. Historically, industrialization concentrated production in large establishments, where coordinating workers, inputs, and information became an important part of production itself (Pollard, 1965; Chandler, 1990). Large firms responded by employing specialized managerial and technical workers and developing systematic solutions to recurring organizational problems. I refer to these functions as producer services and to the skilled workers who perform them as high-skill producer-service workers (HSPS). Because problems such as controlling costs, managing inventories, processing information, and assigning workers to tasks recur across sectors, the knowledge developed to solve them can remain useful as the economy moves into services. Many foundational management practices were developed in and around large industrial firms, including scientific management at Midvale and Bethlehem Steel, return-on-investment accounting at DuPont, the multidivisional organizational structure at General Motors, and statistical process control at Western Electric. These methods later diffused across firms and sectors, often through managers, consultants, and other external sources of organizational knowledge (McKenna, 2006; Mol and Birkinshaw, 2014). The main argument of this paper is therefore that large-scale manufacturing can leave a lasting legacy by creating organizational knowledge that remains valuable as the economy moves toward services.

I provide evidence for this mechanism in two steps. First, across countries since 1990, larger historical manufacturing peaks predict a service transition tilted toward high-skill producer services rather than trade and other consumer-facing services, and faster subsequent GDP-per-capita growth even after manufacturing has declined. Second, using U.S. county data from 1850 to 2000, I show that counties with larger manufacturing peaks accumulated more high-skill producer-service employment within manufacturing and later developed service sectors more specialized in high-skill producer-service industries. A natural concern is that manufacturing simply made counties larger and more urban, independently generating subsequent demand for high-skill producer services. To address this concern, I draw on the historical distinction between high-throughput industries, in which large-scale plants and intensive coordination were technologically important, and low-throughput industries that could operate at smaller scale (Chandler, 1990). While both types of manufacturing predict population growth and urbanization, only high-throughput manufacturing predicts subsequent specialization in HSPS services. I then construct shift-share instruments that interact counties' 1880 manufacturing-industry composition with subsequent leave-county-out national industry growth. In the preferred IV specification, a ten-percentage-point increase in peak manufacturing employment induced by high-throughput national growth raises the high-skill producer-service share of private-service employment in 2000 by 2.83 percentage points, or about 45% of the cross-county average. By contrast, an instrument pooling national

¹ See, among others, Kuznets (1973), Clark (2007), Gollin et al. (2002), and Herrendorf et al. (2014) for classic and modern accounts of structural transformation and economic development.

growth across all manufacturing industries also strongly predicts the manufacturing peak but produces no statistically detectable long-run effect on service composition.

The empirical evidence links large-scale manufacturing to the composition of services, but it cannot quantify how much this mechanism matters for aggregate development, how it propagates through skill accumulation and growth, or what would happen under alternative development paths. I therefore embed the mechanism in a quantitative model of long-run development. The backbone is a three-sector structural-transformation model in which nonhomothetic demand shifts expenditure from agriculture to manufacturing and eventually services as income rises, following [Comin et al. \(2021\)](#). Within each sector, heterogeneous firms enter endogenously and operate under decreasing returns, generating variation in firm size across producers and sectors. I introduce a scale-dependent organizational margin by letting the frequency with which firms encounter recurring organizational problems rise with employment. Each problem can be handled repeatedly as it arises or addressed systematically by paying a fixed skilled-labor cost to develop a reusable procedure. Because developing a procedure entails a fixed cost while its value rises with how often it is used, larger firms develop more procedures and demand more high-skill producer-service labor. Newly developed procedures are initially proprietary but can later diffuse into a public stock of organizational knowledge, lowering the cost of organizational investment for firms in other sectors. A manufacturing phase can therefore leave the subsequent service economy with organizational knowledge that persists after manufacturing employment declines. I close the model by linking the organizational mechanism to long-run productivity growth. New cohorts choose whether to acquire skill in response to the skill premium, so greater demand for high-skill producer services raises the future skilled share. Productivity growth, in turn, rises with the mass of skilled workers, following the basic logic of [Jones \(1995\)](#). Because long-run growth would otherwise remain tied to an exogenous population path, I endogenize fertility using the quantity–quality mechanism of [Galor and Weil \(2000\)](#) and [Lagerlöf \(2006\)](#). Faster technological change raises the return to educating children and lowers fertility. Education also becomes less costly as the economy becomes more skilled, linking the long-run skilled share to the demographic equilibrium. These feedbacks drive fertility to replacement and population to a finite endogenous level, leaving a positive long-run growth rate determined by the limiting skilled share rather than by exogenous population growth.

I calibrate the model to the U.S. transition from 1850 to 2019, initializing the economy in 1790 and disciplining structural change, firm scale, skill composition, producer services, fertility, and productivity growth. I then use the model to ask how economies with different peak manufacturing employment shares diverge in their subsequent development paths. The model reproduces 98 percent of the observed cross-country relationship between peak manufacturing employment share and the contemporary HSPS share of services, and 40 percent of the corresponding relationship with GDP-per-capita growth. Through the lens of the model, economies that pass through a larger manufacturing phase enter the service economy with more organizational knowledge, a more skill-intensive workforce, and a higher long-run growth rate. As an independent validation of the organizational mechanism, the model’s annual additions to public organizational knowledge correlate 0.75 with contemporaneous Library of Congress publications on business organization over 1906–1977, significantly more strongly than with 11 placebo knowledge categories.

I then use the model to ask which policies can reproduce these persistent effects in an economy that has not undergone a large manufacturing phase. I organize the experiments around the supply of skills and the firm-scale conditions that generate demand for them. In an economy calibrated to representative lower-middle-income-country conditions, a permanent reform that removes barriers to firm growth

more than doubles average firm size, raises GDP per capita by 13.6 percent several decades later, and also increases organizational knowledge and the skilled share. Temporary support for skill formation reproduces the large contemporaneous reallocation toward organized employers documented in recent work, but much of this response is transitory: with firm-growth distortions in place, skilled employment is 6.58 percentage points above the no-policy path at withdrawal and only 0.51 percentage points higher by 2089. The picture changes when temporary support for skill formation and organizational investment is combined with an environment in which productive firms can expand. With discounted fiscal expenditure of about 0.6 percent of discounted GDP during the intervention, the mixed package raises GDP per capita by 22.4 percent and leaves the skilled share 4.77 percentage points higher in 2089. In doing so, it creates a lasting stock of organizational knowledge that sustains skill demand after the policy is withdrawn. By contrast, a temporary manufacturing wage subsidy leaves essentially no organizational legacy because manufacturing employment is not itself the source of the externality. The policy implications therefore highlight that temporary skill expansions are most likely to have lasting effects when firms respond by creating organizational knowledge that persists, diffuses, and sustains skill demand by future firms.

The paper is most closely related to recent work connecting firm scale, skills, and the organization of production. [Engbom et al. \(2025\)](#) also build directly on Chandler’s account of the modern enterprise, emphasizing how the supply of educated white-collar workers enables firms to delegate, reorganize, and grow. My mechanism runs in the complementary direction: large-scale production creates demand for managerial and technical workers who solve recurring organizational problems systematically. I then study the long-run consequence of this process: whether the organizational routines and specialized expertise developed inside manufacturing persist and support the emergence of producer services after manufacturing declines. The paper is also closely related to [Peters et al. \(2026\)](#), who study fast-growing developing economies that have expanded services without undergoing broad industrialization and ask whether productivity growth in consumer services can sustain such a development path. I instead ask whether a substantial manufacturing phase changes the service economy that subsequently emerges, and show that large-scale manufacturing promotes high-skill producer services with persistent consequences for skill demand and growth.

More broadly, the paper connects structural transformation to firm organization. Existing work studies the forces behind sectoral reallocation and the changing role of manufacturing in development ([Rondrik, 2016](#); [Comin et al., 2021](#); [Huneus and Rogerson, 2024](#); [Fujiwara and Matsuyama, 2024](#); [Sposi et al., 2026](#)); I instead ask whether the manufacturing phase itself changes the service economy that follows. On the firm side, I build on models of organization capital ([Prescott and Visscher, 1980](#); [Atkeson and Kehoe, 2005](#)) by making the creation and diffusion of organizational knowledge endogenous to firm scale. The policy experiments also relate to recent evidence that temporary industrial interventions can have persistent effects ([Juhász, 2018](#); [Incoronato and Lattanzio, 2025](#); [Choi and Levchenko, 2025](#)); here, persistence operates through organizational knowledge that remains valuable after manufacturing declines.

The remainder of the paper proceeds as follows. Section 2 presents the cross-country evidence. Section 3 studies the mechanism by comparing the development path of U.S. counties. Section 4 develops the model, Section 5 describes its calibration and historical validation, and Section 6 reports the quantitative results. Section 7 studies policy implications and Section 8 concludes.

2 Cross-Country Facts

This section establishes two facts connecting industrialization to the service economy that follows. First, countries that experienced larger manufacturing-employment peaks developed service economies more concentrated in high-skill producer services. Second, larger historical manufacturing peaks are associated with faster GDP-per-capita growth, even decades after the manufacturing peak.

The two exercises use complementary country samples. The service-composition analysis uses the GGDC/UNU–WIDER Economic Transformation Database (ETD), which reports annual employment in 12 broad industries for 51 economies from 1990 through 2018. I define high-skill producer-service (HSPS) industries as business services, finance, and real estate, while generic market services comprise trade and other services. For these 51 economies, historical manufacturing peaks are constructed from the full available GGDC 10-Sector and ETD employment histories, with ETD taking precedence in overlapping years.² The growth exercise uses a broader 173-country sample, with service employment from ILOSTAT and GDP per capita from the World Bank via Gapminder. Appendix B describes the data, historical manufacturing series, concordances, and sample construction in detail.

FACT 1: *Larger historical manufacturing peaks are associated with greater expansion of high-skill producer-service employment.*

I begin by comparing service-employment trajectories across economies with different industrial histories. I split the 51-country ETD sample at the median historical manufacturing peak and compare the evolution of HSPS and generic market services as shares of total employment. To remove initial level differences, I rebase each country’s employment share to zero in 1990 before computing unweighted group means. Because most countries had reached their recorded manufacturing peak by the early 1990s, the exercise compares how service economies subsequently evolved across countries with different degrees of prior industrialization. Figure 1 shows substantially greater HSPS expansion in economies with higher manufacturing peaks. Between 1990 and 2018, the HSPS share of total employment rises by 5.75 percentage points in the higher-peak group, compared with 2.77 percentage points in the lower-peak group. The 2.98-percentage-point difference is statistically significant in a two-sided Welch t -test ($p = 0.001$).

The lower-peak group instead shows greater expansion in generic market services, which are more consumer-facing than HSPS industries. The generic market-service share rises by 9.59 percentage points in the lower-peak group, compared with 5.90 percentage points in the higher-peak group. The approximately 3.7-percentage-point difference is also statistically significant ($p = 0.016$). Thus, the two groups differ in the service industries that absorb employment: higher-peak economies experience greater HSPS expansion, while lower-peak economies experience greater expansion of trade and other services.³ Appendix B.2 shows that this relationship survives controls for contemporaneous GDP-per-capita growth, income-group and region fixed effects, and the overall size of the service sector, and is driven primarily by business services, followed by finance and real estate.⁴

FACT 2: *Larger historical manufacturing peaks are associated with faster growth as economies transition into services.*

² The GGDC 10-Sector Database extends the historical employment series for 32 of the 51 economies; the remaining 19 use their full available ETD histories.

³ The ETD other-services category does not include government services.

⁴ In the ETD data, business services are the service activities most closely connected to firms’ information and specialized professional needs—principally information and communication and professional, scientific, and technical activities.

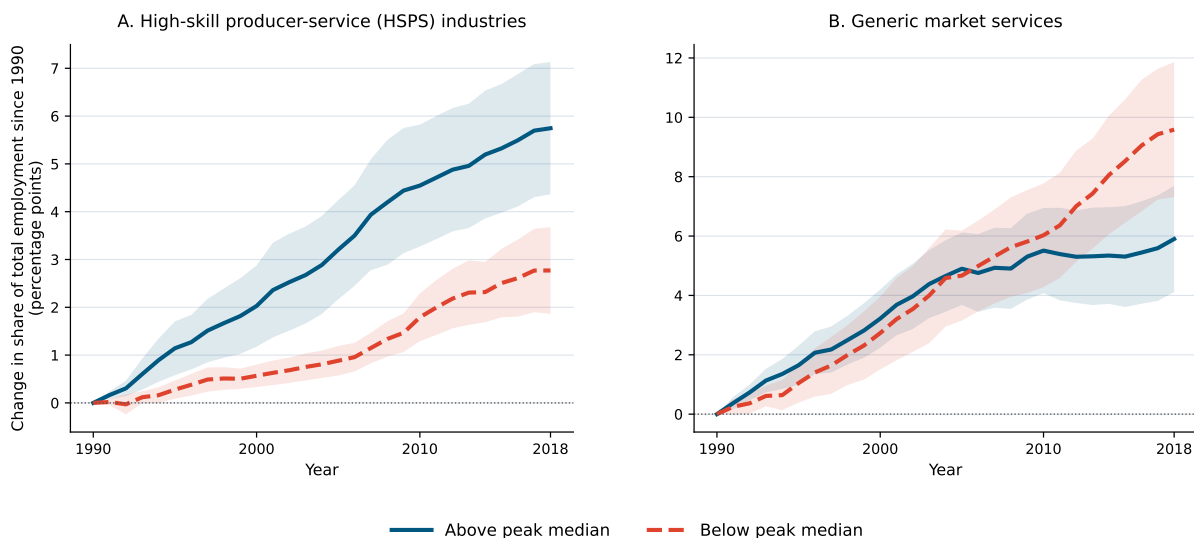


Figure 1: Manufacturing peaks and the subsequent service economy. The figure includes all 51 ETD countries observed annually over 1990–2018. Countries are grouped by the median historical peak manufacturing-employment share, calculated over their full available combined GGDC 10-Sector and ETD histories, with ETD taking precedence in overlapping years. The above-median group includes the country at the median and contains 26 countries; the below-median group contains 25. Panel A reports employment in high-skill producer-service (HSPS) industries—business services, finance, and real estate—as a share of total employment. Panel B reports employment in generic market services, defined as the ETD trade and other services categories, using the same denominator. Trade includes accommodation and food services. Transport and government services are excluded from both numerators. Each country’s share is rebased to zero in 1990 before taking unweighted group means. Shaded regions show the mean plus or minus 1.96 cross-country standard errors.

The rise of services is often associated with slower aggregate productivity growth. In the standard Baumol cost-disease logic, labor reallocates toward sectors with relatively slow productivity growth, reducing aggregate growth as the service share rises (Baumol, 1967; Duernecker and Sanchez-Martinez, 2023; Duernecker et al., 2024). Fact 1 suggests an additional dimension: service economies may differ in their internal composition depending on the development path through which they emerged. Figure 2 plots average annual PPP real GDP-per-capita growth over 2010–2019 against the average service-employment share over the same period for 173 countries. Countries are divided according to whether their historical manufacturing peak lies above or below the cross-country median. Service employment is negatively associated with growth in both groups; conditional on the service-employment share, however, countries with above-median manufacturing peaks tend to grow faster.

The relationship survives more demanding comparisons. Appendix Table B.3 shows that the manufacturing peak remains positively associated with subsequent GDP-per-capita growth after controlling for region and income-group fixed effects and institutional quality. The association also remains positive, with a slightly larger point estimate, among countries where services account for more than half of employment. These facts are consistent with a persistent legacy of industrialization rather than an effect operating only while manufacturing itself remains large.

Economies with larger manufacturing peaks develop service economies more concentrated in high-skill producer services and grow faster. The cross-country evidence does not, however, identify the mechanism behind these associations. Section 3 turns to the U.S. historical transition to test whether

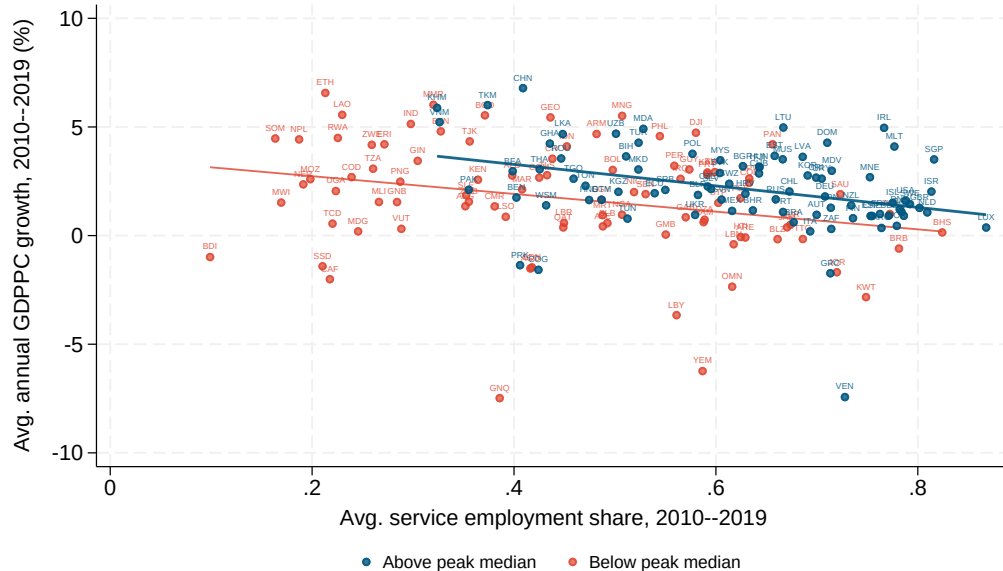


Figure 2: Growth in service economies by peak manufacturing exposure. The vertical axis is average annual growth in PPP real GDP per capita over 2010–2019; the horizontal axis is the average service-employment share over the same period. Countries are divided at the median historical manufacturing peak. Service employment is from ILOSTAT, GDP per capita is from the World Bank via Gapminder, and the historical manufacturing-peak series is described in Appendix B.

they reflect the scale and organizational demands of manufacturing and the managerial and technical capabilities generated in response.

3 U.S. County Evidence

U.S. counties provide a setting in which the proposed mechanism can be observed over the full industrial transition. I first trace the buildup of HSPS capabilities within manufacturing around county-specific industrial peaks and their later emergence in specialized service industries. I then ask whether establishment scale and manufacturing HSPS employment account for the long-run relationship between manufacturing exposure and composition of services. Finally, I exploit counties’ 1880 industry composition and subsequent national industry growth to test whether this legacy is concentrated in high-throughput manufacturing rather than manufacturing expansion more generally. Together, these exercises distinguish the scale-dependent organizational channel from broader consequences of industrialization such as population growth and urbanization.

3.1 Data and measurement

To study the mechanism over the full U.S. structural transformation, I combine county tabulations from the Census of Manufactures with full-count population censuses and postwar administrative employment data, harmonizing counties to fixed 1850 boundaries. The analysis distinguishes three objects: the depth of industrialization, measured by the peak manufacturing employment share; managerial and technical capability within manufacturing, measured by HSPS employment in manufacturing relative to

total classified employment; and the subsequent expansion of specialized HSPS services, measured relative to the same employment base. For the long-run cross-sectional analysis, I additionally measure the HSPS share within private-service employment to isolate the association between industrialization and the composition of the service sector rather than its overall size. The peak manufacturing employment share is manufacturing employment obtained from the Census of Manufactures relative to total classified employment obtained from the Census of Population. Let L_{ct}^C denote total classified employment, defined as the number of employed individuals whose industries can be classified as primary, manufacturing, or services.⁵ I define county c 's manufacturing peak as

$$M_c^{\text{peak}} = \max_{t \in \{1880, 1900, 1920, 1930, 1940\}} \left\{ \frac{L_{ct}^M}{L_{ct}^C} \right\}, \quad (1)$$

where L_{ct}^M is average manufacturing wage earners and L_{ct}^C is total classified employment. I let P_c denote the census year attaining this maximum. The peak window begins in 1880, before the national expansion of manufacturing and at the first major improvement in the quality of U.S. census enumeration.⁶ The years 1890 and 1910 do not feature in the data because most of the 1890 population census was lost and the required county manufacturing tabulations are unavailable in 1910. Manufacturing scale is log manufacturing workers per establishment in the same peak year P_c , also measured in the Census of Manufactures.

To measure the organizational capabilities employed inside manufacturing, I identify workers performing the functions required to organize production at scale and construct

$$MHSPS_{ct} = 100 \times \frac{\text{HSPS-occupation workers in manufacturing}_{ct}}{L_{ct}^C}. \quad (2)$$

The manufacturing HSPS classification contains four functional groups: accounting and record keeping; purchasing, logistics, and coordination; personnel and labor relations; and technical and scientific work, including engineers, draftsmen, statisticians, surveyors, technicians, and inspectors. These occupations capture the information-processing, coordination, and technical functions that are necessary to operate large manufacturing enterprises (Chandler, 1977, 1990).⁷ The service-side measure asks whether employment subsequently shifts toward industries that supply these capabilities on the market. I define

$$SHSPS_{ct} = 100 \times \frac{\text{employment in HSPS service industries}_{ct}}{L_{ct}^C}. \quad (3)$$

In the modern data, HSPS services comprise Information, Professional, Scientific and Technical Services, and Management of Companies—NAICS sectors 51, 54, and 55. These industries provide the closest service-sector counterpart to the organizational capabilities emphasized by Chandler (1977, 1990). NAICS 54 includes accounting, legal, engineering, scientific, technical, and consulting services that supply externally many functions historically organized within large manufacturers; NAICS 55 captures the management and coordination of enterprises; and NAICS 51 captures the production and transmission of

⁵ More specifically, I define total classified employment as employment in primary industries (IND1950 codes 105, 116, and 126), manufacturing (IND1950 codes 306–499), and services (IND1950 codes 506–946). Mining, construction, unclassified employment, and industries outside these categories are excluded.

⁶ Before 1880, federal censuses were conducted by U.S. marshals; the 1880 Census was the first administered by a professional census office and enumerator force.

⁷ The core manufacturing HSPS set contains OCC1950 codes 0, 35, 41–49, 72, 83, 92, 95, 96, 200, 201, 280, 310, 321, 322, 341, 342, and 533. Appendix A reports the full labels and construction.

information. I exclude NAICS 56 because it combines administrative support with many routine activities not closely tied to the mechanism, and I exclude the broad NAICS 52–53 aggregates because they contain substantial financial intermediation, insurance, property operation which are primarily consumer facing.⁸ Appendix A reports the historical IND1950 and SIC concordances used to carry this functional definition backward in time. The county-industry panel within services uses the Population Census through 1950, County Business Patterns in 1960 and 1970, and the harmonized CBP panel of [Eckert et al. \(2020\)](#) thereafter; the manufacturing-HSPS measure uses the Population Census throughout because it requires occupation-by-industry information.⁹ For the 2000 measure of service composition, I use the same HSPS industry definition but divide employment in NAICS 51, 54, and 55 by total private-service employment from the BLS Quarterly Census of Employment and Wages (QCEW), which directly reports the county-level private service-providing aggregate ([U.S. Bureau of Labor Statistics, 2004](#)).¹⁰

3.2 Manufacturing HSPS and the later rise of specialized services

I first trace how managerial and technical employment within manufacturing and employment in specialized HSPS service industries evolve around county-specific manufacturing peaks. Let M_c^{peak} denote the manufacturing employment share at county c 's peak. I estimate by OLS the following specification separately for $MHSPS_{ct}$ and $SHSPS_{ct}$:

$$Y_{ct} = \alpha_c + \gamma_{s(c),t} + \lambda_{t-P_c} + \sum_{k \neq -40} \beta_k \left[\mathbf{1}\{t - P_c = k\} \times \frac{M_c^{\text{peak}}}{0.10} \right] + X'_c \delta Trend_t + \varepsilon_{ct}. \quad (4)$$

The specification absorbs county, state-by-calendar-year, and event-time fixed effects. The vector X_c contains 1850 log population, urbanization, colleges, school enrollment, and log enslaved population, together with 1870 shares of process, machinery, and traditional manufacturing. Each covariate is interacted with a linear calendar-time trend. Event time -40 is omitted. Because repeated county observations may be spatially correlated across neighboring local economies, the confidence intervals use a spatial covariance estimator allowing arbitrary covariance across all observations from counties whose centroids are within 100 kilometers.¹¹ This is a continuous-treatment event-study specification: each β_k measures the differential evolution of the outcome associated with a 10-percentage-point larger manufacturing employment share at the peak. The sample contains 1,116 fixed-geography counties peaking in 1900, 1920, or 1940 for which the peak treatment and a common forty-year pre-period are observed.

Figure 3 establishes a clear sequence. Counties with higher manufacturing peaks accumulate more HSPS employment within manufacturing before the peak, and this differential continues to widen for several decades afterward. The manufacturing coefficient averages 0.063 percentage points over event

⁸ The cross-country ETD analysis necessarily uses a broader sectoral classification. Its business-services category includes information and communication and professional and technical activities, while finance and real estate are observed only as broad sectors.

⁹ Throughout, L_{ct}^C denotes the same broad classified-employment concept. For each outcome and data vintage, it is reconstructed from the same source as the corresponding numerator, avoiding comparisons between workplace-based postwar employment and residence-based Census employment.

¹⁰ Appendix C.2 describes the treatment of suppressed QCEW cells and reports corresponding results using alternative denominators, the harmonized CBP data, and a 2010 endpoint.

¹¹ Clustering standard errors at the state level leaves results unchanged. I cluster spatially because this is the appropriate choice given industrialization historically exploited geographic features, such as the fall-line, coal mines, lakes and rivers, which transcend state boundaries.

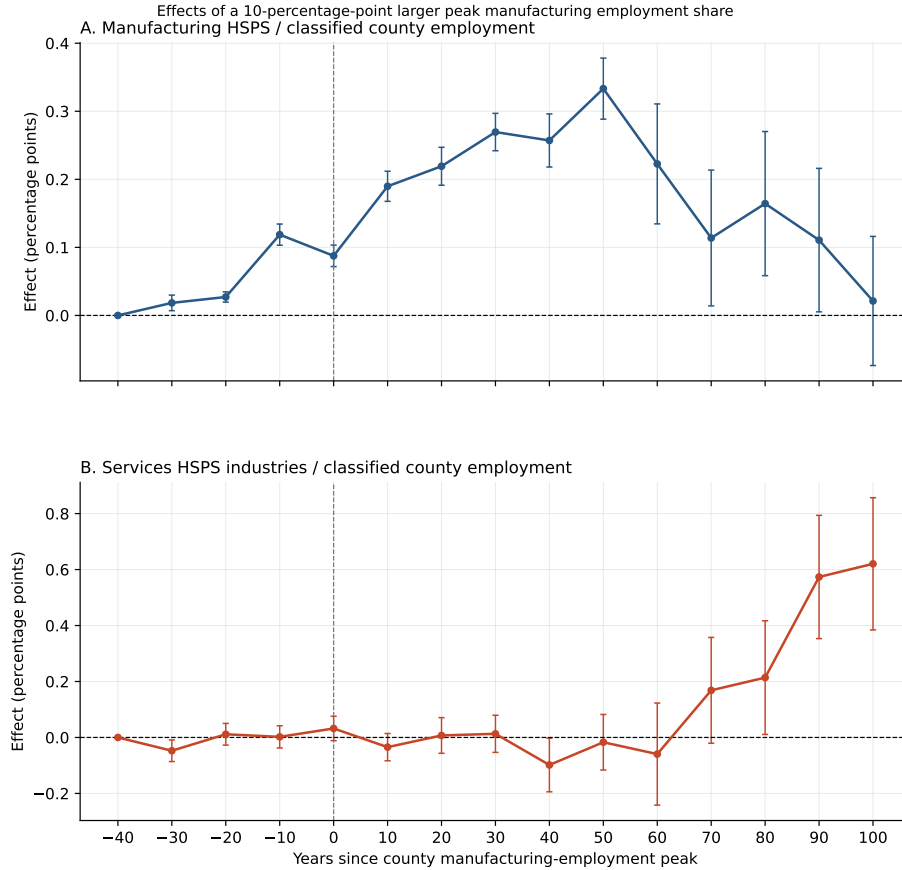


Figure 3: Manufacturing exposure at the peak and the capability sequence. The sample contains 1,116 fixed-geography counties peaking in 1900, 1920, or 1940. Each point is the coefficient associated with a 10-percentage-point larger manufacturing employment share at the county’s peak; event time -40 is omitted. The upper outcome is manufacturing HSPS occupations divided by classified county employment, as defined in equation (2). The lower outcome is employment in HSPS service industries—the historical analogues of NAICS 51, 54, and 55—divided by the broad classified-employment base defined in equation (3). The lower-panel numerator and denominator are constructed from the same employment universe in each vintage: harmonized Census records through 1950, imputed early CBP in 1960 and 1970, and harmonized EFSY CBP thereafter. In the postwar years, both are therefore workplace-based employment counts rather than a workplace numerator divided by employed county residents. The regressions include county, state-by-calendar-year, and event-time fixed effects and baseline-covariate linear trends. Bars are 95-percent confidence intervals using a 100-kilometer spatial covariance estimator. Because the panel ends in 2000, the longest post-peak horizons are supplied by the earlier peak cohorts.

times -30 through zero and 0.254 points over $+10$ through $+50$ ($p < 0.001$ in both periods), before declining thereafter. HSPS services increase later. Their average differential is -0.0004 percentage points over event times -30 through zero ($p = 0.978$) and -0.026 points over $+10$ through $+50$ ($p = 0.439$). It then rises to 0.394 points over $+70$ through $+100$ ($p < 0.001$) in the same period when HSPS employment within manufacturing declines. Appendix C.2.2 shows that this pattern is not a generic consequence of population growth or urbanization. I distinguish high-throughput manufacturing industries, which the historical literature identifies as operating at large scale and requiring intensive managerial and technical coordination, from the remaining manufacturing industries. Peaks in both groups predict subsequent population growth and urbanization, but only high-throughput peaks predict a buildup of HSPS intensity within manufacturing and, later, greater HSPS service employment. This asymmetry also helps reconcile

these findings with evidence that early industrialization can generate substantial short-run gains yet adverse long-run effects (Franck and Galor, 2021).¹²

3.3 Manufacturing scale, HSPS employment, and persistent service composition

I next ask whether the long-run association between manufacturing peaks and service composition is accounted for by the two margins implied by the mechanism: establishment scale and HSPS employment within manufacturing. Larger establishments create greater returns to formalizing organizational tasks, while manufacturing HSPS employment measures the managerial and technical functions developed inside manufacturing. I contrast these margins with a broader local-development account in which manufacturing primarily operates through higher wages, urbanization, or agglomeration.

3.3.1 Channel accounting

Let $Y_{c,2000}$ denote the QCEW HSPS share of private-service employment described above. I estimate

$$Y_{c,2000} = \beta M_c^{\text{peak}} + \theta S_c^{\text{peak}} + \phi H_c^{\text{peak}} + A_c^{\text{peak}'} \kappa + X'_{c,1850} \Gamma + \delta_{s(c)} + \lambda_{P_c} + \varepsilon_c, \quad (5)$$

where S_c^{peak} is log manufacturing workers per establishment and H_c^{peak} is manufacturing HSPS employment divided by total classified county employment, the same underlying fraction as in the event study. The vector $A_c^{\text{peak}'}$ contains average manufacturing wages, the urban population share, and log county population, all observed at the manufacturing peak. Every specification includes state and peak-cohort fixed effects and five 1850 controls: log population, urbanization, school enrollment, colleges, and log enslaved population.

Column 1 establishes the baseline long-run relationship on the common sample: a manufacturing peak larger by 10 percentage points predicts a 0.48-percentage-point higher HSPS share of private-service employment in 2000, approximately 8 percent of the 6.36-percent outcome mean. Adding establishment scale reduces the peak coefficient by 50 percent, while adding manufacturing HSPS reduces it by 37 percent. Including both reduces the coefficient from 0.048 to 0.015, an attenuation of 68 percent, and renders it statistically indistinguishable from zero. Manufacturing HSPS remains quantitatively large and precisely estimated in the joint specification: a one-percentage-point increase in manufacturing HSPS relative to classified employment predicts a 1.93-percentage-point increase in the HSPS share of private services. Establishment scale also remains positive and is significant at the 10-percent level.

Column 5 asks whether similar attenuation arises from broader correlates of local development. Urbanization and population at the peak are positively associated with subsequent HSPS specialization, while manufacturing wages are not. Jointly including the three variables attenuates the peak coefficient by 64 percent, but the remaining coefficient stays positive and significant at the 10-percent level. These controls therefore absorb part of the long-run relationship, but they do not eliminate it. Because all of these variables may themselves respond to industrialization, I interpret Table 1 as descriptive channel accounting rather than causal mediation.

¹² Franck and Galor (2021) study the effects of the steam engine on French industrialization in 1860–65, before the late-nineteenth-century emergence of Chandlerian large industrial enterprise. They find initial income gains followed by a long-run reversal of the effect of industrialization on income and human capital. Repeating their exercise on French data with 1896 establishment-size information yields different results. Industrialization in scale intensive Chandler-type industries predicts higher GDP per capita a century later, while industrialization in traditional industries predicts initial gains in GDP per capita but long-run losses.

Table 1: Manufacturing Exposure, Scale, and Persistent HSPS Services

	Baseline	+ Scale	+ Mfg. HSPS	+ Both	+ Alternative channels
Peak manufacturing employment share	0.048*** (0.010)	0.024** (0.012)	0.030*** (0.011)	0.015 (0.012)	0.017* (0.010)
Log manufacturing establishment scale at peak		0.0060*** (0.0022)		0.0041* (0.0023)	
Manufacturing HSPS / classified employment at peak			2.112*** (0.529)	1.927*** (0.535)	
Average manufacturing wage at peak, per \$100					0.0008 (0.0006)
Urban population share at peak					0.0183** (0.0072)
Log county population at peak					0.0188*** (0.0023)
Attenuation of peak coefficient (%)	0.0	50.3	37.4	68.3	63.9
State fixed effects	Yes	Yes	Yes	Yes	Yes
Peak-cohort fixed effects	Yes	Yes	Yes	Yes	Yes
1850 historical controls	Yes	Yes	Yes	Yes	Yes
Observations	1,487	1,487	1,487	1,487	1,487
R^2	0.235	0.239	0.246	0.247	0.291

Notes: The dependent variable is public numeric QCEW private employment in NAICS 51, 54, and 55 divided by the disclosed private service-providing aggregate less published private-utility employment in 2000. The aggregate denominator retains confidential non-HSPS service employment. Peak manufacturing employment is Census of Manufactures average wage earners divided by total classified county employment and is defined as the maximum of this ratio over 1880, 1900, 1920, 1930, and 1940. Manufacturing HSPS is employment in the core accounting, coordination, personnel, technical, and scientific occupations listed in the text, conditional on employment in manufacturing, divided by total classified county employment. Establishment scale, manufacturing HSPS, and the alternative-channel variables are measured in the county's manufacturing-peak year. Missing peak-year manufacturing wages are replaced with the corresponding peak-cohort median and the specification includes a missing-wage indicator. Every column uses the same 1,487 counties and includes state and peak-cohort fixed effects and 1850 controls for log population, urbanization, school enrollment, colleges, and log enslaved population. Conley standard errors with a 100-kilometer cutoff are in parentheses. The attenuation exercise is descriptive channel accounting. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

3.3.2 National industry shifts

The preceding results suggest that the service-sector legacy of manufacturing is tied to the scale and organizational demands of industrialization. However, industrialization, establishment scale, and HSPS employment are jointly determined, and counties with a pre-existing comparative advantage in managerial or technical activity could exhibit similar development patterns encompassing both the type of manufacturing they develop and the later composition of services. To assuage this concern, I exploit counties' differential exposure to national manufacturing-industry growth over 1880–1900 by interacting their 1880 industry composition with subsequent leave-county-out national growth. Because 1880 specialization may itself reflect persistent local comparative advantage, the design does not require unconditional balance in the initial shares. The exclusion restriction is instead conditional: after accounting for observed initial conditions and complete 1850–1870 histories of manufacturing employment, manufacturing HSPS, and HSPS services, subsequent national industry growth must affect service composition through the local industrialization episode it induces rather than through an independent continuation

of pre-existing county trends. Identification therefore compares counties in the same state and peak cohort that followed similar observed development paths before 1880 but were differently exposed to the subsequent national growth of the industries in which they were specialized.

This industry-level design also permits a sharper test of the mechanism. National growth across manufacturing as a whole may deepen local industrialization regardless of how production is organized and which manufacturing industries expand. The mechanism, however, predicts a persistent change in services specifically when expansion occurs in industries in which operating at scale creates intensive managerial, informational, and technical demands. Following [Chandler \(1990\)](#), I classify continuous-process, large-batch, and complex-assembly industries as high throughput and distinguish them from traditional low-throughput industries that could operate at lower scale. Examples of high-throughput industries include iron and steel, chemicals, machinery, and processed foods, while low-throughput industries include textiles and apparel, wood and furniture, printing, and leather goods.¹³ The classification follows Chandler’s technological taxonomy rather than being constructed from the county outcomes. High-throughput industries are those in which large fixed investments and continuous or high-volume production were necessary to achieve low unit costs, making large establishments and intensive coordination integral to the production technology; low-throughput industries could operate efficiently at substantially smaller scale. For industry group $g \in \{A, H, L\}$, where A denotes all manufacturing, H high-throughput manufacturing, and L traditional low-throughput manufacturing, I construct

$$Z_c^g = \sum_{j \in g} \omega_{cj,1880}^g G_{j,-c,1880-1900}, \quad (6)$$

where $\omega_{cj,1880}^g$ is industry j ’s share of county c ’s 1880 manufacturing employment within group g , and $G_{j,-c,1880-1900}$ is national employment growth in industry j between 1880 and 1900 after removing county c . Each shifter is standardized within the common estimation sample. The within-group normalization makes the exposure shares sum to one and isolates which industries within a production regime a county initially specialized in rather than the overall size of the group in the local economy. Appendix C.2 reports economy-scaled alternatives and specifications that control directly for the initial high-throughput share.

For each shifter Z_c^g , the first and second stages are

$$M_c^{\text{peak}} = \pi_g Z_c^g + X'_{c,1850} \Pi + \mathcal{H}'_{c,<1880} \Omega + \delta_{s(c)} + \lambda_{P_c} + u_c, \quad (7)$$

$$Y_{c,2000} = \beta_g \widehat{M}_c^{\text{peak}} + X'_{c,1850} \Gamma + \mathcal{H}'_{c,<1880} \Psi + \delta_{s(c)} + \lambda_{P_c} + \varepsilon_c, \quad (8)$$

Where the vector $X_{c,1850}$ contains the five baseline controls measured in 1850, while $\mathcal{H}_{c,<1880}$ collects the complete 1850–1870 histories of manufacturing employment, manufacturing HSPS employment, and HSPS-service specialization. The terms $\delta_{s(c)}$ and λ_{P_c} denote state and manufacturing-peak-cohort fixed effects. Table 2 reports the corresponding OLS benchmark and three just-identified 2SLS specifications on a common sample of 1,487 counties.

The all-industry and high-throughput shifters both strongly predict county manufacturing peaks, with Conley first-stage F statistics of 14.67 and 12.87 and Kleibergen–Paap statistics above 20. However,

¹³ Appendix A reports the complete IND1950 concordance, the corresponding Chandler references, and the rationale for every included industry.

Table 2: National Manufacturing Industry Shifts and Persistent HSPS Services

	OLS	All industries IV	High-throughput IV	Low-throughput IV
Peak manufacturing employment share	0.029** (0.012)	0.133 (0.105)	0.283** (0.118)	−1.095 (8.010)
First-stage coefficient		0.0150*** (0.0039)	0.0193*** (0.0054)	0.0006 (0.0042)
First-stage Conley F statistic		14.67	12.87	0.02
Kleibergen–Paap F statistic		20.53	21.94	0.03
Anderson–Rubin p -value		0.203	0.009	0.596
State fixed effects	Yes	Yes	Yes	Yes
Peak-cohort fixed effects	Yes	Yes	Yes	Yes
1850 historical controls	Yes	Yes	Yes	Yes
Complete 1850–1870 histories	Yes	Yes	Yes	Yes
Observations	1,487	1,487	1,487	1,487

Notes: The dependent variable is the QCEW HSPS share of private-service employment described in Section 3.1. Peak manufacturing employment is Census of Manufactures average wage earners divided by total classified county employment and is defined over 1880, 1900, 1920, 1930, and 1940. Column 1 reports OLS. Columns 2–4 instrument the peak manufacturing employment share with standardized shift-share instruments based on, respectively, all manufacturing industries, Chandler high-throughput industries, and traditional low-throughput industries. The instruments combine 1880 county industry-employment shares within the relevant group with leave-county-out national industry growth from 1880 to 1900. All columns use the same 1,487 counties and include state and peak-cohort fixed effects, the five 1850 controls, and complete 1850–1870 histories of manufacturing employment and core manufacturing HSPS relative to total classified employment and of HSPS services relative to private-service employment. Coefficient and first-stage standard errors are Conley with a 100-kilometer cutoff. The first-stage Conley F statistic is the square of the Conley first-stage t statistic. The Kleibergen–Paap statistic is the heteroskedasticity-robust excluded-instrument Wald statistic. The Anderson–Rubin p -value is the corresponding Conley reduced-form test. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

their long-run implications for the composition of services are different. Using national growth across manufacturing as a whole yields a 2SLS coefficient of 0.133, but neither the conventional estimate nor the weak-instrument-robust reduced form is statistically distinguishable from zero ($p_{AR} = 0.203$). By contrast, the high-throughput IV estimate is 0.283 with a Conley standard error of 0.118. A 10-percentage-point increase in the manufacturing peak induced by national high-throughput growth therefore raises the HSPS share of private-service employment in 2000 by 2.83 percentage points—approximately 45 percent of the 6.36-percent cross-county mean. The Anderson–Rubin test rejects zero at the 1-percent level ($p = 0.009$).

The traditional low-throughput shifter has essentially no first stage, so its 2SLS coefficient is not informative. The economically relevant contrast is between broad manufacturing growth, which strongly moves the manufacturing peak without producing a detectable shift in long-run service composition, and high-throughput growth, which moves both.

The contrast points to a scale-dependent organizational legacy of industrialization: persistent changes in service composition arise when manufacturing expansion occurs in production regimes where operating at scale creates intensive managerial, informational, and technical capabilities. Appendix C.2 subjects this result to the principal measurement and inference diagnostics.¹⁴ The result is not driven by any single high-throughput industry, remains positive across alternative classification boundaries and endpoints,

¹⁴ I report extensive robustness to alternative outcome construction, industry classifications, exposure definitions, geography, and inference procedures.

and survives substantially wider spatial-correlation cutoffs. Combined with the event-time and channel-accounting evidence, these findings motivate the quantitative model in Section 4, which asks how this organizational legacy shapes structural transformation, skill accumulation, and long-run growth.

4 Model

4.1 Overview

Time is discrete and one period is one year. The economy contains three final-good sectors—agriculture, manufacturing, and consumer services—indexed by $j \in \mathcal{J} \equiv \{a, m, c\}$. Firms enter endogenously, draw permanent productivity upon entry, and face exogenous exit risk. Final goods are used for consumption and as intermediate inputs, while producer services help firms organize and coordinate production. Nonhomothetic demand reallocates expenditure from agriculture toward manufacturing and services as income rises, generating the economy’s structural transformation path. Households choose consumption, fertility, and education, so population and the supply of skilled labor evolve endogenously.

The central mechanism is that firm scale raises the return to formalizing organizational tasks. Every firm faces a unit measure of organizational tasks required to operate. Each task generates recurring organizational problems, with their frequency increasing linearly in firm size. These tasks are common across sectors: firms must coordinate workers, process information, manage inputs, and organize decision-making regardless of what they produce. Tasks can be handled informally, in which case the underlying problems must be resolved repeatedly, or formalized by installing a reusable procedure. The stock of formalized tasks constitutes the firm’s *organizational capital*. Because a given task generates more problems in a larger firm, formalization avoids more repeated problem solving and is therefore more valuable at larger scale. Larger firms consequently formalize more tasks and demand more high-skill producer-service workers to do so.

Formalization can also create knowledge that extends beyond the firm. When a firm formalizes a task for which no established procedure exists, it must use skilled labor to develop a solution from scratch. The resulting method is initially proprietary, but may subsequently diffuse into a public stock of organizational knowledge, K_t , allowing other firms to formalize the same task at lower cost. The originating firm therefore does not capture the full value of the knowledge it creates. Because all firms, regardless of what they produce, face the same organizational problems, this externality is naturally cross-sectoral: methods developed inside one sector can later lower the cost of organizing production in another.¹⁵

Manufacturing matters in the model not because it uses intrinsically different organizational tasks, but because historically it operated at the scale at which formalization became privately worthwhile. A manufacturing-intensive development path therefore generates more organizational knowledge than an economy which moves from agriculture to a service sector composed of small firms. The service economy that follows manufacturing inherits organizational knowledge, formalizes more activities, and demands more HSPS labor. By raising the return to skill, this organizational legacy also affects education and, ultimately, the economy’s long-run skill composition and productivity growth.

¹⁵ Historical work documents both sides of this mechanism. Large industrial firms developed new systems of cost accounting, return-on-investment control, statistical quality control, budgeting, workflow design, and personnel management as they confronted the organizational demands of operating at scale (Johnson, 1975; Kaplan, 1984; Chandler, 1977; Shewhart, 1931). Many of these methods were subsequently codified, professionalized, and applied outside manufacturing (McKinsey, 1922, 1924; Yates, 1993; McKenna, 2006; Mol and Birkinshaw, 2014). Appendix D.1 discusses several examples in detail.

The productivity-growth block builds on the tradition of [Jones \(1995\)](#). Unlike standard models in this tradition, however, the long-run growth rate is independent of both population scale and population growth and is instead determined by the economy’s *endogenous* share of skilled workers. I obtain this by combining a Jones-inspired productivity technology with the endogenous fertility mechanism of [Galor and Weil \(2000\)](#). This is a natural closure for a model of long-run development in which both population and skill supply evolve endogenously: fertility converges to replacement, population to a finite level, and the limiting skilled share pins down sustained productivity growth. Manufacturing can therefore affect long-run growth insofar as it permanently changes the economy’s skill composition. Proposition 2 characterizes this limiting equilibrium.

The remainder of the section proceeds in four steps. I first characterize firms’ organizational choices and show how scale determines formalization and HSPS demand. I then describe the creation, adaptation, and diffusion of organizational procedures and the entry margin that determines equilibrium firm scale. Next, I embed these decisions in the structural-transformation equilibrium and introduce the household choices that determine education and population growth. Finally, I characterize the limiting equilibrium, including the long-run composition of services, skill supply, and productivity growth. The main text emphasizes the economic mechanisms, while derivations, equilibrium definitions, and proofs are collected in Appendix D.

4.2 Production and organizational choice

Final-good production. A firm f in sector j has permanent ability $a_f \sim G$ and an inherited formalization frontier s_{fjt} . For a generic firm I suppress the f subscript and write its state as (a, s) . Skilled and unskilled labor form a CES composite, and value added exhibits decreasing returns to firm scale:

$$\begin{aligned}\ell_{fjt} &= \left[\alpha_j (h_s L_{fjt}^s)^{\frac{\sigma_L-1}{\sigma_L}} + (1 - \alpha_j) (h_u L_{fjt}^u)^{\frac{\sigma_L-1}{\sigma_L}} \right]^{\frac{\sigma_L}{\sigma_L-1}}, \\ v_{fjt} &= A_{jt} a_f \ell_{fjt}^\eta, \quad 0 < \eta < 1.\end{aligned}\tag{9}$$

Gross output combines value added, agricultural and manufacturing materials, and producer services used directly in production:

$$\begin{aligned}y_{fjt} &= Z_{jt} v_{fjt}^{\theta_{jt}} (m_{fjt}^a)^{\gamma_{jt}^a} (m_{fjt}^m)^{\gamma_{jt}^m} (P_{fjt}^p)^{\gamma_{jt}^p}, \\ \theta_{jt} + \gamma_{jt}^a + \gamma_{jt}^m + \gamma_{jt}^p &= 1.\end{aligned}\tag{10}$$

Let w_{jt}^v denote the unit cost of the labor composite and let $n = \nu_{jt} \ell$ denote the associated employment headcount. Both follow from cost minimization in the CES labor nest. The distinction between ℓ and n matters because production depends on effective labor, whereas organizational problems arise from coordinating workers and therefore depend on employment headcount. The input-output coefficients are initialized to imply value-added production in 1850 and then evolve gradually toward their 2015 BEA input-output shares.¹⁶

¹⁶ Allowing the input-output structure to evolve avoids imposing the modern production network on the 1850 economy. With fixed modern IO coefficients, historical changes in employment generated by the expansion of intermediate-input use would instead be absorbed by the estimated preference and productivity parameters.

Organizational tasks. Every firm faces the same unit mass of organizational tasks. Tasks differ in complexity $z \geq 0$, which determines how difficult they are to formalize and is exponentially distributed:

$$\mathcal{F}(z) = 1 - e^{-z}.$$

A firm's organizational state is summarized by its formalization frontier s : all tasks with $z \leq s$ have been formalized, so its organizational-capital stock can equivalently be represented by the formalized task share

$$q(s) \equiv \mathcal{F}(s).$$

Each task generates $b(n) = n/\bar{N}$ organizational problems, where n is firm employment and $\bar{N}$ normalizes firm scale. Thus, the same organizational task generates more instances requiring resolution in a larger firm. If a task is handled informally, these problems must be repeatedly diagnosed and resolved; once the task is formalized, the firm can instead apply a reusable procedure. Let $r > 1$ denote the current-period-equivalent amount of informal problem solving required relative to operating under such a procedure.¹⁷ Conditional on (n, s) , recurring demand for high-skill formal and low-skill informal producer services is therefore

$$H^o(n, s) = \frac{n}{\bar{N}} q(s), \quad L^o(n, s) = \frac{n}{\bar{N}} r[1 - q(s)]. \quad (11)$$

At competitive prices p_t^H and p_t^L , recurring coordination costs are

$$C_t^o(n, s) = \frac{n}{\bar{N}} [q(s)p_t^H + r[1 - q(s)]p_t^L]. \quad (12)$$

Formalization lowers these recurring costs whenever $rp_t^L > p_t^H$, because it replaces repeated informal problem solving with the application of an established procedure. Since the number of problems rises with employment, the associated saving is larger in bigger firms.

The same organizational state also determines the composition of producer services used directly in production:

$$P^p(s) = \left[q(s)^{1/\sigma_P} (H^p)^{\frac{\sigma_P-1}{\sigma_P}} + [1 - q(s)]^{1/\sigma_P} (L^p)^{\frac{\sigma_P-1}{\sigma_P}} \right]^{\frac{\sigma_P}{\sigma_P-1}}. \quad (13)$$

A higher formalization share therefore shifts producer-service use toward high-skill inputs. Because high-skill producer services are more expensive, formalization can raise the unit cost of production even as it reduces repeated coordination costs. The firm's organizational choice is therefore governed by a single tradeoff: formalization economizes on recurring problem solving but requires a more skill-intensive organization. The next subsection shows how the net return to this tradeoff varies with firm scale.

Static firm problem. After optimizing over labor composition, materials, and productive producer services, the firm's static operating profit can be written as

$$\pi_{jt}^*(a, s) = \max_{\ell \geq 0} \{ p_{jt}^v(q(s)) A_{jt} a \ell^\eta - [w_{jt}^v + c_{jt}^o(q(s))] \ell \}, \quad (14)$$

¹⁷ More precisely, if R denotes the discounted number of informal resolutions associated with the problems generated by a task over the expected life of the firm, then $r = [1 - \beta(1 - \delta_f)]R$. The parameter R is disciplined primarily by sectoral high-skill producer-service employment shares.

where

$$c_{jt}^o(q) = \frac{\nu_{jt}}{\bar{N}} [qp_t^H + r(1-q)p_t^L]$$

is recurring organizational cost per unit of effective labor, and $p_{jt}^v(q)$ is the shadow value of firm value added after materials and productive producer services have been chosen optimally. Its dependence on q captures the fact that using different organizational inputs changes the firm's unit cost in production. Appendix D derives both objects from the gross-output technology.

The firm's optimal effective labor and employment are

$$\ell_{jt}^*(a, s) = \left[\frac{\eta p_{jt}^v(q(s)) A_{jt} a}{w_{jt}^v + c_{jt}^o(q(s))} \right]^{\frac{1}{1-\eta}}, \quad n_{jt}^*(a, s) = \nu_{jt} \ell_{jt}^*(a, s). \quad (15)$$

Thus more productive firms are larger: employment rises with ability with elasticity $1/(1-\eta)$.

The key comparative static concerns the return to formalization. Combining the envelope condition with the firm's employment choice gives

$$\frac{\partial \pi_{jt}^*(a, q)}{\partial q} = n_{jt}^*(a, q) \Psi_{jt}(q), \quad (16)$$

where

$$\Psi_{jt}(q) \equiv \frac{w_{jt}^v + c_{jt}^o(q)}{\eta \nu_{jt}} \frac{\partial \log p_{jt}^v(q)}{\partial q} + \frac{rp_t^L - p_t^H}{\bar{N}}. \quad (17)$$

The first term is the marginal effect of formalization through the productive producer-service input. When high-skill producer services are more expensive than low-skill producer services, $p_t^H > p_t^L$, this term is negative: hiring skilled producer service workers raises unit cost and, in isolation, reduces desired firm scale. The second term is the recurring coordination saving: formalization replaces informal problem solving costing rp_t^L per problem with application of a formal procedure costing p_t^H . Thus $\Psi_{jt}(q)$ is the net marginal operating gain per worker from formalization, and the total operating gain is proportional to firm employment.

Proposition 1 (Firm scale and organizational formalization). *Fix sector j , date t , and organizational state q . Conditional on equilibrium prices, if $\Psi_{jt}(q) > 0$, the marginal operating benefit of formalizing an additional task is strictly increasing in firm employment:*

$$\frac{\partial \pi_{jt}^*(a, q)}{\partial q} = n_{jt}^*(a, q) \Psi_{jt}(q).$$

Hence larger and more productive firms obtain a greater operating return from the same organizational procedure. Moreover, if $rp_t^L > p_t^H$, then $\Psi_{jt}(q) > 0$ implies that greater formalization also raises the firm's desired employment.

Proof. See Appendix D.

The scale dependence in Proposition 1 becomes consequential once formalization requires a one-time installation cost. Because this cost does not scale proportionally with firm employment, larger firms find it profitable to formalize a broader set of tasks. The next subsection introduces the one-time cost of formalization, characterizes firms' dynamic organizational investment decisions, and then shows how newly created procedures can diffuse into public organizational knowledge.

4.3 Organizational investment, diffusion, and entry

This subsection turns the static scale mechanism into a dynamic one. Firms pay a one-time cost to formalize organizational tasks, and the procedures they create can subsequently diffuse into a public stock of organizational knowledge, lowering the cost of formalization for other firms and sectors. Entry determines equilibrium firm size and hence whether these organizational investments become profitable.

Creation and adaptation. Let $K_t \in [0, 1]$ denote the public stock of organizational knowledge, measured as the share of tasks for which a reusable procedure is publicly available. Because tasks are ordered by complexity, public knowledge takes the form of a frontier: a procedure for task z is publicly available whenever $\mathcal{F}(z) \leq K_t$. Formalizing a task requires a one-time organizational investment. If no public procedure exists, the firm must create one from scratch using high-skill producer-service labor; if a procedure is already contained in K_t , the firm can instead adapt it at a fraction of the creation cost:

$$C_t^C(z) = \frac{p_t^H}{\bar{e}_t^O} e^{z-1}, \quad (18)$$

$$C_{jt}^A(z) = \frac{\rho}{\xi_j} C_t^C(z), \quad \xi_m = 1, \quad \rho \in (0, 1). \quad (19)$$

Here $\bar{e}_t^O$ measures the average education embodied in the current workforce, inherited from the education choices made by households one generation earlier. For a given task, a more educated workforce requires fewer skilled-labor resources to produce a reusable procedure. Because education rises gradually over development, this technology also links the historical expansion of education to the timing at which increasingly complex organizational methods become economical to develop. The parameter ρ governs the common cost advantage of adapting an existing procedure, while ξ_j determines how adaptation costs vary across sectors.

A firm moving from inherited formalization frontier s to a new frontier $u \geq s$ therefore incurs the one-time investment cost

$$\begin{aligned} \mathcal{C}_{jt}^O(s, u; K_t) &= \int_s^u \tilde{C}_{jt}(z; K_t) d\mathcal{F}(z), \\ \tilde{C}_{jt}(z; K_t) &= \begin{cases} \min\{C_t^C(z), C_{jt}^A(z)\}, & \mathcal{F}(z) \leq K_t, \\ C_t^C(z), & \mathcal{F}(z) > K_t. \end{cases} \end{aligned} \quad (20)$$

Thus a task beyond the public frontier must be formalized by creating a new procedure, whereas a task inside the frontier can exploit the accumulated organizational knowledge of previous firms.

A firm produces in period t using its inherited frontier s and chooses the frontier $u \geq s$ carried into period $t + 1$. Because the economy is generally away from its limiting equilibrium, this is a nonstationary dynamic problem: the firm's choice depends on the entire perfect-foresight continuation path of aggregate states and prices. I suppress this dependence in the notation. Taking that continuation path as given, the firm's value satisfies

$$W_{jt}(a, s) = \pi_{jt}^*(a, s) + \max_{u \geq s} \{ -\mathcal{C}_{jt}^O(s, u; K_t) + \beta(1 - \delta_f)W_{j,t+1}(a, u) \}. \quad (21)$$

Because the organizational investment cost is fixed and Proposition 1 shows that the benefit to formalization rises with firm scale, larger firms therefore find it profitable to formalize a broader range of tasks and incidentally create more organizational knowledge.

Diffusion. Newly created procedures are initially proprietary. Let $\mu_{jt}(a, s)$ denote the distribution of active firms in sector j , let

$$N_{jt} = \int n_{jt}^*(a, s) d\mu_{jt}(a, s), \quad N_t = \sum_{j \in \mathcal{J}} N_{jt},$$

and let $c_{jt}(a, s; z) \in \{0, 1\}$ indicate that a firm creates at date t a procedure for task z that lies beyond the current public frontier. Economy-wide employment-weighted exposure to newly created proprietary procedures is

$$\mathcal{E}_t = \frac{1}{N_t} \sum_{j \in \mathcal{J}} \int n_{jt}^*(a, s) \left[\int_{\mathcal{F}(z) > K_t} c_{jt}(a, s; z) d\mathcal{F}(z) \right] d\mu_{jt}(a, s).$$

Thus $\mathcal{E}_t$ is larger when more procedures are created and when those procedures are created in firms employing more workers. Public organizational knowledge evolves according to

$$K_{t+1} = \min\{1, K_t + \delta_K \mathcal{E}_t\}, \quad (22)$$

where δ_K governs the rate at which proprietary procedures diffuse into the public stock. Because K_t is irreversible, a temporary period of large-scale production can therefore leave a persistent change in equilibrium skill-demand.

Entry. A potential entrant pays its setup cost before drawing permanent ability $a \sim G$ and enters with frontier $s = 0$. Let I_{jt} denote entry. Convex setup-input supply gives the marginal entry cost $F_j I_{jt}^{\phi_E}$. Free entry and the law of motion for establishment counts are summarized by

$$\begin{aligned} V_{jt}^E &= \int W_{jt}(a, 0) dG(a), & M_{jt} &= (1 - \delta_f) M_{j,t-1} + I_{jt}, \\ I_{jt} &\geq 0, & F_j I_{jt}^{\phi_E} - V_{jt}^E &\geq 0, \\ I_{jt} &\left[F_j I_{jt}^{\phi_E} - V_{jt}^E \right] &= 0. \end{aligned} \quad (23)$$

The shifter F_j governs equilibrium entry and therefore average firm size at the sectoral level. Through Proposition 1, it consequently determines the strength of firms' incentives to invest in formal organization.

4.4 Households and aggregate dynamics

The household block connects firms' organizational choices to the economy's structural transformation and long-run growth path. Nonhomothetic demand reallocates expenditure from agriculture through manufacturing and into services. Endogenous skill acquisition allows the resulting changes in firms' demand for high-skill labor to alter the skill composition of the workforce, while fertility and education

determine population dynamics. Together, these choices close the feedback between structural transformation, skill supply, population, and productivity growth. The household side builds on the quantity–quality mechanism of [Galor and Weil \(2000\)](#) and [Lagerlöf \(2006\)](#) and the nonhomothetic demand system of [Comin et al. \(2021\)](#); the productivity-growth technology introduced below follows the tradition of [Jones \(1995\)](#).

Fertility and education. Following [Galor and Weil \(2000\)](#) and [Lagerlöf \(2006\)](#), households choose fertility and educational investment in their children. Adults differ by employment origin $i \in \{a, m, s\}$, with s pooling all service activities and $\lambda_{i,t}$ denoting the corresponding population shares. Origin affects only the fixed time cost of raising a child, $\tau_i \equiv \tau\tau_i^n$; labor income and education costs are otherwise pooled across households. Preferences and child quality are

$$U_{i,t} = (1 - \gamma) \log C_{i,t} + \gamma \log[n_{i,t}h_{i,t}], \quad (24)$$

$$h_{i,t} = \frac{e_{i,t} + \tau_h}{e_{i,t} + \tau_h + g_t} \quad 0 < \tau_h \ll 1 \quad (25)$$

Where $n_{i,t}$ and $h_{i,t}$ denote fertility and child human capital, respectively. Here $g_t \equiv \sum_{i \in \{a, m, s\}} \lambda_{i,t} g_{i,t}^A$ is aggregate productivity growth. The small constant $\tau_h > 0$ keeps child quality strictly positive at the zero-education corner. As in [Galor and Weil \(2000\)](#), faster technological change raises the return to education, inducing households to invest more in their children and reduce fertility. This quantity–quality response provides the demographic feedback that stabilizes population in the long run: population becomes an endogenous adjustment margin rather than an exogenous source of sustained growth. Raising and educating a child requires $\tau_i + \tau_e(S_t)e_{i,t}$ units of parental time, where

$$\tau_e(S) = 1 - S(1 - \tau_e^s), \quad \tau_e^s \in (0, 1).$$

Thus, a more skilled economy makes educational investment less costly in parental time. This reduced-form complementarity captures the greater availability of educated parents, teachers, peers, and educational institutions in a more skilled environment and is consistent with education technologies featuring dynamic complementarity ([Cunha and Heckman, 2007](#); [Cunha et al., 2010](#)). It also links the terminal skilled share to long-run growth. At a given rate of technological change, a higher S_t lowers the time cost of educating children and raises fertility; restoring replacement fertility therefore requires faster technological change, which induces sufficiently greater educational investment. Proposition 2 shows that this makes the limiting growth rate increasing in the terminal skilled share. Let $\bar{w}_t$ denote pooled potential labor earnings per adult and d_t net dividends per adult. The household budget constraint is

$$\sum_{j \in \mathcal{J}} p_{j,t}^c C_{ij,t} = [1 - n_{i,t} \{\tau_i + \tau_e(S_t)e_{i,t}\}] \bar{w}_t + d_t.$$

Final demand and skill acquisition. Final consumption follows the nonhomothetic CES system of [Comin et al. \(2021\)](#):

$$\sum_{j \in \mathcal{J}} \omega_j^{1/\sigma} \left(\frac{C_{ij,t}}{C_{i,t}^{\epsilon_j}} \right)^{\frac{\sigma-1}{\sigma}} = 1, \quad 0 < \sigma < 1. \quad (26)$$

Income and relative-price effects reallocate expenditure away from agriculture, generate a hump-shaped manufacturing share, and eventually shift activity toward services. This provides the structural-transformation path along which firm scale, organizational investment, and knowledge diffusion evolve.

Members of the entering cohort also choose whether to acquire skill. I summarize this decision by

$$p_t^S = \mathcal{S}(\Pi_{t+1}), \quad \mathcal{S}'(\Pi) > 0, \quad (27)$$

where Π_{t+1} is the expected equilibrium skilled-wage premium next period and expectations are rational. In the quantitative model, $\mathcal{S}$ is parameterized as a smooth logistic choice probability.¹⁸ Greater formalization raises demand for high-skill producer-service labor and therefore affects the skilled composition of subsequent cohorts through the equilibrium skill premium.

Laws of motion. Let Δ denote generation length. Following the annualization in Lagerlöf (2006), a fraction $1/\Delta$ of the workforce is replaced each year.¹⁹ Let $\bar{n}_t$ denote average fertility. Population and the skilled share evolve according to

$$L_{t+1} = L_t \bar{n}_t^{1/\Delta}, \quad S_{t+1} = \left(1 - \frac{1}{\Delta}\right) S_t + \frac{1}{\Delta} p_t^S. \quad (28)$$

Sectoral value-added productivity growth increases with the economy's skilled-worker mass:

$$g_{j,t}^A \equiv \frac{A_{j,t+1}}{A_{j,t}} - 1 = \left[1 + \lambda_A (S_t L_t)^{\phi_j}\right]^{1/\Delta} - 1, \quad \phi_j \in (0, 1). \quad (29)$$

Consumer services and both producer-service industries share the service exponent ϕ_c , while gross-output productivity grows exogenously at rate g^{GO} .

Equation (29) builds on the central insight of Jones (1995) that a larger supply of workers raises productivity growth. In the standard semi-endogenous formulation, diminishing returns to the accumulated stock of knowledge—the idea that new ideas become progressively harder to find—eliminate the population-scale effect, but in doing so make the long-run growth rate depend on exogenous population growth. I take a different route. I omit the scale effect through the demographic equilibrium: technological change affects education and fertility, fertility converges to replacement, and population converges to a finite endogenous level. This formulation is natural in the present setting because over an extended time window population and the supply of skilled workers are themselves outcomes of the development process rather than exogenous inputs into growth. The resulting limiting growth rate is therefore fully endogenous and remains positive even when population growth is zero. Proposition 2 characterizes this equilibrium formally.

Equilibrium. Manufacturing is the numeraire. Given the initial state of the economy, a perfect-foresight competitive equilibrium is a sequence of prices, household choices, firm policies, entry, firm

¹⁸ Specifically, $\mathcal{S}(\Pi) = \left[1 + \exp\left(-\frac{\beta_\pi \log \Pi - c^{\text{skill}}}{\sigma_{\text{skill}}}\right)\right]^{-1}$.

¹⁹ Time is measured annually throughout the model, while Δ denotes generation length. Household fertility and cohort skill choices are generational objects, which I map into annual population and workforce dynamics following the annualization in Lagerlöf (2006).

distributions, and aggregate states such that households optimize; firms solve (14) and (21); entry satisfies (23); final-good, producer-service, and labor markets clear; and the firm distribution, public organizational knowledge, population, skills, and sectoral productivities follow their respective laws of motion. Aggregate uncertainty is absent, although firms face idiosyncratic exit risk and entrants draw heterogeneous ability. A formal definition is provided in Appendix D.²⁰

4.5 Limiting growth and the manufacturing legacy

I characterize the model's long-run behavior in two steps. First, I show that the terminal skilled share pins down the economy's limiting productivity growth rate, while population scale adjusts to the finite level required to sustain it. Second, I show how a temporary manufacturing phase can permanently raise the terminal skilled share, and thus long-run growth, by expanding the stock of public organizational knowledge inherited by the service economy. For the analytical results, I assume that broad services, including consumer and producer services, absorb the workforce in the limit; in the calibrated economy they account for 92 percent of terminal employment.²¹

Proposition 2 (Endogenous population scale and limiting growth). *Fix a terminal skilled share $S \in (0, 1)$ and let*

$$\zeta \equiv 1 + \frac{d^\infty}{\bar{w}^\infty}$$

denote the full-income ratio in the terminal service equilibrium. Suppose $\gamma\zeta > \tau_s$ and, for this analytical result, $\tau_h = 0$. Population converges from every $L_0 > 0$ to a unique steady state $L^\infty(S; \zeta)$, and fertility converges to replacement. At that steady state,

$$g^\infty = \mathcal{G}(S; \zeta) \equiv \frac{(\gamma\zeta - \tau_s)^2}{\tau_s} \times \frac{1}{\tau_e(S)}, \quad (30)$$

$$L^\infty(S; \zeta) = \frac{1}{S} \left[\frac{[1 + \mathcal{G}(S; \zeta)]^\Delta - 1}{\lambda_A} \right]^{1/\phi_c}. \quad (31)$$

Holding ζ fixed,

$$\frac{\partial \mathcal{G}(S; \zeta)}{\partial S} > 0.$$

Proof. See Appendix D.

Proposition 2 follows from a stabilizing feedback between population, technological change, and fertility. Conditional on the terminal skilled share, a larger population raises the mass of skilled workers SL and hence productivity growth. Faster technological change increases the return to educating children, raising educational investment and reducing fertility. Consequently, when population is below L^∞ , productivity growth is too low to induce the education required for replacement fertility and population expands; when population is above L^∞ , the reverse occurs. This feedback selects a unique finite population at which fertility equals replacement. The dependence of limiting growth on the skilled share follows from $\tau'_e(S) < 0$: a higher S makes education more effective, so at the previous growth rate children require less parental time and fertility rises above replacement. Restoring replacement fertility

²⁰ Computationally, I solve the perfect-foresight transition using the extended-path logic of Fair and Taylor (1983). I conjecture the aggregate path, solve firms' dynamic decisions backward along that path, propagate the firm distribution and aggregate states forward, and iterate until the conjectured and implied aggregate sequences coincide.

²¹ The remaining nonservice employment in the calibrated economy reflects input-output linkages: even at high income levels, service production continues to require agricultural and manufacturing intermediates.

requires faster technological change, which induces sufficiently greater educational investment. Thus the terminal skilled share determines the limiting growth rate, while population adjusts endogenously to the level required to sustain it.

I next ask when passing through manufacturing places the economy on a permanently higher growth path. Proposition 2 establishes that limiting productivity growth is determined by the terminal skilled share. Manufacturing therefore raises long-run growth only if it leaves the subsequent service economy sufficiently more skill intensive to raise S^∞ . In the model, this occurs through organizational knowledge: large firms create reusable procedures, those procedures diffuse, and service firms later formalize more tasks and demand more high-skill producer-service labor than they would have without the manufacturing phase. To characterize when this chain of events occurs in the model, consider two economies beginning from the same state (S_1, L_1, K_1) . Economy A passes through manufacturing before entering services, whereas economy B moves directly from agriculture into services. I isolate the mechanism using a three-stage comparison with symmetric static firms, common prices and operating-surplus terms, and a common terminal full-income ratio.²² I additionally require that the service economy is not organizationally saturated, so that access to an additional public procedure induces some service firms to formalize an additional task. Appendix D states this condition formally.

Proposition 3 (Manufacturing and long-run growth). *Under the preceding restrictions,*

$$g_A^\infty > g_B^\infty \iff S_A^\infty > S_B^\infty \iff K_{A,3} > K_{B,3} \iff \underbrace{\frac{\tau_m}{\tau_s}}_{\text{education}} \underbrace{(S_1 L_1)^{\phi_m - \phi_c}}_{\text{productivity growth}} \underbrace{\left(\frac{F_m}{F_c}\right)^{\frac{2}{1+\phi_E}}}_{\text{firm scale}} > 1. \quad (32)$$

Proof. See Appendix D. Proposition 3 makes the model's central mechanism explicit. Manufacturing raises long-run growth when it leaves the subsequent service economy with a larger stock of public organizational knowledge: cheaper adaptation induces service firms to formalize more tasks, raising HSPS demand, the return to skill, and ultimately S^∞ and g^∞ . Condition (32) shows when this occurs. The terms τ_m/τ_s and $(S_1 L_1)^{\phi_m - \phi_c}$ capture the education force along the transition, through differences in household education choices and sectoral productivity growth, respectively; both affect the availability of educated labor able to develop new procedures. The term $(F_m/F_c)^{2/(1+\phi_E)}$ captures the complementary scale force: through entry, it determines relative firm size and hence the private return to formalization. The condition therefore summarizes the interaction between the capacity to create organizational knowledge and the scale at which creating it becomes worthwhile. The quantitative model relaxes the simplifying restrictions behind the proposition and determines the strength of these forces in full general equilibrium.

5 Calibration

I calibrate the model at an annual frequency to the U.S. development path from 1850 to 2019. The economy is initialized in 1790, allowing the endogenous state variables to evolve before the first targeted observations, and is solved through a terminal horizon well beyond 2019. Parameters fixed by normalization,

²² For this analytical comparison I also set $\tau_h = 0$ and suppress the annualization.

timing conventions, direct empirical evidence, or prior literature are assigned externally, while the remaining 19 parameters are estimated jointly by simulated method of moments (SMM). Appendix D.4 describes the data, external parameter choices, numerical solution, parameter estimates, and local identification analysis.

Let $m_{bk}(\Theta)$ and $\hat{m}_{bk}$ denote the model and data values of moment k in block b . I estimate Θ by minimizing the block-weighted SMM objective

$$r_{bk}(\Theta) = \frac{m_{bk}(\Theta) - \hat{m}_{bk}}{\max\{|\hat{m}_{bk}|, 0.01\}}, \quad \mathcal{L}(\Theta) = \frac{\sum_b \omega_b K_b^{-1} \sum_{k=1}^{K_b} r_{bk}(\Theta)^2}{\sum_b \omega_b}. \quad (33)$$

The normalization expresses discrepancies relative to the magnitude of each target, with the 0.01 floor preventing moments close to zero from receiving disproportionate weight. Averaging within each block prevents long time series from mechanically dominating scalar targets.²³

The set of moment blocks disciplines the main quantitative margins of the model. Sectoral employment paths, normalized GDP per capita, and the height and timing of the manufacturing employment peak discipline structural transformation and aggregate development. Fertility paths and the agriculture-to-services fertility ratio discipline the demographic transition. The aggregate skilled share, skilled earnings premium, and sectoral value-added skill shares discipline skill supply and demand. Sectoral and aggregate high-skill producer-service (HSPS) shares, together with sectoral firm size, discipline the scale-formalization mechanism. Finally, sectoral value-added productivity growth disciplines the productivity-growth technology. Scalar targets are measured in 2019 in both the model and the data unless otherwise noted; the agriculture-to-services fertility ratio is measured over its historical window, while manufacturing peak timing is treated as an interval target, with dates between 1930 and 1960 receiving no penalty. Appendix D.4 reports the data sources, measurement windows, block weights, and construction of each target.

5.1 Measurement

Producer and consumer services. The model distinguishes services by their use rather than by the industry of the supplier. This distinction matters because transportation, wholesale trade, finance, insurance, real estate, legal services, and other business services sell to both firms and final consumers. I combine 2019 IPUMS USA microdata with the 2019 BEA Summary Use Table. Within agriculture, manufacturing, and consumer-service industries, occupations are divided between sectoral value-added work and organizational or technical producer-service tasks. Within each group workers are then divided by education, with high skill defined as at least a bachelor’s degree.

Employment in industries that primarily supply intermediate services is allocated across destination sectors using BEA intermediate-use expenditure shares; the portion sold to final consumers is assigned to consumer-service value added. This construction yields sector-specific HSPS and skilled value-added employment shares without imposing a mutually exclusive producer-service industry classification. The IPUMS–BEA concordance is described in Appendix D.4. The allocation applies a common skill composition and labor requirement to an industry’s sales across buyers, the identifying restriction required because worker records cannot be linked to the purchaser of the industry’s output.

²³ I assign a larger weight to the manufacturing peak share, manufacturing peak timing, sectoral productivity growth, and sectoral HSPS shares. The additional weight is placed on moments that directly discipline the paper’s central structural-transformation and organizational mechanisms. Appendix D.4 reports the complete weighting scheme.

Firm scale. Modern firm-size targets come from the 2019 Census Statistics of U.S. Businesses and are measured as employees per employer firm. Average firm sizes are 7.435 workers in agriculture, 49.694 in manufacturing, and 20.454 in broad services. I calibrate the model’s scale dimension using firm size rather than establishment size as an establishment size measure would disproportionately lower scale in services, where multi-establishment firms are especially common (Hsieh and Rossi-Hansberg, 2023). A common initial scale of 10.7 workers is anchored in both manufacturing and services to the 1880 Census of Manufactures; the subsequent historical manufacturing observations, excepting the 2019 targets, are used to assess the model’s performance in generating the timing of firm size growth and decline over the development path.

5.2 Parameters

Parameters governing standard preferences, firm dynamics, substitution elasticities, and units are fixed before estimation using conventional values or normalizations; Appendix D.4 reports the complete list and their sources. I discuss here the assignments that are most specific to the mechanism.

The dispersion of idiosyncratic skill-acquisition costs, σ_{skill} , determines the elasticity of skill supply to the expected skilled earnings premium. With the normalization $\beta_{\pi} = 1$, the logit rule in equation (27) implies

$$\sigma_{\text{skill}} = \frac{\Delta \log \Pi}{\Delta \log \{p^S / (1 - p^S)\}}.$$

I discipline this parameter using the long-run U.S. joint evolution of college attainment and the college earnings premium. Using bachelor-or-higher attainment and the corresponding skilled–unskilled earnings premium gives an implied value close to 0.10, which I use as the baseline. This follows the quantitative education-choice approach of Heathcote et al. (2010), who discipline heterogeneity in college costs using aggregate changes in educational attainment and the college wage premium; the logit random-utility structure follows McFadden (1974). The location parameter c_{col} is instead estimated jointly and primarily disciplines the level of the skilled share.

The knowledge-diffusion parameter δ_K governs how quickly exposure to newly created proprietary procedures expands the public stock of organizational knowledge. I set $\delta_K = 0.09$. Mansfield (1961) reports a mean time of 7.8 years from first commercial use to 50 percent adoption across twelve industrial innovations, corresponding to an annual closing rate of $1 - 2^{-1/7.8} = 0.085$. Appendix D.4 reports the full mapping and shows that the headline results are robust to significant variations in δ_K .

Two additional organizational parameters are assigned rather than estimated directly. The scale normalization $\bar{N} = 25$ implies that recurring coordination absorbs approximately five percent of employment in the calibrated economy, an order of magnitude consistent with the organization-capital evidence in Atkeson and Kehoe (2005). The adaptation-cost ratio is set to $\rho = 0.40$, implying a substantial but incomplete cost advantage to adapting an existing procedure rather than developing it from scratch. This is consistent with evidence that established management practices can be transferred across firms through consulting, training, and managerial knowledge flows, and that informational barriers are an important obstacle to their adoption (Bloom et al., 2013; Giorcelli, 2019). Because these studies do not identify the adaptation-to-creation cost ratio directly, I treat ρ as a maintained benchmark and report sensitivity to it in Appendix D.4.

The remaining 19 parameters are chosen jointly by simulated method of moments. Table D.4 reports their estimates and principal empirical disciplines. Fertility moments primarily discipline (τ, τ_a^n) ; sectoral

productivity growth and the GDP-per-capita path discipline $(\phi_a, \phi_m, \phi_c, g_{GO})$; aggregate and sectoral skill shares discipline $(c_{col}, \alpha_a, \alpha_m, \alpha_c)$; structural-change paths discipline (ϵ_c, w_a, w_c) ;²⁴ sectoral HSPS shares discipline (R, ξ_a, ξ_c) ; and sectoral firm-size moments discipline (F_a, F_m, F_c) . These mappings describe the principal sources of variation rather than one-parameter-one-moment identification: all 19 parameters are estimated jointly.

The sectoral adaptability parameters are the least precisely identified objects in the calibration. In particular, $\xi_c = 6.8774$ places consumer services near the saturation region of the adaptation technology, where further increases have little effect on model moments. The HSPS targets strongly reject low adaptability but are less informative about its precise value once service firms are able to exploit most procedures developed elsewhere.²⁵ Appendix Figure D.2 reports the corresponding parameter profiles and singular-value diagnostics. Two additional parameters lie close to economically meaningful bounds. The estimated service nonhomotheticity parameter, $\epsilon = 1.0105$, implies only a modest direct luxury-good effect for services. The estimate is lower than the service nonhomotheticity typically obtained in models using CLM-type preferences, but the comparison is not one-to-one: here ϵ is estimated jointly with sector-specific productivity growth, time varying input-output linkages, and endogenous firm organization. As emphasized by [Herrendorf et al. \(2013\)](#), the apparent importance of income effects in the structural change literature depends importantly on production structure of the model and how it is mapped to the data. In this model, the estimate therefore implies that the U.S. service transition is not being generated primarily by a large income effect. The agricultural growth elasticity, $\phi_a = 0.9984$, is similarly close to its upper bound and implies nearly linear sensitivity of agricultural productivity growth to skilled-worker mass. This reflects the high historical rate of agricultural productivity growth relative to manufacturing and services, which the common growth shifter must reconcile through sectoral elasticities.

5.3 Fit and untargeted validation

Figure 5 compares the model with the U.S. employment transition. The model reproduces the decline of agriculture, the rise of services, and the hump-shaped manufacturing path. Manufacturing peaks in 1928 at 24.93 percent of employment, compared with 28.28 percent in 1950 in the data. Table D.4 reports all the calibrated parameters next to the fit associated with their principal empirical discipline. The moments most directly tied to the paper’s mechanism—sectoral firm size and HSPS shares—are matched almost exactly, with the largest deviation below 1 percent.

Four untargeted implications evaluate the calibrated transition. First, decadal GDP-per-capita growth is computed from the targeted GDP-level series but does not itself enter the objective. Across the 22 decades in Figure 4, the model matches the hump-shaped pattern from low growth in the early nineteenth century, to faster growth during the industrial expansion, and then to a gradual slowdown as the economy transitions toward services. Second, I compare the model’s household education choice with the average of primary and secondary enrollment rates over 1825–2020. The correlation between the model’s education choice and enrollment is 0.94, so the quantity–quality household block captures the secular expansion of education remarkably well despite the historical education path being untargeted. Third, the model generates substantial cross-firm dispersion in formalization and a secular increase in

²⁴ I set ϵ_a to 0.01 consistent with significant subsistence constraints, [Kongsamut et al. \(2001\)](#); [Herrendorf et al. \(2014\)](#).

²⁵ Given observed service-sector firm size, matching the consumer-service HSPS share requires firms to make extensive use of procedures created outside the sector. Once this margin is nearly exhausted, higher values of ξ_c have little additional effect.

Table 3: Parameter-Level Local Identification Diagnostics

Parameter	Estimate	Unique share (%)	HC1 relative SE (%)
τ	0.03236	17.92	4.86
τ_a^n	0.44017	30.66	5.92
ϕ_a	0.99840	15.49	10.15
ϕ_m	0.91919	14.04	10.38
ϕ_c	0.69046	15.97	10.86
g_{GO}	0.00318	61.33	24.31
c_{col}	0.56821	35.90	5.36
α_a	0.24338	81.42	2.81
α_m	0.33001	84.90	2.96
α_c	0.40995	82.84	2.62
ϵ_c	1.01048	24.87	30.15
w_a	0.65105	34.49	12.44
w_c	3.82326	25.57	39.53
R	22.78696	33.06	5.33
ξ_a	3.56402	64.17	88.56
ξ_c	6.87739	70.22	491.31
F_a	25.872	80.00	48.09
F_m	187.625	69.56	60.27
F_c	85.117	91.00	22.85

Notes: The unique shares use the exact, smooth, nonredundant 82-by-19 Jacobian. HC1 standard errors are computed in log-parameter space and translated to levels by the delta method.

the dispersion of firm size, both untargeted and well-known empirical features of firm dynamics.²⁶ Finally, the model captures the rise and subsequent decline of average manufacturing firm size and the growth of the HSPS employment share even though only the terminal observations of these series are targeted.²⁷ The next subsection provides the strongest untargeted validation of the model’s novel mechanism by showing that the accumulation of public organizational knowledge in the model closely tracks the historical production of books and manuals codifying new methods of business organization and management in the data.

5.4 Organizational knowledge in the historical record

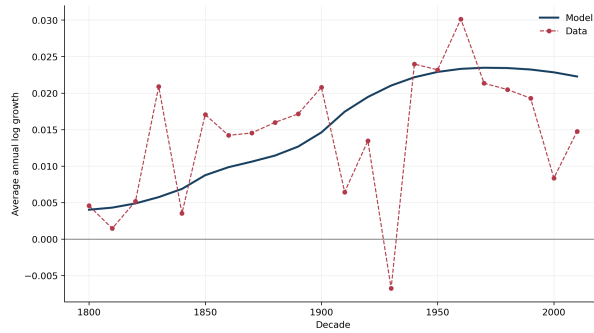
I assess the timing of public organizational knowledge using an independent historical series. The model’s annual increment,

$$\Delta K_t \equiv K_{t+1} - K_t, \quad (34)$$

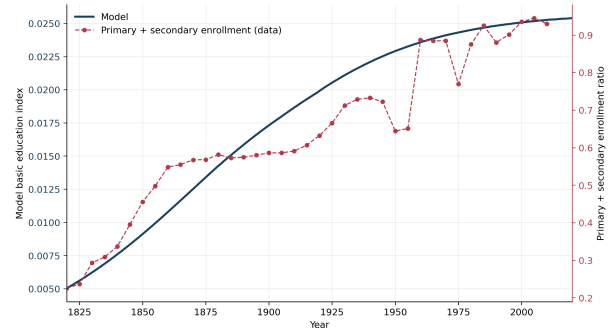
measures additions to the stock of public knowledge concerning business organization which firms can draw on to formalize production. This object has a natural empirical counterpart: the publication of books and manuals through which methods of business organization and management became codified

²⁶ See, for instance, [Poschke \(2018\)](#) for the secular rise in firm-size dispersion and [Bloom et al. \(2012\)](#) for dispersion in management practices across firms. The employment-weighted standard deviation of firm-level formalization in the model is 0.176 in agriculture, 0.187 in manufacturing, and 0.049 in services. I construct data-based proxies of 0.229, 0.205, and 0.211 using the DEFRA Farm Business Survey 2024/25 for agriculture and the ONS Management and Expectations Survey 2019 for manufacturing and services. None of these moments enters the calibration objective.

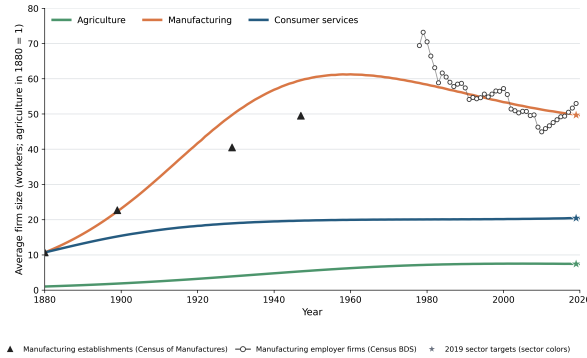
²⁷ For firm size, the initial level is normalized to its value in the data, so the untargeted validation comes from the subsequent path, in particular the decline in manufacturing firm size after the manufacturing peak.



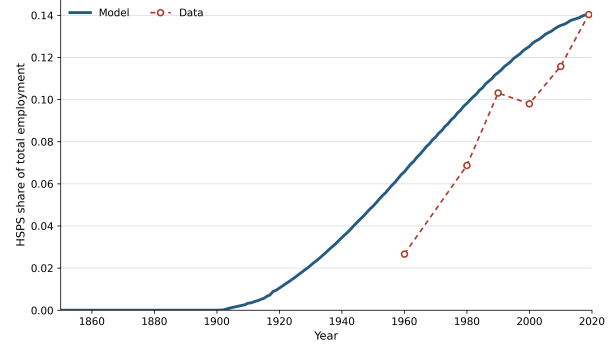
(a) Decadal GDP-per-capita growth



(b) Basic education and school enrollment



(c) Sectoral firm size



(d) High-skill producer-service employment

Figure 4: Historical-path validation. Panel (a) reports annualized GDP-per-capita growth by decade; the model line is the seven-decade quadratic low-frequency trend, with the unsmoothed series available in the replication files. Panel (b) compares the model’s basic-education index with the primary-plus-secondary enrollment ratio. Because the two education measures have different units, panel (b) evaluates timing and shape. Panel (c) compares the model’s sectoral firm-size paths with historical manufacturing establishment sizes, later manufacturing employer-firm sizes, and the 2019 sectoral calibration targets. Panel (d) compares the model-implied employment share of high-skill producer services with its empirical counterpart, constructed using harmonized IPUMS occupations and industries together with BEA input-use weights.

and publicly available. I construct this series from digitized volumes of the Library of Congress *Catalog of Copyright Entries*. Individual titles are parsed and classified using terms associated with accounting and record keeping, office and business systems, management methods, and the organization of sales, personnel, and multi-unit firms. The classification includes general Taylorist concepts such as “scientific management,” “time study,” and “motion study,” and excludes titles containing explicit manufacturing or industrial language.²⁸ This produces a measure of organizational knowledge that is not mechanically driven by references to manufacturing. Using the same underlying data source and classification method, I also construct analogous series for 11 different placebo topics ranging from children’s literature to fields which are more plausibly connected to manufacturing such as chemistry and applied physics.

²⁸ For instance titles including the words “industrial, manufacturing, factory, etc.” are excluded.

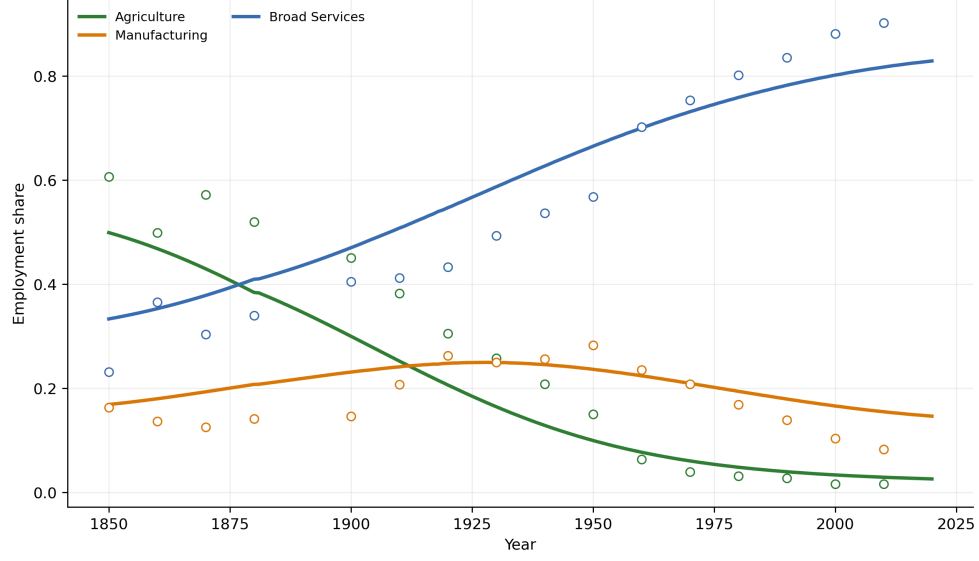


Figure 5: Fit to U.S. structural change. Lines are model outcomes and circles are historical employment shares. The sectoral employment paths enter the calibration objective.

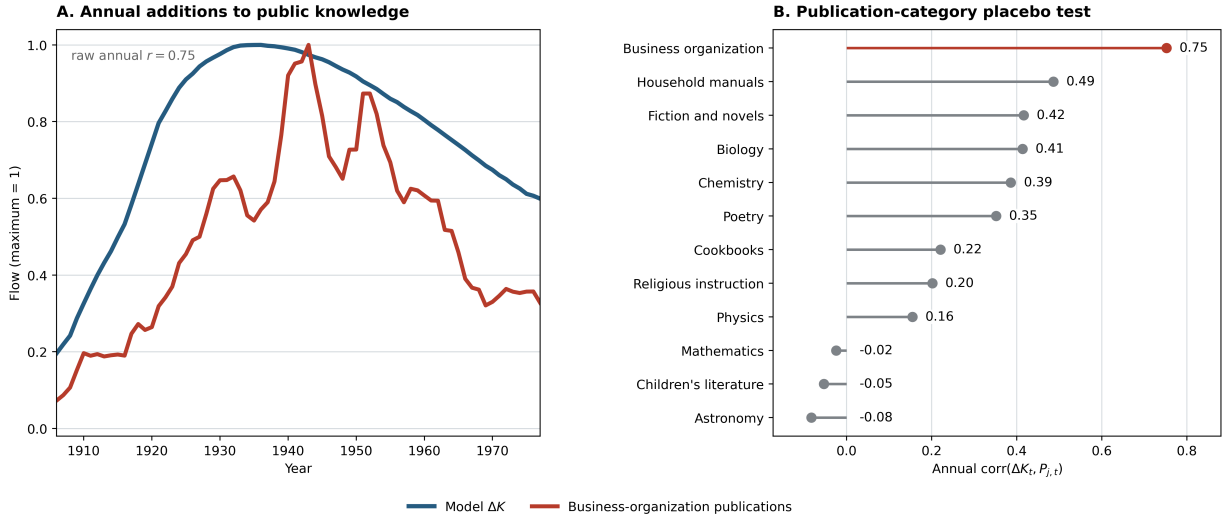


Figure 6: Public organizational knowledge and copyright publications. Panel A compares the annual increase in the model's public knowledge stock with business-organization publication intensity. The displayed series are centered five-year averages normalized by their own maxima; the reported correlation uses unsmoothed annual observations. Panel B reports the corresponding annual correlation for the business-organization category and eleven placebo publication categories. All publication series use identical processing and the common 1906–1977 support.

Because the number and size of digitized catalog volumes vary across years, I measure publication intensity rather than raw counts. For publication category g , define

$$P_{gt} \equiv \frac{\text{matched copyright records}_{gt}}{\text{OCR words}_t}. \quad (35)$$

All categories use the same annual support and OCR denominator. The common sample contains 72 annual observations from 1906 through 1977.

Figure 6 presents the comparison. Panel A compares ΔK_t with business-organization publication intensity. The displayed lines are centered five-year averages normalized by their respective maxima; the reported correlation uses the unsmoothed annual observations. The correlation is 0.752. Both series rise during the first half of the twentieth century, peak around mid-century, and subsequently decline.

Panel B evaluates the specificity of this relationship using eleven placebo categories spanning scientific, technical, household, and literary subjects. Business-organization publications rank first, with an annual correlation of 0.752. The closest placebo is household manuals, with a correlation of 0.486; all remaining correlations are 0.416 or lower. In one-sided comparisons using Newey–West standard errors with four annual lags, the business correlation exceeds each placebo correlation. The largest comparison p -value is 0.069. The ranking does not depend solely on the common hump: using five-year log changes, business organization remains first ($r = 0.543$, compared with 0.355 for fiction and novels), and the difference is significant using Newey–West standard errors with five lags (one-sided $p = 0.033$). During the period in which the model predicts rapid additions to public organizational knowledge, the historical record shifts disproportionately toward works about business organization. This pattern reflects the co-movement of manufacturing employment and organizational-knowledge production that appears in both the model and the data. To the best of my knowledge, this is the first study to document that the production of codified business-organization knowledge moved systematically with the rise and decline of manufacturing employment and to relate that historical comovement to the knowledge flow generated by a structural model.

6 Results

6.1 Manufacturing exposure and development outcomes

I use the calibrated model to quantify how the historical importance of manufacturing shapes subsequent development. Starting from the U.S. calibration, I vary only the manufacturing weight in final demand, w_m . This traces an equilibrium locus of otherwise identical economies with different endogenous peak manufacturing employment shares.²⁹ The eight model economies in Figure 7 span manufacturing peaks from 6.35 to 38.80 percent of employment. Larger manufacturing peaks are associated with greater tertiary education, a composition of services tilted towards HSPS employment, higher 2019 GDP per capita, and faster GDP-per-capita growth over 2010–2019. They are also associated with lower population growth over 1850–2019, as in the data. In the model, deeper industrialization raises productivity growth, which increases the return to education and lowers fertility. The model performs best on the outcomes closest to the organizational mechanism: it accounts for 98 percent of the empirical HSPS gradient and 32 percent of the tertiary-education gradient. It also accounts for 40 percent of the 2010–2019

²⁹ The experiment also has a price-wedge interpretation. Suppose the manufacturing demand weight changes from w_m to $f_m w_m$, and let $k_j \equiv \varepsilon_j / \varepsilon_m$. Holding w_m fixed, the same relative final-demand schedules are obtained by applying wedges to the prices of agriculture and consumer services such that

$$1 + \tau_a = f_m^{-k_a/(1-\sigma)}, \quad 1 + \tau_c = f_m^{-k_c/(1-\sigma)}.$$

Varying w_m is therefore isomorphic, for household final demand, to varying relative price wedges on agricultural and service consumption. The exercise can be read as a parsimonious change in the relative pull of manufacturing, rather than as a structural change in preferences. Appendix D.4 derives this equivalence.

GDP-per-capita growth gradient, 17 percent of the 2019 GDP-per-capita gradient, and 17.6 percent of the population-growth gradient. Variation in manufacturing history therefore generates a substantial share of the observed cross-country differences in service composition, higher education, and growth in income per capita, while accounting for a smaller, but still substantial, share of differences in income levels and demographic change.

The cross-country accounting exercise describes differences observed around the end of the calibration period, but the model implies that the consequences of industrial history persist beyond that point. Because a deeper manufacturing phase leaves the service economy with a larger skilled share, Proposition 2 implies a permanently higher limiting productivity-growth rate. Table 4 illustrates this mechanism by comparing two otherwise identical economies with different manufacturing histories. The low-peak economy reaches a manufacturing employment share of 11.19 percent, close to the average peak among low- and middle-income economies, while the high-peak economy reaches 29.88 percent. The high-peak economy enters the service era with a persistently larger skilled-employment share and therefore grows faster thereafter. The annual growth gap is modest—0.25 percentage points in 2030 and 0.17 points in 2070—but compounds over time: GDP per capita in the high-peak economy rises from 39 percent above the low-peak economy in 2030 to 51 percent above it by 2070. In the model, industrial history therefore affects not only the level of income reached by the end of industrialization, but also the subsequent growth path of the service economy.

Table 4: Manufacturing history and subsequent growth paths

Economy	Outcome	2030	2050	2070
High-peak economy	GDP per capita	50.36	78.43	121.80
	Skilled-employment share (%)	34.18	34.64	35.02
	GDP-per-capita growth (% p.a.)	2.24	2.22	2.20
Low-peak economy	GDP per capita	36.12	53.88	80.72
	Skilled-employment share (%)	28.74	29.37	29.83
	GDP-per-capita growth (% p.a.)	1.99	2.01	2.03
High-low comparison	GDP per capita (low peak = 100)	139.44	145.57	150.89
	Skilled-employment-share gap (pp)	5.44	5.27	5.19
	GDP-per-capita growth gap (pp p.a.)	0.25	0.21	0.17

Notes: The low-peak economy sets $w_m = 0.60$ and reaches a manufacturing-employment peak of 11.19 percent in 1941. The high-peak economy sets $w_m = 3.49$ and reaches a peak of 29.88 percent in 1927. GDP-per-capita levels equal nominal GDP divided by the consumption price index and population. The skilled-employment share is the model stock S_t . GDP-per-capita growth is the annualized log change in chain-weighted real GDP per capita over the preceding ten years. In the GDP-per-capita comparison, the low-peak economy is normalized to 100. The skill and growth comparisons report high-minus-low differences in percentage points.

6.2 Which mechanisms matter?

To understand which source of sector-heterogeneity is driving the results, I repeat the manufacturing-demand experiment after removing one source of heterogeneity at a time. Each experiment equalizes one of the following: entry barriers F_j , sectoral productivity growth ϕ_j , adaptability ξ_j , or value-added skill intensity α_j . The final experiment sets $\delta_K = 0$, eliminating the diffusion of newly formalized routines into public organizational knowledge. All other parameters remain at their calibrated U.S. values.

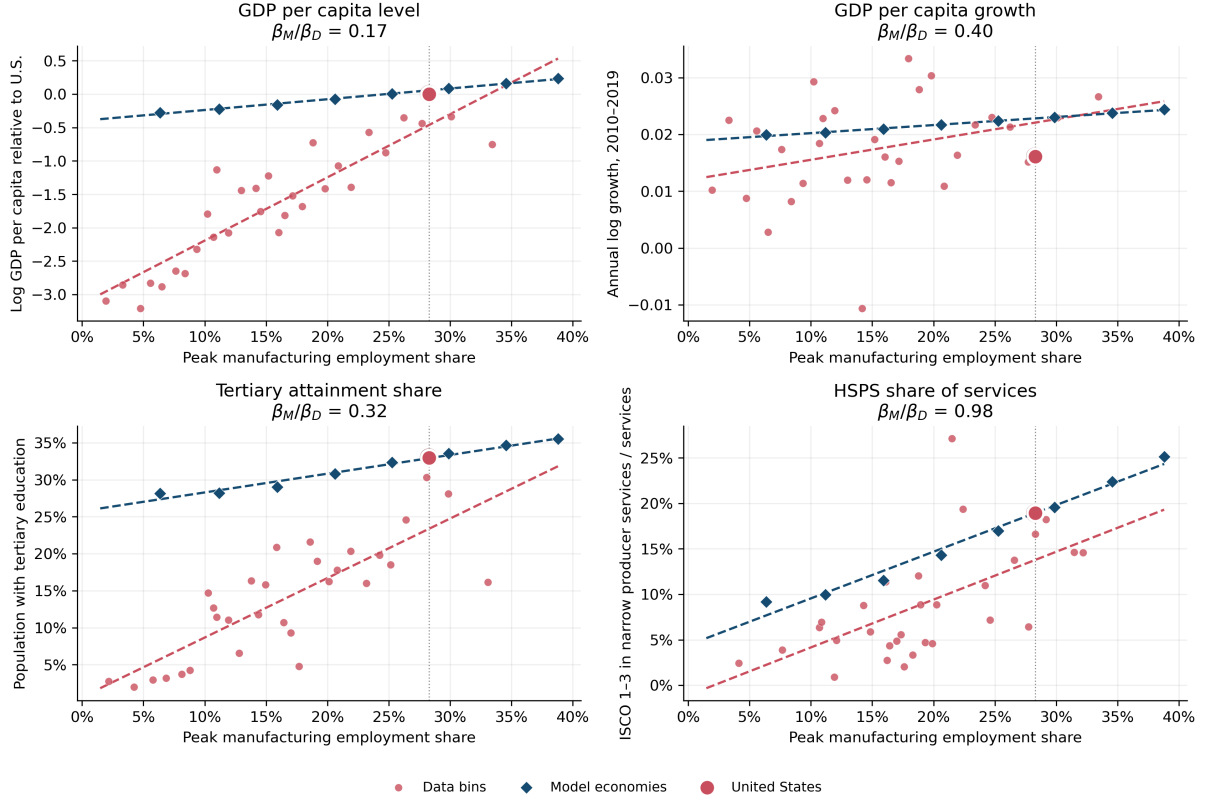


Figure 7: Manufacturing peaks and development outcomes. Red circles show means within 30 approximately equal-sized country bins; blue diamonds show model economies. Dashed lines are OLS fits to the underlying observations; β_M/β_D denotes the ratio of model to empirical slopes. The large red circle identifies the United States. Manufacturing peaks use the harmonized historical-employment series described in Appendix A.1. GDP levels (2019, in logs relative to the United States) and annual log growth (2010–2019) use Gapminder’s PPP GDP-per-capita series. Completed tertiary attainment averages Barro–Lee observations for 2010 and 2015. HSPS is employment in ISCO-08 groups 1–3 within ISIC Rev. 4 divisions 62–66 and 69–75, divided by employment in market services (sections G–N and R–S), using ILOSTAT data averaged over 2010–2019. Samples comprise 172 countries for GDP, 137 for education, and 48 for HSPS. U.S. manufacturing and education observations use the empirical calibration targets.

Because each intervention changes the mapping from w_m into structural transformation, every coefficient is estimated against the counterfactual economy’s realized manufacturing peak. Table 5 shows the results.

The results in Table 5 isolate one source of cross-sector heterogeneity as the main force behind the paper’s quantitative cross-country accounting results. Equalizing F_j leaves 49.0 percent of the GDP-level slope and 40.7 percent of the growth slope, *reverses* the tertiary-attainment slope, and reduces the HSPS slope to 12.7 percent of baseline. Thus, manufacturing exposure raises the skill-share and producer-service employment primarily when sectoral entry conditions allow manufacturing firms to operate at scale. Knowledge diffusion propagates this effect. Setting $\delta_K = 0$ preserves 71.3 percent of the GDP-level slope and 67.6 percent of the growth slope, but only 26.5 percent of the tertiary slope and 47.2 percent of the HSPS slope. Private formalization still depends on manufacturing exposure through the model’s

Table 5: Mechanism accounting for the model’s cross-country slopes

Outcome	Baseline	Equal entry barriers	Equal productivity growth	Equal adaptability	Equal VA skill intensity	No knowledge spillover
GDP-per-capita level, 2019	1.720 [100.0]	0.843 [49.0]	1.314 [76.4]	1.594 [92.7]	1.871 [108.8]	1.225 [71.3]
GDP-per-capita growth, 2010–2019	0.015 [100.0]	0.006 [40.7]	0.018 [118.7]	0.013 [89.7]	0.015 [99.8]	0.010 [67.6]
Population growth, 1850–2019	−0.678 [100.0]	−0.244 [36.0]	−0.010 [1.5]	−0.630 [93.0]	−0.832 [122.8]	−0.460 [67.9]
Tertiary attainment, 2010–2019	0.301 [100.0]	−0.116 [−38.4]	0.246 [81.8]	0.235 [77.9]	0.340 [113.0]	0.080 [26.5]
HSPS share of services, 2010–2019	0.580 [100.0]	0.074 [12.7]	0.691 [119.1]	0.485 [83.6]	0.579 [99.7]	0.274 [47.2]

Notes: Each cell reports a model slope; brackets report its percentage of the baseline slope. Outcomes match Figure 7. All columns use the converged common support $w_m \in \{1.16, 1.8, 2.56, 3.49, 4.69\}$ and realized manufacturing peaks. Counterfactuals equalize F_j , ϕ_j , or α_j across sectors; set $\xi_a = \xi_c = \xi_m = 1$; or set $\delta_K = 0$, as indicated.

other channels, but the effect of the manufacturing legacy is substantially reduced. The population-growth relationship is driven more directly by sectoral productivity growth: equalizing ϕ_j reduces the population-growth slope to 1.5 percent of baseline. This is consistent with the demographic mechanism in the model, through which faster productivity growth raises the return to education and lowers fertility.

Equalizing value-added skill shares leaves all four slopes essentially unchanged or slightly larger on account of the large value-added skill-share of consumer services. Equal adaptability retains between 77.9 and 92.7 percent of the baseline slopes. Equal productivity growth reduces the GDP-level and tertiary slopes while increasing the growth and HSPS slopes. Sectoral entry conditions therefore generate the scale margin which makes manufacturing especially important and knowledge diffusion transmits its effects to the broader economy.

6.3 Robustness

The results are stable across a broad sensitivity battery. Across 18 economic variants all 90 cross-country slopes retain their benchmark sign. Median absolute changes are 2.9 percent for GDP levels, 8.5 percent for GDP growth, 10.2 percent for tertiary education, 2.4 percent for HSPS, and 4.3 percent for population growth. Excluding the two deliberately radical exercises—eliminating knowledge spillovers and replacing employment weights with equal sector weights—the largest absolute changes are 15.1, 17.5, 27.3, 16.4, and 13.7 percent, respectively. A separate exact grid over $\delta_K \in [0.01, 0.15]$ also preserves every sign. Appendix D.4 reports the complete sensitivity results.

7 Industrial Policy: Skill Supply and Firm Scale

The policy experiments separate the supply of skills from firms’ demand for them. I consider interventions that operate on one or both of these margins and ask a simple question:

Can temporary development policies leave a persistent effect after they are withdrawn?

Recent work shows that expanding skill supply can induce firms to reorganize production and increase employment in larger, more organized firms (Engbom et al., 2025; Cox, 2025). Whether these effects persist after the expansion in skill supply ends is less clear. In general equilibrium, the skill premium adjusts as skilled workers become more abundant, so the initial increase in skills need not generate the demand required to sustain itself. The framework in this paper makes this distinction explicit. Proposition 2 implies that a temporary intervention can permanently raise the growth rate of the economy only if it

changes the limiting skilled share. In the model, this can occur when the intervention induces firms to create organizational procedures that diffuse into the public stock of knowledge. The resulting organizational knowledge survives the policy itself, sustaining firms' demand for skilled labor after the original increase in skill supply has disappeared. If the skill expansion does not induce additional organizational investment, however, the increase in skilled employment and organized-firm activity is largely transitory: once the policy is withdrawn, both converge back toward their no-policy paths.

7.1 A LMIC policy laboratory

To study this interaction in a developing-country setting, I construct a lower-middle-income version of the benchmark economy. I introduce productivity-correlated firm distortions in the spirit of [Bento and Restuccia \(2017, 2021\)](#), which reduce the scale at which productive firms operate, and recalibrate the model to representative LMIC moments. Specifically, I re-estimate the organizational-scale parameter, the manufacturing and service setup-cost shifters, the sectoral productivity-growth parameters, the manufacturing-demand shifter, and the skill-acquisition cost to match average firm size, sectoral productivity growth, the manufacturing-employment peak, the skilled share, and the HSPS share. The remaining structural parameters are inherited from the benchmark economy or fixed at their externally disciplined LMIC values. Appendix D.5 describes the distortion technology, parameterization, and recalibration in detail.

I first remove the productivity-correlated distortion without introducing any temporary skill or organizational policy. Beginning in 2026, the distortion is phased out linearly through 2035 while all other recalibrated LMIC parameters are held fixed. This experiment asks whether the increase in firm scale generated by reform is sufficient, on its own, to activate the organizational mechanism.

Table 6: Permanent distortion reform: scale without temporary policy

Outcome in 2089	Distortions remain	Full reform
Average manufacturing firm size	5.71	13.42
Average service firm size	1.89	4.48
GDP per capita relative to unreformed path (%)	0.00	13.64
Formalized task-share change (pp)	0.00	9.50
Skilled employment-share change (pp)	0.00	1.84
Public organizational knowledge, K	0.129	0.203

Notes: Full reform removes the excess productivity-correlated distortion linearly during 2026–2035. Firm-size and public-knowledge entries are levels. GDP, formalization, and skilled employment are measured relative to the unreformed LMIC path. No temporary policy is active in either economy. The institutional or administrative cost of the permanent reform is not modeled.

Table 6 shows that removing the distortion substantially raises firm scale and income. Average firm size more than doubles in manufacturing and services, and GDP per capita is 13.64 percent higher by 2089. The increase in scale also activates part of the organizational mechanism: the formalized-task share rises by 9.50 percentage points, the skilled-employment share by 1.84 percentage points, and public organizational knowledge increases from 0.129 to 0.203. Removing barriers to firm growth therefore does more than increase output; it also leaves the economy with greater organizational knowledge and a more skill-intensive workforce, persistently raising its growth rate.

7.2 Temporary policies

I next consider four temporary interventions operating from 2026 through 2045. A *college scholarship* lowers the private cost of acquiring skill. *Formalization support* reimburses part of the one-time cost of creating an organizational procedure. A *mixed package* combines the two instruments. Finally, a *manufacturing wage subsidy* reimburses part of the skilled and unskilled manufacturing wage bill and provides a benchmark for a policy targeted directly at manufacturing employment. Each intervention is evaluated both with the calibrated LMIC distortions left in place and together with the permanent distortion reform described above.

The scholarship experiment first provides an external validation of the mechanism emphasized in recent work. In the distorted LMIC economy, the scholarship raises the skilled share of recent cohorts by 7.30 percentage points by 2045 and increases the nonagricultural employment share of establishments with at least six workers that use skilled organizational tasks by 4.91 percentage points. The implied response, $4.905/7.302 = 0.672$, is close to the 0.654 response obtained by Engbom et al. (2025) and the 0.730 estimate in Cox (2025).³⁰

The distinction in this paper is what happens after the support for skill formation disappears. The Cox (2025) and Engbom et al. (2025) establish that additional skill supply can induce firms to reorganize and shift employment toward larger employers while the skill expansion is taking place. It does not by itself determine whether that reorganization survives once the intervention ends. In the model, persistence depends on whether firms use the temporary abundance of skilled labor to create organizational knowledge that remains valuable after the policy is withdrawn.

Policy rates are chosen so that discounted government expenditure is approximately 0.6 percent of discounted GDP over 2026–2045, corresponding to an average fiscal effort of roughly 0.6 percent of GDP while the programs are active. Temporary programs are financed through contemporaneous lump-sum household taxes. Withdrawal is unexpected: through 2045 households and firms behave as if the policy will continue, and the subsidy is removed at the start of 2046. The post-withdrawal path therefore isolates persistence generated by the economic state accumulated while the intervention was in place rather than anticipation of a pre-announced termination date.

7.3 Persistence after policy withdrawal

Table 7 compares the effects of each intervention in 2045, the final year before withdrawal, with its long-run effects in 2089. All effects are measured relative to the no-temporary-policy path under the same distortion environment.

When the LMIC distortions remain, the scholarship reproduces the large organized-employer response documented by Cox (2025) and Engbom et al. (2025) while the policy is active, but most of that response subsequently disappears once the subsidy is withdrawn. Skilled employment is 6.58 percentage points above the no-policy path in 2045 and only 0.51 percentage points above it in 2089; employment inside firms with at least six workers similarly falls from 4.89 percent above control to 0.15 percent. A strong contemporaneous response to skill supply therefore need not imply a persistent change in firm

³⁰ Cox estimates the change in the share of formal employment in firms with more than six workers per percentage-point increase in college attainment among 25–34-year-olds. Engbom et al. simulate the same education expansion and measure the corresponding change in employment in model firms with more than six workers. I measure the change in nonagricultural employment in establishments with at least six workers and a complete block of skilled organizational tasks per percentage-point increase in the recent skilled share. The three exercises therefore capture closely related employment reallocations toward larger, organized employers following an expansion in college-type skills.

Table 7: Temporary policies at withdrawal and in the long run

Temporary policy	CEV (%)	GDPpc 2089 (%)	Skilled emp. 2045 → 2089 (pp)	Emp. in 6+ firms 2045 → 2089 (%)	ΔK 2089
<i>Panel A: LMIC distortions remain</i>					
College scholarship	6.75	9.35	6.58 → 0.51	4.89 → 0.15	0.063
Formalization support	2.81	11.77	3.59 → 2.45	7.59 → 2.92	0.203
Mixed package	2.00	4.80	5.16 → 0.48	1.69 → 0.39	0.018
Manufacturing wage subsidy	-5.56	-3.69	0.02 → -0.02	5.75 → 0.33	0.000
<i>Panel B: Full permanent distortion reform</i>					
College scholarship	7.93	17.38	8.34 → 3.36	7.86 → 5.26	0.193
Formalization support	4.68	20.71	5.57 → 5.15	10.72 → 8.12	0.340
Mixed package	8.67	22.37	9.19 → 4.77	9.99 → 7.53	0.265
Manufacturing wage subsidy	-0.05	-0.21	0.01 → 0.09	0.88 → 0.04	0.003

Notes: Effects are measured relative to the no-temporary-policy path under the same distortion environment. Programs operate through 2045 and are withdrawn unexpectedly at the start of 2046. CEV is evaluated over 2026–2089. Policy rates are chosen to generate discounted government expenditure of approximately 0.6 percent of discounted GDP over 2026–2045. Formalization support under the permanent reform is evaluated at the highest stable reimbursement rate and spends 0.423 percent of GDP.

structure. Direct support for formalization behaves differently, leaving public organizational knowledge 0.203 above control in 2089 and sustaining substantially larger effects on skilled employment and employment in large firms after withdrawal.

Removing the firm-growth distortion strengthens this propagation mechanism. Under a distortion removal scenario, the scholarship has substantially more persistent effects, leaving skilled employment 3.36 percentage points and employment in six-plus-worker firms 5.26 percent above control in 2089. Formalization support produces the largest organizational legacy, whereas the mixed package generates the largest long-run GDP and welfare gains. Larger firms therefore do more than increase contemporaneous demand for skill: by increasing the return to reusable organizational procedures, they exploit a temporary expansion in skills to change the organizational technologies they employ, leaving behind a stock of knowledge which persists once the policy is withdrawn.

Manufacturing wage subsidies do not generate welfare gains or persistent changes in skill demand. They temporarily raise employment in larger firms but leave essentially no organizational knowledge and almost no long-run effect on skills or firm structure. The policy lesson is therefore that lasting gains come not from subsidizing manufacturing employment itself, but from policies that induce firms to reorganize production, changing their organizational technology and generating knowledge which leaves a persistent effect once support for skill expansion is withdrawn.

8 Conclusion

This paper asks whether economies that move directly from agriculture into services develop the same productive capabilities as economies that first pass through large-scale manufacturing. The evidence suggests that they do not. Across countries, larger historical manufacturing peaks are followed by service economies more concentrated in high-skill producer services and by faster subsequent growth. Across U.S. counties, larger manufacturing peaks are associated with greater managerial and technical employment within manufacturing and specialization in high-skill producer services decades later. The results

are concentrated in manufacturing industries where production in large establishments was technologically important. These findings point to a persistent organizational legacy of production at scale.

I formalize this mechanism in a growth model in which larger firms have stronger incentives to replace recurring organizational problems with reusable procedures, which can subsequently diffuse across firms and sectors. Calibrated to the U.S. transition, variation in manufacturing history generates almost all of the observed cross-country gradient in the HSPS share of services and substantial shares of the corresponding gradients in tertiary education and subsequent growth. Firm scale is central to the creation of organizational capabilities, while knowledge diffusion transmits its effects into the service economy that follows.

However, the fact that manufacturing historically generated these capabilities does not imply that manufacturing itself should be subsidized. Manufacturing wage subsidies are poorly targeted to the underlying externality. Persistent gains instead arise when reforms that allow productive firms to operate at scale are combined with temporary support for skill formation and organizational formalization. Development without industrialization is possible, but economies that bypass industrialization must still find other ways to build the capabilities that large-scale manufacturing historically created.

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A Data

This appendix defines the cross-country and U.S. county data, variables, and harmonizations. Data used only in the quantitative model are described in Appendix D.4.

A.1 Cross-country evidence

The facts in Section 2 use complementary samples. Service composition requires detailed sectors and uses the balanced Economic Transformation Database (ETD) panel; growth requires longer manufacturing histories and uses the broader combined GGDC–ILOSTAT sample.

A.1.1 Detailed sectoral employment and the composition of services

The service-composition evidence uses employment from the GGDC/UNU–WIDER ETD (de Vries et al., 2021; Kruse et al., 2022). The database reports annual persons engaged and value added in 12 sectors for 51 economies over 1990–2018: agriculture, mining, manufacturing, utilities, construction, trade, transport, business services, finance, real estate, government services, and other services.

High-skill producer-service (HSPS) employment comprises ETD business services, finance, and real estate; generic market services comprise trade and other services. Figure 1 plots

$$HSPS_{ct} = 100 \times \frac{EMP_{ct}^{Business} + EMP_{ct}^{Finance} + EMP_{ct}^{RealEstate}}{EMP_{ct}^{Total}}, \quad (36)$$

$$GenericServices_{ct} = 100 \times \frac{EMP_{ct}^{Trade} + EMP_{ct}^{Other}}{EMP_{ct}^{Total}}. \quad (37)$$

Both outcomes use total employment as the denominator. Trade includes accommodation and food services; transport and government services enter neither numerator.

For each of the 51 ETD economies, the historical manufacturing peak uses the full combined GGDC 10-Sector–ETD employment history, with ETD taking precedence in overlapping years. Splitting at the median peak places 26 economies, including the median country, in the above-median group and 25 in the below-median group.

The dynamic comparison subtracts each country’s 1990 employment share before taking unweighted group means. Shaded bands in Figure 1 are the mean plus or minus 1.96 cross-country standard errors.

A.1.2 Income, growth, population, education, and institutions

PPP real GDP per capita and population come from Gapminder’s harmonized series, combining modern World Bank estimates with historical income series. Average annual log GDP-per-capita growth over 2010–2019 is

$$g_c = \frac{100}{9} [\log y_{c,2019} - \log y_{c,2010}]. \quad (38)$$

The service-employment regressor is the mean ILOSTAT service share over 2010–2019; later observations are excluded. Income-quintile fixed effects use terminal GDP per capita, and years since the manufacturing peak equal $2019 - P_c$.

Institutional quality is the World Bank Worldwide Governance Indicators control-of-corruption percentile rank. Initial population and GDP per capita in the service-composition robustness regressions are measured in 1990; initial tertiary education is World Bank gross tertiary enrollment in, or closest to,

1990. The separate education figure instead uses completed tertiary attainment from Barro–Lee ([Barro and Lee, 2013](#)). These data are observed every five years; the post-1960 long-difference exercise requires at least five observations and at least ten years between the first and last jointly observed education and income years.

A.2 U.S. county evidence

Section 3 combines Census of Manufactures employment and establishments, population-census classified employment and manufacturing occupations, and Census and County Business Patterns (CBP) HSPS-service employment for the long event study. The Quarterly Census of Employment and Wages (QCEW) independently supplies the 2000 service-composition outcome in Tables 1 and 2. Unless stated otherwise, additive counts are aggregated to fixed geography before ratios are formed.

A.2.1 Fixed county geography

Historical county identifiers and boundary crosswalks come from IPUMS NHGIS ([Schroeder et al., 2026](#)). For historical censuses and Census of Manufactures tabulations, I assign each source-year county to the 1850 county with the greatest geographic overlap and sum additive counts. This accommodates microdata that locate individuals only by county of residence. Modern establishment counts are allocated across fixed 1850 counties using polygon-overlap shares. Numerators and denominators are mapped separately before forming ratios; employment shares are never interpolated directly.

A.2.2 Classified employment and the manufacturing peak

Manufacturing employment, establishments, wage bills, population, and urban population come from the historical county files of [Haines et al. \(2010\)](#), which compile population censuses and Census of Manufactures tabulations. Manufacturing employment is the average number of wage earners in the Census of Manufactures. Total classified employment comprises employed residents consistently assignable to primary industries, manufacturing, or services through harmonized IND1950:

primary industries: {105, 116, 126},
 manufacturing: 306–499,
 services: 506–946.

Mining, construction, unclassified employment, and industries outside these groups are excluded. This denominator provides the common economic base for peak manufacturing exposure and manufacturing HSPS.

For county c in year t , define

$$M_{ct} = \frac{L_{ct}^M}{L_{ct}^C}, \quad (39)$$

where L_{ct}^M is manufacturing employment and L_{ct}^C is classified employment. The manufacturing peak and its date are

$$M_c^{\text{peak}} = \max_{t \in \{1880, 1900, 1920, 1930, 1940\}} M_{ct}, \quad P_c = \min\{t : M_{ct} = M_c^{\text{peak}}\}. \quad (40)$$

Ties are assigned to the earliest peak year. The largely destroyed 1890 population census cannot supply classified employment, and the required county manufacturing tabulations are unavailable for 1910, so neither year enters the peak search. Establishment scale is log manufacturing workers per establishment in P_c ; peak exposure, scale, and manufacturing HSPS therefore refer to the same county-year.

The alternative-channel variables in column 5 of Table 1 are also measured in P_c : average manufacturing wages are the wage bill divided by average wage earners, urbanization is urban population divided by population, and county size is log population. Peak-cohort fixed effects absorb common nominal-wage differences across census years. The 48 counties with unavailable 1940 peak-year wage bills receive their cohort median and a replacement indicator.

A.2.3 HSPS capability inside manufacturing

Occupation-by-industry counts use IPUMS Ancestry Full Count censuses through 1940 and harmonized IPUMS USA files thereafter (Ruggles et al., 2024, 2025). All employed individuals enter regardless of household role, as counts in full-count years and with person weights in sample years. Industries and occupations use harmonized IND1950 and OCC1950.

Manufacturing HSPS counts the accounting, coordination, personnel, technical, and scientific occupations in Table A.1 among workers in IND1950 industries 306–499:

$$MHSPS_{ct} = 100 \times \frac{\text{core HSPS-occupation workers in manufacturing}_{ct}}{L_{ct}^C}. \quad (41)$$

The event study and channel-accounting table use this definition. The occupation set is fixed over time by function within manufacturing, independently of census-year educational attainment.

Table A.1: Occupation-based manufacturing HSPS classification

Function	Exact occupation content	OCC1950 codes
Accounting and record keeping	Accountants and auditors; bookkeepers; bill and account collectors; office-machine operators; shipping and receiving clerks	0, 310, 321, 341, 342
Purchasing, logistics, and coordination	Buyers and department heads in stores; buyers and shippers of farm products; purchasing agents and buyers not elsewhere classified; dispatchers and starters of vehicles	200, 201, 280, 322
Personnel administration	Personnel and labor-relations workers	72
Technical and scientific work	Draftsmen; engineers by specialty; statisticians and actuaries; surveyors; testing technicians; technicians not elsewhere classified; inspectors not elsewhere classified	35, 41–49, 83, 92, 95, 96, 533

Notes: Manufacturing is defined by IND1950 codes 306–499. The four groups jointly form the core manufacturing-HSPS measure used throughout the main analysis. Legal occupations and commercial intermediaries are not included.

A.2.4 HSPS service industries and the long event-study panel

Modern HSPS services are industry based: Information; Professional, Scientific, and Technical Services; and Management of Companies and Enterprises (NAICS 51, 54, and 55). These sectors supply information, professional and technical expertise, and organizational direction. NAICS 54 includes accounting,

legal, engineering, scientific, technical, and consulting suppliers; NAICS 55 identifies enterprise management and coordination; and NAICS 51 covers information production and transmission. NAICS 56 is excluded because its two-digit aggregate combines business support with routine administration, cleaning, security, and waste management. NAICS 52–53 are excluded because they include substantial financial intermediation, insurance distribution, property operation, and rental activity outside this organizational and technical core.

The manufacturing and service measures are functional analogues: occupations identify capabilities within manufacturing firms, while industries identify specialized establishments supplying related information, technical, and organizational services. The latter is the narrow U.S. analogue of the business-and-professional component of the cross-country HSPS measure in Figure 1.

To measure when HSPS services become quantitatively important in the county economy, the event study divides their employment by classified employment:

$$SHSPS_{ct} = 100 \times \frac{\text{employment in HSPS service industries}_{ct}}{L_{ct}^C}. \quad (42)$$

The service numerator uses population-census industries through 1950, imputed early CBP in 1960 and 1970, and the harmonized, suppression-imputed CBP panel of [Eckert et al. \(2020\)](#) (EFSY) thereafter. EFSY supplies a common postwar industry classification and imputes confidential cells subject to CBP hierarchy adding-up restrictions. Table A.2 gives the concordance.

Table A.2: HSPS service-industry measurement by data vintage

Event-study year	Source and classification	HSPS service-industry numerator
1850–1950	Population-census microdata; harmonized IND1950	Communications, advertising, accounting, business, legal, engineering, and professional and technical services: codes 578, 579, 806, 807, 808, 856, 857, 879, 898, and 899.
1960	Imputed 1959 CBP; SIC	Communications, business services, legal services, and engineering, accounting, research, and related professional services: SIC groups 4800, 7300, 8100, and 8900.
1970	Imputed 1970 CBP; SIC	The same SIC analogue used for 1960.
1980–2000	EFSY harmonized and imputed CBP; common 2012 NAICS	All mutually exclusive EFSY employment cells beginning with NAICS 51, 54, or 55.

Notes: The early CBP files are from [Eckert et al. \(2022\)](#); the suppression imputations and common-NAICS panel are from [Eckert et al. \(2020\)](#). The EFSY long panel stores employment in mutually exclusive cells at the most detailed assignable industry level, so residual aggregate cells do not duplicate their descendants. Management companies are not separately identified in the historical IND1950 and SIC classifications, and those years use the closest functional analogue. IND1950 code 808 includes some activities later assigned to NAICS 56 that cannot be separated in the historical classification.

A.2.5 The QCEW long-run service-composition outcome

Tables 1 and 2 measure HSPS employment within total private services. The 2000 endpoint uses public BLS QCEW files ([U.S. Bureau of Labor Statistics, 2004](#)), retaining the modern HSPS numerator. QCEW directly publishes a county private service-providing aggregate; the EFSY denominator instead requires summing mutually exclusive imputed CBP components.

For county c in year t , define

$$Y_{ct}^{QCEW} = \frac{E_{c,51,t}^{\text{public}} + E_{c,54,t}^{\text{public}} + E_{c,55,t}^{\text{public}}}{E_{c,102,t}^{\text{public}} - E_{c,22,t}^{\text{public}}}, \quad (43)$$

where ownership code 5 restricts all components to private employment, code 102 is QCEW's service-providing aggregate, and code 22 is private utilities. The denominator includes wholesale and retail trade, private transportation and warehousing, information, finance, real estate, professional and business services, private education and health, leisure, hospitality, and other private services. Private ownership excludes public administration and the U.S. Postal Service.

The main-text numerator sums public numeric records in NAICS 51, 54, and 55, without estimating latent confidential employment. In the reconstructed 2000 file, records failing disclosure are absent; later files identify suppressed records with an N flag and a zero numeric field. Unpublished contributions therefore enter this sum at zero even though the disclosed aggregate denominator includes them. Section C.2.4 uses the first explicit-flag year for QCEW-specific hierarchical imputation; Section C.2.5 compares QCEW with EFSY at county and commuting-zone levels.

The unweighted outcome mean is 6.36 percent, reflecting equal county weights and greater disclosure losses in small cells. Private-service-employment weighting yields 13.97 percent. Summing available public county records before forming the ratio yields 14.02 percent, versus 14.58 percent from official national QCEW aggregates. Thus most national HSPS employment is observed and disclosure gaps concentrate in small counties. This aggregate comparison does not establish innocuous county measurement error; the flag-aware and commuting-zone checks address that concern.

A.2.6 Historical controls and estimation samples

The baseline county covariates are measured in 1850 and come from the Haines files: log population, the urban population share, the school-enrollment rate, the number of colleges, and log enslaved population plus one. The event study additionally includes predetermined 1870 employment shares in process, machinery-producing, and traditional manufacturing. Each baseline covariate is interacted with a linear calendar-time trend in the event-study specification.

The event-study sample contains 1,116 fixed-geography counties peaking in 1900, 1920, or 1940. These cohorts supply a common forty-year pre-period and have the treatment and baseline controls observed. Event time -40 is omitted. Outcome availability varies across county-event-time cells because the manufacturing and service series draw on different source years. Confidence intervals use a spatial covariance estimator allowing arbitrary covariance among observations whose county centroids lie within 100 kilometers.

Tables 1 and 2 use the same 1,487 counties. The common sample requires the QCEW 2000 outcome, the manufacturing peak, peak-year establishment scale and manufacturing HSPS, the 1850 covariates, county coordinates, all three shift-share measures, and the complete pre-1880 histories used in the IV specifications. Holding this sample fixed ensures that differences across columns do not reflect changes in county composition. Cross-sectional inference uses Conley standard errors with a 100-kilometer cutoff.

A.3 The Chandler throughput shift-share

The shift-share design begins with county-by-manufacturing-industry employment in the 1880 full-count Census of Population, harmonized to IND1950. Workers with unclassifiable industries do not enter the county-industry counts. Following [Chandler \(1990\)](#), I distinguish industries according to the production regime through which national expansion occurred. High-throughput industries are those in which continuous-process or large-batch production, or the fabrication and assembly of complex standardized machinery, required large fixed investments and extensive managerial coordination. Traditional low-throughput industries comprise wood products, textiles and apparel, printing, and leather, where increasing output remained more readily divisible across smaller establishments. Table A.3 reports the concordance and source for every broad industry. The all-industry instrument uses every classifiable manufacturing industry in IND1950 codes 306–499.

Table A.3: Manufacturing industries in the Chandler throughput design

Group	Broad industry	IND1950 codes	Chandler source and classification rationale
High throughput	Glass and cement	316–317	Chandler (1990, pp. 20, 113–120) . Chandler identifies plate and flat glass as scale intensive and describes continuous-process technologies and substantial scale economies in glass and cement.
High throughput	Primary iron, steel, and nonferrous metals	336–338	Chandler (1990, pp. 20, 23, 70–71, 121–140) . Refining and smelting, including the Bessemer and open-hearth processes and electrolytic refining, required capital-intensive plants operating at high throughput.
High throughput	Nonelectrical machinery	356–358	Chandler (1990, pp. 23, 65–68, 194–205) . Agricultural, office, and industrial machinery exemplify complex standardized machinery fabricated and assembled from interchangeable parts and supported by specialized distribution and service organizations.
High throughput	Electrical machinery	367	Chandler (1990, pp. 20, 68–69, 212–220) . Electrical-equipment producers concentrated production in large plants and required technically trained organizations to design, manufacture, install, and service complex equipment.

Continued on next page

Table A.3 – continued from previous page

Group	Broad industry	IND1950 codes	Chandler source and classification rationale
High throughput	Transportation equipment	376–379	Chandler (1990, pp. 20, 205–212) . These codes concord to automobiles, aircraft, ships, and railroad equipment. Production required large-scale fabrication and assembly and extensive managerial coordination.
High throughput	Photographic equipment	387	Chandler (1990, pp. 20, 67) . Chandler identifies mass-produced cameras as scale intensive and discusses Eastman Kodak's mass production of cameras and photographic film.
High throughput	Processed foods	406–409	Chandler (1990, pp. 23, 62–65) . Meatpacking, dairy processing, canning, and grain milling relied on mechanized processing and packaging, high-volume plants, and coordinated purchasing, distribution, and national marketing.
High throughput	Beverages	418	Chandler (1990, pp. 23, 64–65) . Chandler places brewing and distilling among refining and processing industries and emphasizes the production and refrigerated-distribution networks constructed by national brewers.
High throughput	Tobacco	429	Chandler (1990, pp. 20, 23, 63–65) . Mechanized processing and packaging permitted high-volume cigarette production for national markets.
High throughput	Pulp and paper mills	456	Chandler (1990, pp. 20, 112–113) . Chandler identifies newsprint and describes pulp, newsprint, and paperboard as acquiring substantial scale economies after the transition to capital-intensive machinery.
High throughput	Chemicals, synthetic fibers, drugs, and paint	466–469	Chandler (1990, pp. 20, 23–26, 69–70, 170–193) . Chemical production is a canonical continuous- and large-batch activity in which interdependent processes generated economies of scale and scope.

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Table A.3 – continued from previous page

Group	Broad industry	IND1950 codes	Chandler source and classification rationale
High throughput	Petroleum refining and coal products	476–477	Chandler (1990, pp. 20, 23–26, 94–104) . Petroleum refining is Chandler’s canonical illustration of throughput economies: large sunk investments were profitable only when managerial coordination maintained continuous flows from inputs through refining and distribution.
High throughput	Rubber products	478	Chandler (1990, pp. 20, 105–111) . Chandler singles out tires and documents the large-batch processing, integrated distribution, and organizational investments of leading rubber producers.
Traditional low throughput	Logging, wood products, and furniture	306–309	Chandler (1990, pp. 20, 22–23) . Chandler places lumber and furniture among traditional labor-intensive industries in which larger establishments obtained limited cost advantages over smaller ones.
Traditional low throughput	Textiles and apparel	436–439, 448–449	446, Chandler (1990, pp. 20, 22–23) . Textiles from natural fibers and apparel are labor intensive: increasing output primarily required adding machines and workers rather than exploiting large throughput economies.
Traditional low throughput	Printing and publishing	459	Chandler (1990, pp. 20, 22–23) . Chandler includes printing and publishing among labor-intensive industries in which scale advantages were present but not substantial.
Traditional low throughput	Leather and footwear	487–489	Chandler (1990, pp. 20, 105–106) . Chandler contrasts concentrated rubber footwear with leather footwear, where scale economies remained limited and production was dispersed across establishments.

Notes: Chandler states the general distinction on pp. 22–24: high-throughput industries used capital-intensive continuous or large-batch processes, or fabricated and assembled complex standardized machinery, whereas traditional industries remained comparatively labor intensive and obtained smaller cost advantages from greater plant size. Chandler’s categories are expressed in the historical classifications used in *Scale and Scope*; the IND1950 cells are the paper’s concordance to those categories. The baseline retains only cells that can be matched unambiguously. Ambiguous adjacent industries are excluded from both groups. Page references use the printed pagination of [Chandler \(1990\)](#).

For production regime $g \in \{A, H, L\}$, let $L_{cj,1880}$ denote county c 's employment in manufacturing industry j in 1880 and define

$$\omega_{cj,1880}^g = \frac{L_{cj,1880}}{\sum_{\ell \in g} L_{c\ell,1880}}. \quad (44)$$

Here A denotes all manufacturing industries, H the high-throughput group, and L the traditional low-throughput group. The leave-county-out national shift is

$$G_{j,-c,1880-1900} = \frac{L_{-c,j,1900}}{L_{-c,j,1880}} - 1, \quad (45)$$

and the instrument is

$$Z_c^g = \sum_{j \in g} \omega_{cj,1880}^g G_{j,-c,1880-1900}. \quad (46)$$

Within-group normalization removes mechanical variation in the initial size of the high- and low-throughput regimes: the instruments compare initial industry composition rather than initial regime size. All three use the same base year, horizon, leave-county-out rule, and normalization and are standardized within the estimation sample.

The 1880 base predates the observed manufacturing peak for 89 percent of counties and coincides with the earliest observed peak for the remainder. Removing county c from each national shift prevents its own later employment growth from entering its instrument mechanically. The preferred specification includes state and peak-cohort fixed effects; the five 1850 covariates defined above; and complete 1850, 1860, and 1870 histories of manufacturing employment and manufacturing HSPS relative to classified employment and of HSPS services relative to private-service employment. The identifying assumption is conditional: after accounting for location, initial development, and these observed pre-1880 trajectories, differential national growth across the industries represented in a county's 1880 manufacturing mix affects service composition through the local manufacturing expansion it induces.

A.4 Treatment of missing, confidential, and structurally zero observations

I distinguish a structural zero from an unpublished positive count. In full-count Census tabulations, the absence of workers in an otherwise observed county-industry cell contributes zero. In QCEW files with disclosure flags, an N-coded record is confidential and its numeric employment field is not treated as an observed zero in the flag-aware exercises below. An absent county-industry macro record is treated as a structural absence under the BLS publication convention. I do not replace an unavailable county outcome or denominator with zero. Additive counts are aggregated before ratios are calculated, and a ratio is missing whenever its denominator is unavailable or nonpositive.

The main-text 2000 QCEW numerator sums released numeric values without allocating employment to omitted detailed cells. EFSY instead uses constrained CBP imputations. The QCEW robustness exercises preserve disclosed counts, identify unknown cells through explicit flags, and allocate only the residual required by disclosed parents. The sole sample-preserving imputation for historical regressors replaces 48 unavailable peak-year manufacturing wages with peak-cohort medians and includes a replacement indicator.

B Additional Cross-Country Evidence

This section relates recorded manufacturing peaks to HSPS employment, service composition, and subsequent income growth, including among countries whose peaks preceded the outcome period.

B.1 When do countries reach their manufacturing-employment peaks?

Recorded peaks are maxima of manufacturing-employment shares over each country's available annual history, with ties assigned to the earliest year. Table B.1 reports counts and unweighted mean peak shares for the 174-country growth-by-decade sample, whose histories combine GGDC 10-sector and Gapminder data. Service-composition regressions use a separate 51-country GGDC-ETD sample.

Mean peak shares decline from 15.76 percent in the 1990s to 12.29 percent in the 2000s and 11.25 percent in the 2010s; 140 of 174 countries (80.5 percent) peak before 2010. Coverage partly explains the 1990s concentration: 79 of its 108 peaks occur at the first observation. Recorded maxima therefore need not identify historical turning points.

B.2 Manufacturing peaks and HSPS employment

The HSPS analysis covers business services, finance, and real estate in all 51 ETD countries in Figure 1. Manufacturing peaks use full GGDC 10-sector and ETD histories, preferring ETD in overlapping years. GGDC extends 32 countries' histories; the other 19 use their full ETD histories.

Let H_{ct} denote HSPS employment as a percentage of total employment, $\Delta H_c = H_{c,2018} - H_{c,1990}$, and P_c the historical manufacturing peak in percentage points. The most extensive change and terminal-level specifications are

$$\Delta H_c = \alpha + \beta(P_c/10) + \gamma \Delta \ln y_c + \theta S_{c,1990} + \delta_{r(c)} + \lambda_{g(c,1990)} + \varepsilon_c, \quad (47)$$

$$H_{c,2018} = a + b(P_c/10) + \vartheta S_{c,2018} + d_{r(c)} + \ell_{g(c,1990)} + u_c. \quad (48)$$

Here $\Delta \ln y_c = \ln y_{c,2018} - \ln y_{c,1990}$, and S_{ct} is non-government service employment as a percentage of total employment. GDP per capita comes from the Maddison Project Database 2023. Region effects distinguish Africa, the Americas, and Asia. Initial income-group effects use the four World Bank categories based on 1990 GNI per capita (FY1992). The two sets of fixed effects enter additively.

Table B.2 introduces income growth in column (2), both sets of fixed effects in column (3), and the 1990 service share in column (4). For terminal HSPS employment, column (6) adds both sets of fixed effects and column (7) adds the 2018 service share. All regressions use equal country weights and HC3 standard errors.

A ten-percentage-point larger manufacturing peak is associated with a 2.87-percentage-point greater increase in HSPS employment over 1990–2018. The estimate is 2.28 after including income growth, region and initial income-group effects, and the initial service share. For the 2018 HSPS share, the coefficient falls from 5.96 without controls to 2.12 with fixed effects and the contemporary service share. Both final estimates remain significant at the five-percent level.

Disaggregating service-employment growth.

Table B.1: Timing and Size of Recorded Manufacturing-Employment Peaks

Peak decade	Number of countries	Mean peak share (%)
1950s	3	25.97
1960s	4	29.35
1970s	7	20.67
1980s	4	23.01
1990s	108	15.76
2000s	14	12.29
2010s	34	11.25
All countries	174	15.45

Notes: The sample comprises the 174 countries in Table B.4. The manufacturing-employment share uses total employment as its denominator. Peak shares are maxima over each country's available history; exact ties are assigned to the earliest year. Means give equal weight to each country. Recorded peak dates need not identify historical turning points: 79 of the 108 countries assigned to the 1990s peak at their first available observation, and 88 have a recorded peak in 1991. The 173-country baseline growth sample excludes South Sudan because its initial World Bank income classification is unavailable.

Table B.2: Manufacturing peaks and HSPS employment: changes and terminal levels

	HSPS share change, 1990–2018				HSPS share, 2018		
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Manufacturing peak (10 pp)	2.867*** (0.507)	2.937*** (0.501)	2.173*** (0.782)	2.283** (0.949)	5.962*** (0.729)	3.255*** (0.648)	2.116** (1.003)
GDP-per-capita gain ($\Delta \ln y$)		−1.202 (0.732)	−0.176 (1.029)	−0.145 (1.079)			
Service share, 1990 (pp)				−0.021 (0.068)			
Service share, 2018 (pp)							0.172* (0.093)
Region and income-group FE	No	No	Yes	Yes	No	Yes	Yes
Observations	51	51	51	51	51	51	51
R^2	0.530	0.550	0.617	0.618	0.647	0.817	0.842

Notes: All columns use the same 51 ETD economies. HSPS is business services, finance, and real estate employment as a percentage of total employment. Columns (1)–(4) use the 1990–2018 change; columns (5)–(7) use the 2018 level. The manufacturing peak is the maximum employment share over the full available GGDC 10-sector–ETD history, with ETD preferred in overlapping years. Its coefficient is reported per 10-percentage-point larger peak. GDP-per-capita gain is $\ln(y_{2018}) - \ln(y_{1990})$, using the Maddison Project Database 2023. The service-share controls use 1990 in column (4) and 2018 in column (7). They measure employment in trade/accommodation/food, transport, business services, finance, real estate, and other services as a percentage of total employment; government services are excluded from their numerator. Their coefficients are per one-percentage-point higher service share in the indicated year. Region and income-group effects are included together as additive fixed effects. Region effects use Africa, Americas, and Asia; initial income-group effects use the World Bank low-, lower-middle-, upper-middle-, and high-income classifications based on 1990 GNI (FY1992). All regressions include an intercept. HC3 standard errors are in parentheses; * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

For each service industry j , I estimate $\Delta s_{cj} = \alpha_j + \beta_j (M_c^{\max}/0.10) + \varepsilon_{cj}$, where Δs_{cj} is the 1990–2018 change in its share of total employment, in percentage points, and $M_c^{\max}$ is the historical manufacturing

peak expressed as a fraction. Each regression uses all 51 countries and no additional controls. Government services are excluded from the industries examined; total employment remains the denominator.

Business services have the largest coefficient: 1.59 percentage points per ten-percentage-point larger peak, followed by finance (0.67) and real estate (0.61). Business services account for approximately 56 percent of the aggregate HSPS association. The ETD category includes information and communication, professional, scientific, and technical activities, and administrative and support services; business-organization services are not separately identified. The remaining coefficients are negative: -0.13 for other services, -0.59 for transport, and -2.00 for trade, accommodation, and food services.

B.3 GDP-per-capita growth and peak manufacturing employment

The growth regressions relate manufacturing history to average annual GDP-per-capita growth over 2010–2019. The most extensive specification controls for both contemporary sectoral shares:

$$g_c = \alpha + \alpha_{r(c)} + \alpha_{i(c)} + \beta^m(M_c^{\max}/0.10) + \beta^s(\bar{S}_{c,2010:2019}/0.10) + \delta(\bar{M}_{c,2010:2019}/0.10) + \gamma I_c + \varepsilon_c. \quad (49)$$

The outcome g_c is mean annual log GDP-per-capita growth multiplied by 100. Employment shares are fractions of total employment, averaged over 2010–2019 where indicated. The fixed effects distinguish regions and the four World Bank income groups based on 2009 GNI per capita (FY2011). Institutional quality I_c is the World Bank control-of-corruption percentile rank.

Columns (1)–(5) of Table B.3 successively add the service share, both sets of fixed effects, institutional quality, and the contemporary manufacturing share to the peak-only regression. Columns (6)–(7) repeat columns (4)–(5) for economies with an average service-employment share above 50 percent.

In the full sample, controlling for contemporary manufacturing reduces the peak coefficient from 0.63 to 0.40, with $p = 0.356$ for the latter. Among economies where services employ a majority of workers, the corresponding coefficient rises from 0.64 to 1.11 and remains significant ($p = 0.020$). In this group, a ten-percentage-point larger historical peak is therefore associated with 1.11 percentage points faster annual growth after accounting for both contemporary sectoral shares. The difference from the full-sample coefficient has not been established statistically; alternative service-share thresholds were also examined.

B.4 Service composition and growth after earlier manufacturing peaks

To examine whether these associations persist among countries with earlier peaks, let D_{cd} indicate the decade of country c 's recorded peak. For each outcome Y_c , I estimate

$$Y_c = a + \sum_{d \in \mathcal{D}} \beta_d D_{cd}(M_c^{\max}/0.10) + X_c' \Gamma + e_c. \quad (50)$$

Each regression has a common intercept and decade-specific manufacturing-peak slopes. Table B.4 presents the same two specifications for each outcome: one without additional controls and one adding log GDP per capita in 2018 and region fixed effects. The regressions include neither separate decade intercepts nor a linear peak-year control. All coefficients are scaled per ten-percentage-point higher manufacturing peak, and HC3 standard errors are used throughout.

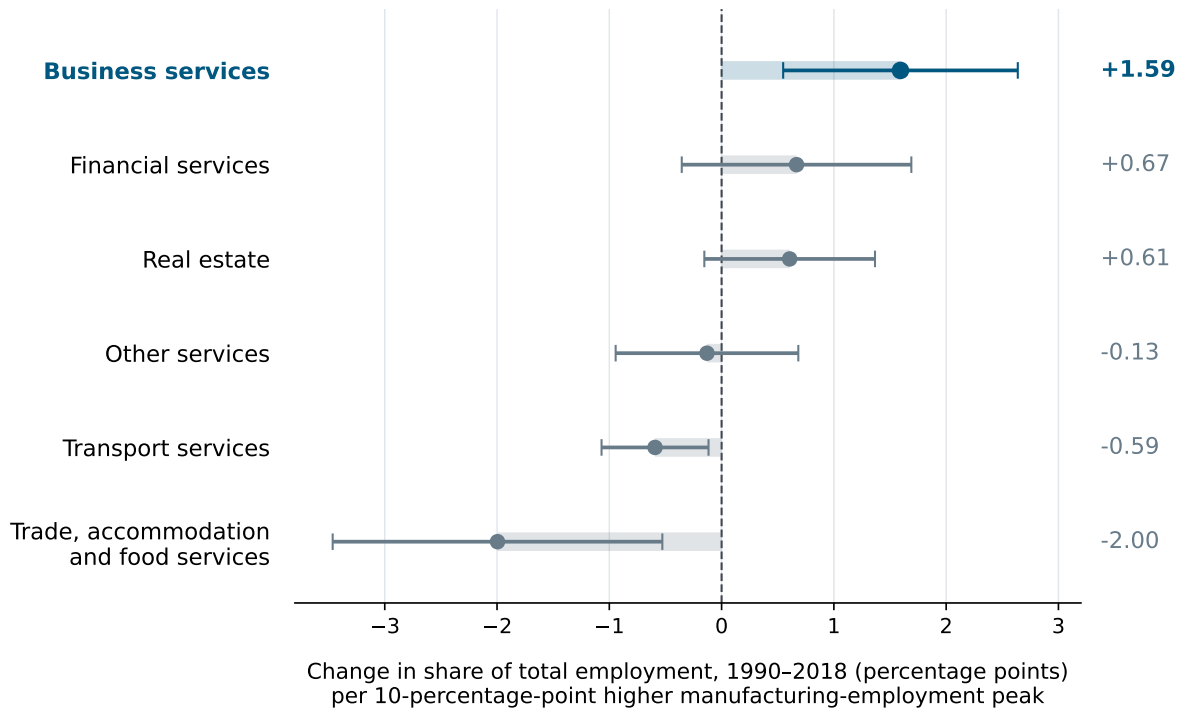


Figure B.1: Manufacturing peaks and service-industry employment growth. Points report coefficients from separate regressions of the 1990–2018 change in each industry’s share of total employment on historical peak manufacturing employment. Coefficients are expressed in percentage points per ten-percentage-point larger manufacturing peak. Industries are ordered by their estimated coefficients. Manufacturing peaks use the full available GGDC 10-sector–ETD histories, with ETD preferred in overlapping years. All regressions include an intercept, use the same 51 economies, and assign equal weight to each country. Bars show pointwise 95 percent confidence intervals based on HC3 standard errors and the residual-degrees-of-freedom t distribution. Government services are excluded from the displayed industries.

Columns (1)–(2) use average annual GDP-per-capita growth over 2010–2019 for 174 countries. Columns (3)–(4) use the 2018 HSPS share of non-government service employment for the 51 ETD countries. For the latter outcome, the numerator is business, finance, and real-estate employment; the denominator additionally includes trade, accommodation and food services, transport, and other services. Government is excluded from both. Unlike the earlier HSPS regressions, this outcome measures composition within services rather than a share of total employment.

For countries with recorded peaks in the 1990s, a ten-percentage-point larger manufacturing peak is associated with 0.56 percentage points faster annual GDP-per-capita growth without controls ($p = 0.008$) and 0.63 points after accounting for 2018 income and region ($p = 0.044$). For these 108 countries, average manufacturing employment fell from 10.92 to 10.45 percent of total employment between 2010 and 2019, while services rose from 52.75 to 56.64 percent. The positive growth association is therefore present among countries whose recorded peaks preceded the growth window and whose employment continued, on average, to shift toward services. The 2010s coefficients are larger, at 1.28 and 1.05, with $p = 0.008$ and $p = 0.051$, respectively.

The service-composition results show a related pattern. Among the 11 ETD countries with 1990s peaks, the estimated association with the 2018 HSPS share is 9.10 percentage points without controls ($p < 0.001$) and 5.11 points with income and region controls ($p = 0.099$). For the 20 countries with

Table B.3: Manufacturing Peaks and Growth in Service Economies

	All countries					Services > 50%	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Peak manufacturing share (10 pp)	0.305* (1.67)	0.953*** (4.54)	0.733*** (2.70)	0.629** (2.53)	0.398 (0.92)	0.645** (1.99)	1.113** (2.36)
Service-employment share (10 pp)		-0.538*** (-4.70)	-0.311 (-1.41)	-0.572*** (-2.96)	-0.540*** (-2.68)	-1.045** (-2.59)	-1.313** (-2.41)
Manufacturing-employment share (10 pp)					0.354 (0.70)		-0.924 (-1.16)
Observations	173	173	173	173	173	104	104
R^2	0.012	0.137	0.236	0.360	0.362	0.278	0.288
Region and income-group FE	No	No	Yes	Yes	Yes	Yes	Yes
Institutional quality control	No	No	No	Yes	Yes	Yes	Yes

Notes: The dependent variable is average annual GDP-per-capita growth over 2010–2019, in percentage points. Column (1) includes only the manufacturing peak and an intercept. Columns (1)–(5) use the same 173-country sample; columns (6)–(7) use the 104 countries whose average service-employment share over 2010–2019 exceeds 50 percent. The service-employment share and the manufacturing-employment share are averages over 2010–2019; all 173 countries have manufacturing observations in each of these ten years. The peak manufacturing share is the maximum over the available historical series. All employment shares use total employment as the denominator, and their coefficients are reported per ten-percentage-point increase. Columns (5) and (7) add the average manufacturing-employment share to the specifications in columns (4) and (6), respectively. Region and income-group fixed effects enter together as separate additive sets in columns (3)–(7). Income groups are the World Bank classifications based on 2009 GNI per capita (FY2011); the full sample contains 40 low-, 50 lower-middle-, 37 upper-middle-, and 46 high-income countries. South Sudan is omitted because its 2009 classification is unavailable. Columns (4)–(7) control for the World Bank control-of-corruption percentile rank. All regressions include an intercept. HC1 robust t statistics are in parentheses; tests use the residual-degrees-of-freedom t distribution. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

2010s peaks, neither coefficient is statistically significant. Income and region adjustment thus weakens the earlier-peak composition association, while the corresponding growth association remains significant at the five-percent level.

These results complement the contemporary-share controls in Table B.3: positive associations are also present for substantially earlier recorded peaks. Growth in 2010–2019 occurs 11–29 years after a 1990s peak, and 2018 service composition is measured 19–28 years later. Interpretation remains subject to the coverage limits in Table B.1 and the small samples in some decade groups.

C Additional U.S. County Evidence

C.1 Timing of county manufacturing peaks

County manufacturing peaks follow equation (1): the maximum manufacturing-employment share of total classified employment over 1880, 1900, 1920, 1930, and 1940. The years 1890 and 1910 are excluded because the required population and manufacturing tabulations, respectively, are unavailable.

Peak dates vary from the turn of the twentieth century to the interwar period, providing the staggered timing used in the event-study analysis.

Table B.4: Historical Manufacturing Peaks, Subsequent Growth, and Service Composition

Peak decade	GDP-per-capita growth, 2010–2019		HSPS share of services, 2018	
	(1)	(2)	(3)	(4)
1950s	0.085 (0.299)	0.340 (0.406)	6.078*** (1.176)	3.846 (2.297)
1960s	0.062 (0.139)	0.174 (0.289)	—	—
1970s	0.169 (0.293)	0.226 (0.380)	8.751*** (0.956)	5.147* (3.027)
1980s	−0.146 (1.027)	−0.120 (1.056)	7.822*** (1.353)	3.717 (2.383)
1990s	0.559*** (0.208)	0.625** (0.308)	9.098*** (2.211)	5.107* (3.022)
2000s	0.179 (0.471)	0.167 (0.568)	10.425*** (3.307)	5.787 (4.404)
2010s	1.278*** (0.475)	1.054* (0.535)	2.603 (1.634)	−0.045 (2.247)
Log GDP per capita, 2018	No	Yes	No	Yes
Region fixed effects	No	Yes	No	Yes
Observations	174	174	51	51
R^2	0.076	0.115	0.701	0.754

Notes: Each column is a pooled, unweighted OLS regression with a common intercept and decade-specific slopes on the historical manufacturing-employment peak. Coefficients are per ten-percentage-point higher peak. Columns (1) and (3) include no additional controls; columns (2) and (4) add log GDP per capita in 2018 and region fixed effects. No linear peak-year control, separate decade intercepts, service-share control, or other covariates are included. Income is measured in 2018 in both controlled columns, using Gapminder for the growth sample and Maddison Project Database 2023 for the ETD sample, consistent with their underlying datasets. Growth is the mean annual log GDP-per-capita change over 2010–2019, multiplied by 100. HSPS composition is business, finance, and real-estate employment as a percentage of non-government service employment in 2018; government is excluded from numerator and denominator. Growth columns use 174 countries and GGDC–Gapminder manufacturing histories; composition columns use 51 ETD countries and GGDC–ETD histories. Both specifications within each pair retain the same countries. No ETD country has a recorded peak in the 1960s. HC3 standard errors are in parentheses throughout; inference uses residual-degrees-of-freedom t distributions and pointwise significance levels. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

C.2 Robustness of the U.S. county evidence

The following exercises assess outcome measurement and identification: endpoint and peak-window choices; separate high- and low-throughput event studies; industry-share diagnostics; QCEW confidentiality and EFSY comparisons; national-shock inference; historical controls and spatial cutoffs; and the role of corporate headquarters.

C.2.1 QCEW endpoint and peak-window robustness

Table C.1 moves the public-record QCEW endpoint from 2000 to 2001 and 2010. The baseline peak coefficient remains positive. Adding manufacturing scale and HSPS attenuates it by 68 percent in 2000, 68 percent in 2001, and 57 percent in 2010; manufacturing HSPS remains large and precisely estimated.

Table C.2 reports the IV results. The high-throughput estimate is 0.241 in 2010, with a conventional p -value of 0.045 and an Anderson–Rubin p -value of 0.051. Excluding counties whose observed peaks occur at the 1880 or 1940 boundaries raises the high-throughput estimate to 0.318 in 2000 and 0.273 in 2010. The result is therefore not generated by counties whose true peak may lie outside the observed window.

The disclosed service aggregate retains confidential non-HSPS employment in the denominator; summing public two-digit records instead removes confidential employment from both sides of the ratio. Table C.3 shows that including private utilities is immaterial and reproduces the public-record result in 2001, when explicit suppression flags become available. Section C.2.4 uses those flags for imputation.

Table C.1: Channel accounting across QCEW endpoints

	2000	2001	2010
<i>Panel A. Baseline</i>			
Peak manufacturing employment share	0.048*** (0.010)	0.043*** (0.009)	0.042*** (0.009)
<i>Panel B. Add manufacturing scale and HSPS</i>			
Peak manufacturing employment share	0.015 (0.012)	0.014 (0.011)	0.018* (0.010)
Log manufacturing establishment scale at peak	0.0041* (0.0023)	0.0033 (0.0021)	0.0024 (0.0025)
Manufacturing HSPS / classified employment at peak	1.927*** (0.535)	1.884*** (0.508)	1.671*** (0.506)
Attenuation of peak coefficient (%)	68.3	67.8	56.9
Observations	1,487	1,511	1,509

Notes: Every outcome uses the QCEW public-record numerator and disclosed service-aggregate denominator. Each column uses the maximum available common sample for that endpoint. Specifications otherwise match the baseline and joint-channel columns of Table 1. Standard errors are Conley with a 100-kilometer cutoff. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table C.2: Outcome endpoint and peak-window robustness

	Full sample 2000	Interior peaks 2000	Full sample 2010	Interior peaks 2010
All-industries IV	0.133 (0.105)	0.198* (0.111)	0.092 (0.090)	0.188* (0.105)
First-stage Conley F	14.67	16.14	13.77	14.36
Anderson–Rubin p -value	0.203	0.093	0.330	0.100
High-throughput IV	0.283** (0.118)	0.318*** (0.121)	0.241** (0.120)	0.273** (0.117)
First-stage Conley F	12.87	12.57	11.00	10.29
Anderson–Rubin p -value	0.009	0.008	0.051	0.028
Observations	1,487	1,013	1,509	1,031

Notes: The 2000 and 2010 outcomes use the same QCEW public-record construction. Interior-peak samples exclude counties whose observed manufacturing peak occurs at either boundary of the 1880–1940 window. All other variables and specifications match Table 2. Standard errors are Conley with a 100-kilometer cutoff. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table C.3: QCEW denominator and public-record bridge

Year	Construction	Estimate	SE	p -value	AR p -value
2000	Public numerator; disclosed service aggregate	0.283	0.118	0.016	0.009
2000	Public numerator; sum of public components	0.188	0.123	0.126	0.117
2000	Public numerator; aggregate including utilities	0.281	0.117	0.017	0.009
2001	Public numerator; disclosed service aggregate	0.279	0.116	0.016	0.013
2010	Public numerator; disclosed service aggregate	0.241	0.120	0.045	0.051
2010	Public numerator; sum of public components	0.138	0.127	0.275	0.289
2010	Public numerator; aggregate including utilities	0.239	0.119	0.045	0.052

Notes: Each row reports the high-throughput IV with otherwise identical treatment, controls, sample rule, and 100-kilometer Conley inference. The component-sum construction adds only public two-digit values. The aggregate-denominator construction retains all employment incorporated by BLS in the disclosed private service-providing total. The final construction in each endpoint shows that subtracting published private utilities is immaterial.

C.2.2 High- and low-throughput manufacturing peaks

To distinguish general local development from HSPS formation, I partition manufacturing into Chandler high-throughput industries and all remaining industries, labeled low throughput. Each group's peak is its maximum employment share over 1880, 1900, 1910, 1920, 1930, and 1940. Separate event studies align counties on each type-specific peak and interact its magnitude with event time, conditioning on the other type's magnitude interacted with the same windows. This compares the population, urbanization, manufacturing-HSPS, and service-HSPS paths associated with each type's own peak, rather than its share at the county-wide peak:

$$Y_{ct} = \alpha_c^g + \lambda_{s(c),t}^g + \mu_{k_{ct}}^g + \sum_{b \in \mathcal{B}} \mathbf{1}\{k_{ct}^g \in b\} \left[\beta_b^g \frac{P_c^g}{0.10} + \rho_b^g \frac{P_c^{\bar{g}}}{0.10} \right] + X_c' \Gamma^g \frac{t - 1900}{10} + \varepsilon_{ct}^g, \quad (51)$$

Here α_c^g , $\lambda_{s(c),t}^g$, and $\mu_{k_{ct}}^g$ are county, state-by-calendar-year, and exact decennial event-time fixed effects. The predetermined controls X_c are 1850 population, urbanization, schooling, colleges, and enslaved population, plus 1870 process-, machinery-, and traditional-industry shares; Γ^g gives each a linear calendar-decade trend, and ε_{ct}^g is the error. Each type and outcome is estimated separately. The plotted β_b^g associates a 10-percentage-point larger type- g peak with outcome Y_{ct} in window b , conditional on the other type's magnitude through ρ_b^g ; the other type's peak year does not set the clock. Standard errors use the 100-kilometer spatial estimator. Figure C.1 uses uniform 20-year windows and ends at +60, the last horizon supported by every eligible peak cohort.³¹ Service numerators and denominators share an employment universe within each vintage: harmonized Census records through 1950, imputed early CBP in 1960 and 1970, and EFSY CBP thereafter, as in Figure 3.

Both manufacturing types predict rising population and urbanization before and after their peaks. Over +10 to +40, a 10-percentage-point larger high-throughput peak is associated with 0.375 log points more population and a 6.33-percentage-point larger urban share; low-throughput associations are 0.143 log points and 2.32 percentage points, both distinguishable from zero. HSPS outcomes differ: the manufacturing-HSPS share is 0.831 percentage points higher after a larger high-throughput peak and 0.396 points lower after a larger low-throughput peak. At +50/+60, the service-HSPS coefficients are 0.429 points ($p = 0.006$) and -0.095 points ($p = 0.362$), a difference of 0.525 points ($p = 0.003$). This sequence is consistent with high-throughput industrialization developing HSPS work first within manufacturing and later in services. Population growth and urbanization alone cannot explain the pattern: low-throughput peaks predict both without a positive HSPS response.

C.2.3 Identification from initial industry shares

The shift-share design admits shares and shocks interpretations. Under [Goldsmith-Pinkham et al. \(2020\)](#), national growth rates combine predetermined 1880 county industry shares into one instrument, and 2SLS is a weighted combination of share-specific just-identified estimates. Figure C.2 reports these Rotemberg weights after residualizing treatment, shares, and outcome on the complete Table 2 conditioning set.

Twenty-six high-throughput shocks are usable. Blast furnaces and steel works, grain milling, tobacco, primary nonferrous metals, and other primary iron and steel have the largest absolute weights. The largest accounts for 19.8 percent, the top three for 49.8 percent, and the top five for 70.4 percent;

³¹ For the high-throughput service regression, support falls from 1,417 county observations at +60 to 226 at +70 and six at +100.

negative weights account for 10.2 percent of absolute weight. The inverse-Herfindahl effective number of Rotemberg weights is 8.6, and the exposure-weighted effective number of national shocks is 4.7. Identification is concentrated but does not rest on one industry.

Figure C.3 removes each high-throughput industry and rebuilds both stages. All second-stage coefficients remain positive, and 27 of the 28 reported concordance estimates remain significant at 5 percent. Removing grain milling reduces first-stage precision but leaves a positive estimate of 0.287. Removing primary metals, steel, or transportation equipment does not eliminate the result.

The within-group weights in equation (44) sum to one wherever the instrument is defined. The benchmark shifter therefore asks which high-throughput industries a county initially specialized in, conditional on having a classifiable high-throughput base; it does not use the initial size of that base as an omitted exposure share. Table C.4 nevertheless reports economy-scaled alternatives in which the incomplete high-throughput share is included separately, as required under the quasi-experimental exposure framework of [Borusyak et al. \(2022\)](#). It also alters the mapping of debatable sectors. The estimate remains positive and statistically significant when processed foods, machinery, or transportation equipment are removed separately, under a narrower continuous-process classification, and under a broader classification. Removing all processed foods weakens the first stage and makes the estimate imprecise; that specification discards several industries that Chandler explicitly associates with high-throughput production.

The comparison that most directly tests the mechanism holds the instrument construction fixed and changes the production regime supplying the national shifts. Table C.5 enters the high-throughput and comparison-group shifters jointly. High-throughput national growth predicts both a larger manufacturing peak and a larger long-run HSPS-service share. Growth in all remaining manufacturing industries or in the narrower traditional low-throughput set does not predict the service outcome. Equality of the second-stage coefficients is rejected at 10 percent against all non-high-throughput industries and at 5 percent against the traditional group. A generic local-demand or agglomeration account would need to explain why national growth outside high-throughput manufacturing generates industrial exposure without the same long-run service-composition response.

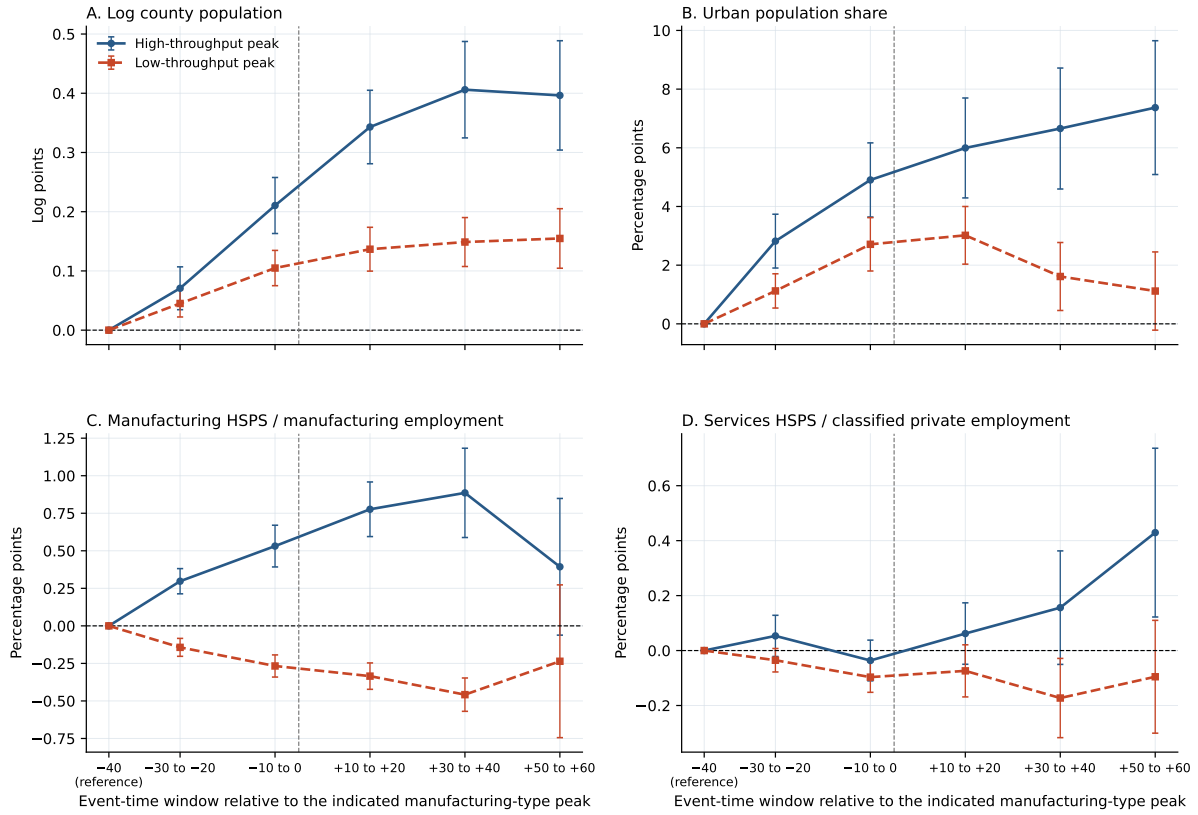


Figure C.1: Local development and HSPS employment around separate manufacturing-type peaks. The blue path is aligned on the county's high-throughput manufacturing-employment peak and the red path on its separately calculated low-throughput peak. Each point is the coefficient associated with a 10-percentage-point larger peak share in the indicated manufacturing type, conditional on the magnitude of the other type's peak interacted with every event-time window. The windows are $-30/-20$, $-10/0$, $+10/+20$, $+30/+40$, and $+50/+60$; event time -40 is omitted. Panels A and B report log county population and the urban population share. Panel C reports manufacturing HSPS employment divided by manufacturing employment; Panel D reports HSPS-service employment divided by classified private employment. The high- and low-peak regressions use different eligible county sets because each requires an observation at -40 relative to its own peak: 1,419 and 1,158 counties, respectively, for the service outcome. Both specifications include county, state-by-calendar-year, and exact decennial event-time fixed effects and predetermined baseline-covariate linear trends. Bars are 95-percent confidence intervals based on a 100-kilometer spatial covariance estimator.

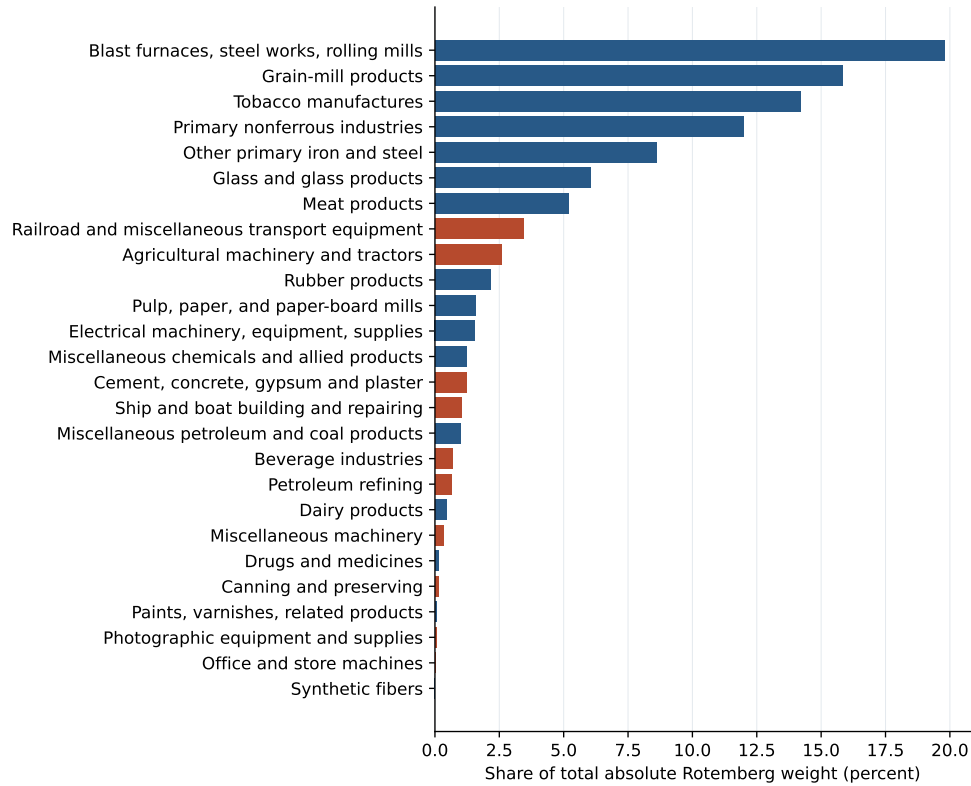


Figure C.2: Rotemberg weights in the high-throughput design. Absolute Rotemberg-weight shares are calculated on the QCEW-2000 estimation sample after partialling out the state and peak-cohort fixed effects, five 1850 controls, and complete 1850–1870 histories used in Table 2. Colors distinguish positive from negative weights. The underlying machine-readable table reports signed weights, first stages, and just-identified estimates for every usable industry.

Table C.4: Alternative throughput classifications and exposure normalizations

Specification	IV estimate	SE	p-value	First-stage F	AR p-value
<i>Panel A. Throughput classification</i>					
Benchmark Chandler set	0.283	0.118	0.016	12.87	0.009
Exclude processed foods	0.325	0.257	0.207	3.63	0.159
Exclude machinery	0.189	0.098	0.055	28.98	0.041
Exclude processed foods and machinery	0.366	0.175	0.036	9.19	0.011
Exclude transportation equipment	0.251	0.102	0.014	16.38	0.008
Narrow continuous/process set	0.183	0.078	0.018	29.69	0.011
Broader Chandler set	0.267	0.120	0.026	12.45	0.014
<i>Panel B. Exposure normalization</i>					
Within-high shares, benchmark	0.283	0.118	0.016	12.87	0.009
Within-high shares; control high-throughput share of manufacturing	0.322	0.135	0.017	11.36	0.009
Economy-scaled exposure; control included-share sum	0.206	0.080	0.010	40.29	0.015
Within-high shares; control high-throughput share of county employment	0.822	0.674	0.223	1.59	0.016
Economy-scaled exposure; county-employment denominator	−0.419	1.313	0.749	0.38	0.645

Notes: Every row instruments the same peak-manufacturing treatment and uses the QCEW-2000 outcome, sample rule, fixed effects, and predetermined controls in Table 2. The within-high specifications normalize 1880 industry employment by the county’s total employment in the relevant high-throughput set. The economy-scaled specification normalizes by the stated broader base and includes the sum of included exposure shares as a separate control. First-stage F statistics and conventional inference use a 100-kilometer Conley covariance estimator; AR entries are the corresponding reduced-form tests.

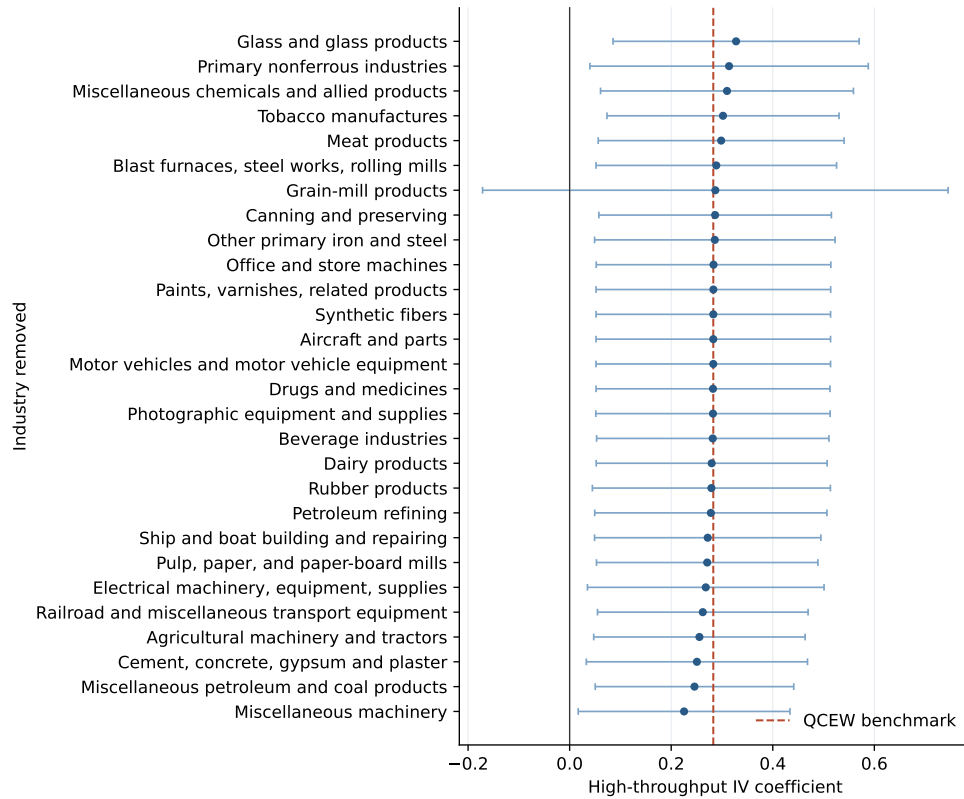


Figure C.3: Leave-one-industry-out high-throughput estimates. Each row removes the labeled IND1950 industry from the high-throughput set, reconstructs the leave-county-out shifter and its within-group weights, and re-estimates the first and second stages on the Table 2 sample. Points are 2SLS estimates; bars are 95-percent confidence intervals based on 100-kilometer Conley standard errors. The vertical line marks the benchmark estimate.

Table C.5: High-throughput and comparison-industry contrasts

Joint instrument set	High throughput	Comparison group	<i>p</i> -value, equality
<i>Panel A. First stage: peak manufacturing employment share</i>			
High throughput versus all non-high-throughput industries	1.928	0.193	0.014
High throughput versus traditional low-throughput industries	1.928	0.041	0.011
<i>Panel B. Reduced form: 2000 HSPS share of private services</i>			
High throughput versus all non-high-throughput industries	0.546	−0.049	0.053
High throughput versus traditional low-throughput industries	0.547	−0.074	0.021

Notes: Entries are percentage-point effects of a one-standard-deviation shifter from regressions that include both standardized group shifters simultaneously. Every regression uses the QCEW-2000 sample and the complete Table 2 conditioning set. The final column tests equality of the two coefficients using a 100-kilometer Conley covariance estimator. Panel B reports reduced forms because each group shifter predicts the common manufacturing-peak treatment with a different strength.

C.2.4 Flag-aware QCEW imputation

The public-record outcome is well-known to suppress data on employment from smaller counties to protect confidentiality of businesses. To address whether this may systematically bias the estimates, I perform two additional robustness checks: imputing suppressed data and aggregating to commuting zones. First I use the flags available beginning in 2001 which provide employment bins for suppressed data in QCEW to estimate the outcome variable in data suppressed counties. QCEW is preferable to EFSY for this specific purpose because it reports, in addition to a denominator capturing all private service employment, an aggregate employment count of NAICS industries 54-55 and 56 (parent code 1024). This permits a more precise estimate for the dependent variable than would be possible from EFSY. The disclosed county private service-providing total, code 102, supplies the top adding-up constraint, and disclosed service supersectors 1021–1029 supply the next set of margins. Two of these supersectors are especially informative for the outcome. Code 1022 equals Information (NAICS 51), while code 1024 is the higher-level Professional and Business Services aggregate that equals NAICS 54, 55, and 56. Consequently, whenever NAICS 56 is public, NAICS 54 plus 55 is observed exactly as code 1024 less NAICS 56 even if the separate 54 and 55 records are confidential. When NAICS 56 is also confidential, the disclosed 1024 total still tightly constrains the joint residual that can be assigned to 54, 55, and 56. Public three-digit descendants provide additional lower bounds whenever a two-digit parent is confidential.

Within each disclosed parent, I allocate only the residual using state-industry shares, with national-industry and equal-share sensitivities. Allocations never use the instrument, manufacturing peak, long-run outcome, or regression residuals. Zero- and one-worker lower bounds for confidential macro cells give virtually identical results because parent margins determine nearly all allocated employment. QCEW and CBP differ in administrative systems, reference periods, and treatment of auxiliary establishments. EFSY estimates CBP employment, rather than missing QCEW records, and therefore provides an independent-source check; the QCEW hierarchy reconstructs the latent main-outcome numerator.

Across 3,194 located counties, the preferred reconstruction implies an employment-weighted HSPS share of 14.38 percent, compared with 14.53 percent in the official national QCEW aggregates. The state- and national-share allocations differ by less than 0.01 percentage point in the aggregate, and changing the lower bound from zero to one worker changes the high-throughput estimate by less than 0.001. Unallocated hierarchy margins and published-child overages total 1,504 and 2,680 workers, respectively, and are negligible relative to national service employment.

The high-throughput coefficient is 0.187 (SE 0.100, $p = 0.061$, AR $p = 0.066$) under state-industry shares and 0.185 (SE 0.099, $p = 0.063$, AR $p = 0.066$) under national-industry shares. Equal splitting ignores observed state and national composition and gives 0.159 (SE 0.100, $p = 0.113$, AR $p = 0.121$). State shares are preferred because they reflect regional composition while remaining external to county treatment and outcome.

Table C.7 starts with the main Table 2 result. EFSY's fully imputed native-2000 CBP outcome gives a positive but imprecise high-throughput estimate of 0.103 (SE 0.080, AR $p = 0.202$). The QCEW flag-aware estimate is 0.187 (SE 0.100, conventional $p = 0.061$, AR $p = 0.066$). QCEW's separate code-1024 parent for NAICS 54–55–56 supplies an additional constraint on confidential employment, helping explain the source difference. Its all-industry first stage remains strong (Conley $F = 13.55$), but the second-stage estimate is 0.062 (AR $p = 0.519$). Averaging flag-aware outcomes over 2001–2003 preserves this contrast: the all-industry estimate is an insignificant 0.065, while the high-throughput estimate is 0.164 and significant at 10 percent under conventional and Anderson–Rubin tests.

Table C.6: Disclosure flags and QCEW-imputation audit, 2001

Sector	County records	Confidential records	Confidential with public local parent	Confidential with public three-digit descendant
NAICS 51	3,129	583	0	36
NAICS 54	3,197	948	826	0
NAICS 55	2,160	1,153	1,124	0
<i>Aggregate reconciliation</i>				
Preferred state-share allocation	14.3825%		employment-weighted HSPS share	
National-share allocation	14.3788%		employment-weighted HSPS share	
Equal-share allocation	14.4228%		employment-weighted HSPS share	
Official national QCEW aggregates	14.5272%		employment-weighted HSPS share	

Notes: A public local parent is QCEW supersector 1022 for Information or 1024 for Professional and Business Services. The latter equals NAICS 54, 55, and 56. Published cells remain fixed. State, national, and equal shares are alternative priors used only to split an otherwise unidentified residual among confidential children of a disclosed parent. Official national employment is 12,179,566 in NAICS 51, 54, and 55 and 83,839,631 in private services net of utilities.

The last two rows hold the year, denominator, controls, instruments, and 1,501-county sample fixed and change only the numerator source. The QCEW reconstruction gives 0.180, while the EFSY numerator gives 0.138 and is not statistically significant. The EFSY-minus-QCEW difference is -0.043 with a standard error of 0.098 and a p -value of 0.662. The data do not reject equality of the two estimates.

The flag-aware outcome also preserves the channel result. At the county level, the baseline peak coefficient is 0.029. Adding manufacturing scale and manufacturing HSPS reduces it to 0.012, an attenuation of 59 percent; manufacturing HSPS remains positive and precisely estimated at 1.260. At the commuting-zone level, the point estimate on the peak is already less precise, but the joint specification again leaves a small residual peak coefficient while both scale and manufacturing HSPS remain positive. EFSY gives a 75-percent attenuation at the commuting-zone level in both 2000 and 2001.

Table C.7: County IV results under alternative disclosure treatments

Outcome construction	OLS	All-industry IV	F_{all}	AR p_{all}	High-throughput IV	F_{high}	AR p_{high}
QCEW public records, 2000	0.029** (0.012)	0.133 (0.105)	14.67	0.203	0.283** (0.118)	12.87	0.009
EFSY-imputed CBP, native 2000	0.028*** (0.010)	0.031 (0.094)	14.67	0.747	0.103 (0.080)	12.87	0.202
Flag-aware QCEW, state shares, 2001	0.018* (0.009)	0.062 (0.093)	13.55	0.519	0.187* (0.100)	11.25	0.066
Flag-aware QCEW, 2001–2003 average	0.018** (0.009)	0.065 (0.085)	14.30	0.458	0.164* (0.091)	13.00	0.075
Flag-aware QCEW, matched 2001 sample	0.017* (0.009)	0.065 (0.095)	13.17	0.511	0.180* (0.100)	11.07	0.076
EFSY numerator / QCEW denominator, matched 2001 sample	0.034*** (0.011)	−0.013 (0.113)	13.17	0.910	0.138 (0.102)	11.07	0.186
Observations	1,487; 1,487; 1,511; 1,497; 1,501; 1,501, respectively						

Notes: Every specification matches Table 2. The matched 2001 rows use the same QCEW service denominator and differ only in the NAICS 51, 54, and 55 numerator. F statistics use 100-kilometer Conley first-stage standard errors; AR entries are the corresponding Conley reduced-form tests. The EFSY-minus-QCEW high-throughput coefficient in the matched sample is −0.043 (SE 0.098, $p = 0.662$). *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table C.8: Channel accounting under imputation and commuting-zone aggregation

	QCEW public county, 2000	QCEW flag-aware county, 2001	QCEW flag-aware CZ, 2001	EFSY native CZ, 2000	EFSY harmonized CZ, 2001
<i>Panel A. Baseline</i>					
Peak manufacturing employment share	0.048*** (0.010)	0.029*** (0.008)	0.022 (0.017)	0.038** (0.016)	0.037** (0.017)
<i>Panel B. Add manufacturing scale and HSPS</i>					
Peak manufacturing employment share	0.015 (0.012)	0.012 (0.011)	−0.012 (0.020)	0.010 (0.022)	0.009 (0.023)
Log manufacturing establishment scale at peak	0.0041* (0.0023)	0.0016 (0.0021)	0.0050* (0.0030)	0.0028 (0.0033)	0.0028 (0.0034)
Manufacturing HSPS / classified employment at peak	1.927*** (0.535)	1.260*** (0.433)	1.382* (0.814)	1.729** (0.834)	1.693** (0.837)
Attenuation of peak coefficient (%)	68.3	58.7	153.3	74.8	74.6
Observations	1,487	1,511	391	391	391

Notes: The first two columns use fixed 1850 counties. The last three aggregate additive historical and modern employment counts to 1990 commuting zones before forming shares. CZ fixed effects use the dominant historical state and the CZ peak cohort. The QCEW flag-aware columns use the preferred state-share allocation. Specifications otherwise reproduce the baseline and joint-channel columns of Table 1. The attenuation percentage exceeds 100 when the residual peak coefficient changes sign. Standard errors are Conley with a 100-kilometer cutoff. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

C.2.5 EFSY imputation and commuting-zone aggregation

Commuting-zone aggregation reduces the influence of confidentiality in small county-industry cells. I map 2000 and 1850 counties to commuting zones using weighted area overlap, then recompute outcomes, controls, and instruments with leave-commuting-zone-out national growth.

Table C.9 compares the 2001 outcomes on the same 391 commuting zones. Flag-aware QCEW gives 0.324 (SE 0.141, $p = 0.022$, AR $p = 0.005$), versus EFSY's 0.204 (SE 0.125, conventional $p = 0.102$, AR $p = 0.094$). Holding the QCEW denominator fixed and changing only the numerator raises EFSY's estimate to 0.248. The EFSY-minus-QCEW difference is −0.076 (SE 0.080, $p = 0.343$), so equality is not rejected. The source discrepancy is smaller at this geography, where confidentiality is less consequential.

The EFSY endpoint checks give 0.279 in native-2000 data (SE 0.127, $p = 0.028$, AR $p = 0.012$), 0.248 in harmonized-2000 data (SE 0.130, $p = 0.056$, AR $p = 0.035$), and 0.279 in harmonized-2010 data (SE 0.135, $p = 0.039$, AR $p = 0.009$). Thus EFSY recovers the high-throughput result after commuting-zone aggregation despite its imprecise county estimate, consistent with concentrated small-cell confidentiality and HSPS activity serving wider labor markets.

The difference between QCEW and EFSY is concentrated in the two sectors for which institutional coding and local margins differ most. In the 2000 county bridge, QCEW and EFSY give similar high-throughput coefficients for NAICS 51, 0.072 and 0.077. The QCEW coefficients for NAICS 54 and 55 are 0.141 and 0.070, whereas the corresponding EFSY coefficients are 0.034 and 0.003. EFSY's harmonization is built to recover CBP employment consistently across industry vintages; the QCEW allocation is built to recover the QCEW concept used by the main outcome. Because the two systems differ in their reference period, establishment frame, treatment of auxiliary establishments, and available hierarchy margins, neither is a mechanical correction of the other.

The QCEW reconstruction directly audits the main result by retaining its employment concept and annual-average reference period, identifying confidentiality through explicit flags, and exploiting the disclosed code-1024 parent for NAICS 54, 55, and 56 as a local constraint on the numerator.

Table C.9: QCEW and EFSY IV estimates at the commuting-zone level

Outcome construction	OLS	All-industry IV	F_{all}	AR p_{all}	High-throughput IV	F_{high}	AR p_{high}
Flag-aware QCEW, 2001	0.004 (0.020)	−0.274 (1.176)	0.17	0.736	0.324** (0.141)	9.03	0.005
EFSY harmonized, 2001	0.021 (0.020)	−0.537 (1.632)	0.17	0.541	0.204 (0.125)	9.03	0.094
EFSY numerator / QCEW denominator, 2001	0.044* (0.026)	−1.015 (2.804)	0.17	0.314	0.248* (0.141)	9.03	0.070
EFSY native classification, 2000	0.028 (0.018)	−0.640 (1.877)	0.17	0.428	0.279** (0.127)	9.03	0.012
EFSY harmonized, 2000	0.023 (0.020)	−0.362 (1.309)	0.17	0.669	0.248* (0.130)	9.03	0.035
EFSY harmonized, 2010	0.018 (0.018)	0.020 (0.839)	0.17	0.981	0.279** (0.135)	9.03	0.009

Commuting zones

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Notes: All rows use the same standard 1990 commuting zones, historical treatment, controls, fixed effects, and commuting-zone shift-share instruments. The QCEW row aggregates the preferred county-level state-share reconstruction. The EFSY 2001 row uses an EFSY numerator and EFSY private-service denominator; the following row changes only the numerator while retaining the QCEW denominator. The all-industry instrument is weak after aggregation and its CZ second-stage estimates are not interpreted. Coefficient and first-stage standard errors are Conley with a 100-kilometer cutoff. AR entries report the corresponding reduced-form test. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

C.2.6 Identification and inference from national industry shocks

Under [Borusyak et al. \(2022\)](#), identification instead rests on national industry-growth shocks, so sampling variation is organized by industry. The exact implementation retains county–industry leave-county-out shocks and clusters equivalent shock-level scores by industry. A common-shock approximation replaces the small leave-out differences with one national shock per industry and estimates the collapsed regression. Both account for the shared disturbances facing counties with similar industry shares ([Adão et al., 2019](#)).

In Table C.10, the exact transformation preserves the county 2SLS coefficient but changes uncertainty. The public-record QCEW estimate is 0.283 with an industry-clustered SE of 0.159; conventional and Anderson–Rubin tests reject at 10 percent ($p = 0.088$ and $p = 0.062$). The common-shock approximation yields 0.284 (SE 0.132, AR $p = 0.047$). Flag-aware 2001 and averaged 2001–2003 outcomes give smaller positive estimates, with exact-shock AR p -values of 0.065 and 0.082.

Precision is limited by 26 usable high-throughput shocks and an exposure-weighted effective count of approximately 4.7. Exact shock-clustered first-stage F statistics are 5.26 for the main outcome and 5.02 for the flag-aware outcome, below county-level Conley values. The all-industry shock-level first stage is also weak, and neither its reduced form nor second stage is distinguishable from zero. The shocks interpretation supports a positive high-throughput effect at 10 percent but is less precise than inference organized around 1,487 counties.

Table C.10: Industry-shock inference for the shift-share estimates

Outcome and inference	IV estimate	SE	<i>p</i> -value	First-stage <i>F</i>	AR <i>p</i> -value	Industries
<i>Panel A. High-throughput instrument</i>						
QCEW public, 2000: county-level Conley	0.283	0.118	0.016	12.87	0.009	26
QCEW public, 2000: exact leave-out shocks	0.283	0.159	0.088	5.26	0.062	26
QCEW public, 2000: common national shocks	0.284	0.132	0.041	3.94	0.047	26
Flag-aware QCEW, 2001: county-level Conley	0.187	0.100	0.061	11.25	0.066	26
Flag-aware QCEW, 2001: exact leave-out shocks	0.187	0.121	0.135	5.02	0.065	26
Flag-aware QCEW, 2001: common national shocks	0.188	0.106	0.089	3.83	0.034	26
Flag-aware QCEW, 2001–2003: county-level Conley	0.164	0.091	0.069	13.00	0.075	26
Flag-aware QCEW, 2001–2003: exact leave-out shocks	0.164	0.108	0.142	5.39	0.082	26
<i>Panel B. All-industry instrument, QCEW public outcome</i>						
County-level Conley	0.133	0.105	0.202	14.67	0.203	57
Exact leave-out shocks	0.133	0.223	0.551	1.59	0.623	57
Common national shocks	0.133	0.141	0.349	3.29	0.422	57
<i>Panel C. Concentration diagnostics for high-throughput shocks</i>						
Exposure-weighted effective number of shocks				4.71		
Inverse-Herfindahl effective number of Rotemberg weights				8.62		
Largest / three largest / five largest absolute weights				19.8% / 49.8% / 70.4%		
Negative share of total absolute weight				10.2%		

Notes: County-level rows use the 100-kilometer Conley estimator reported elsewhere. Exact leave-out rows retain the county-specific leave-out shock and cluster the equivalent shock-level score by IND1950 industry. Common-shock rows use one leave-sample-out national growth shock per industry. Their first-stage *F* is the shock-level statistic. *p*-values use finite-industry degrees of freedom; AR entries are the corresponding reduced-form tests. The bounded 95-percent exact-shock AR confidence set for the main high-throughput estimate is $[-0.023, 2.108]$; for the flag-aware outcome it is $[-0.014, 2.181]$.

C.2.7 Predetermined balance, controls, and spatial inference

Table C.11 tests pre-1880 outcomes against the standardized high-throughput shifter, five 1850 covariates, state fixed effects, and the other two outcome families' histories. Exposure predicts the 1860–1870 change in manufacturing and, at 10 percent, HSPS services; it does not predict 1850–1860 changes in manufacturing, manufacturing HSPS, or services, or the 1860–1870 manufacturing-HSPS change. Nonzero level correlations further motivate conditioning on complete observed 1850–1870 paths rather than asserting unconditional balance of the 1880 mix.

Table C.12 then varies the conditioning set and the spatial cutoff. Omitting the pre-1880 histories reduces the high-throughput coefficient from 0.283 to 0.184 rather than creating it. Adding proximity to coal, waterpower, ports, the Ohio and Mississippi rivers, and the Great Lakes raises the estimate to 0.320. Adding peak-year wages, population, and urbanization lowers first-stage precision and produces a positive but imprecise estimate of 0.242. Adding the manufacturing employment share in 2000 leaves a positive estimate of 0.196 that is significant at 5 percent. This last result rules out the mechanical explanation that high-throughput exposure changes service composition simply because it leaves more manufacturing employment in place at the endpoint. These variables are post-1880 and can be caused by industrialization, so both rows are channel-blocking exercises rather than preferred identification specifications. The point estimate is invariant to the Conley radius by construction, while the 95-percent uncertainty widens at long cutoffs. It remains significant at 5 percent through 300 kilometers and at 10 percent at 500 kilometers; the AR test likewise rejects at 10 percent throughout.

Table C.11: High-throughput exposure and pre-1880 changes

Pre-period outcome	Coefficient	SE	<i>p</i> -value
Manufacturing share, 1850–1860 change	−0.0027	0.0018	0.126
Manufacturing share, 1860–1870 change	0.0047	0.0017	0.007
Manufacturing HSPS, 1850–1860 change	0.000012	0.000007	0.100
Manufacturing HSPS, 1860–1870 change	0.000006	0.000007	0.399
HSPS services, 1850–1860 change	0.151	0.171	0.378
HSPS services, 1860–1870 change	−0.225	0.122	0.064

Notes: Each row regresses the indicated change on the standardized high-throughput shifter, state fixed effects, the five 1850 controls, and the complete histories of the other two outcome families, using the 1,487-county QCEW-2000 sample. The tested outcome’s own history is omitted from the conditioning set. Standard errors are Conley with a 100-kilometer cutoff. Peak-cohort fixed effects are excluded because the peak is realized after the 1880 exposure shares.

Table C.12: Conditioning-set and spatial-inference robustness

Specification	IV estimate	SE	<i>p</i> -value	First-stage <i>F</i>	AR <i>p</i> -value
Preferred controls, 100 km	0.283	0.118	0.016	12.87	0.009
No peak-cohort fixed effects	0.298	0.115	0.010	13.46	0.004
Omit 1850–1870 histories	0.184	0.047	< 0.001	45.26	< 0.001
Add historical geography controls	0.320	0.120	0.008	13.43	0.002
Add peak-year wages, urbanization, and population	0.242	0.197	0.218	5.45	0.188
Add 2000 manufacturing employment share	0.196	0.096	0.042	14.81	0.042
Preferred controls, 200 km	0.283	0.114	0.013	8.37	0.018
Preferred controls, 300 km	0.283	0.134	0.034	5.20	0.038
Preferred controls, 500 km	0.283	0.170	0.096	2.83	0.067

Notes: Every row uses the QCEW-2000 outcome and high-throughput instrument. Historical-geography controls are log distance to coal, waterpower, the nearest port, the Ohio River, and the Mississippi River and an indicator for Great Lakes access. The peak-year row includes average manufacturing wages, the urban population share, log population, and the missing-wage indicator. The endpoint-manufacturing row adds manufacturing employment divided by total classified employment in 2000. Rows 1–6 use a 100-kilometer cutoff; rows 7–9 vary the spatial radius. AR entries are the reduced-form tests computed with the same covariance estimator.

These exercises preserve the main empirical pattern. Manufacturing scale and HSPS account for most of the peak-industrialization–service-composition relationship. Both manufacturing types predict population growth and urbanization, but only high throughput predicts later HSPS services; higher manufacturing peaks predict HSPS employment rather than a larger overall service sector. Endpoint, peak-window, classification, leave-one-industry-out, exposure-normalization, geography, and spatial checks support this pattern. Flag-aware QCEW preserves the high-versus-all contrast at 10 percent; EFSY is imprecise at county level but positive and significant at commuting-zone level in 2000 and 2010, with estimates statistically indistinguishable from QCEW. Exact industry-shock Anderson–Rubin inference also rejects at 10 percent despite only 26 high-throughput shocks. The evidence therefore links national growth in production requiring large-scale coordination to deeper local industrialization and persistent service reallocation toward organizational functions; broad manufacturing growth alone does not show the same pattern.

C.2.8 Corporate headquarters and NAICS 55

NAICS 55 includes headquarters and management establishments of firms operating in manufacturing. Such establishments supply the managerial services central to the mechanism, but could nevertheless account for the measured service response. I therefore decompose the flag-aware numerator into NAICS 51, 54, and 55 and re-estimate the high-throughput IV for each.

Table C.13: Excluding management of companies and enterprises from HSPS services

	Counties	Commuting zones
Full HSPS numerator: NAICS 51, 54, and 55	0.187* (0.100)	0.324** (0.141)
Excluding NAICS 55: NAICS 51 and 54	0.147 (0.090)	0.254** (0.121)
NAICS 51: Information	0.024 (0.034)	0.079* (0.046)
NAICS 54: Professional, scientific, and technical services	0.122 (0.086)	0.175** (0.089)
NAICS 55: Management of companies and enterprises	0.040* (0.021)	0.069* (0.036)
First-stage Conley F statistic	11.25	9.03
Anderson–Rubin p -value, excluding NAICS 55	0.111	0.012
Observations	1,511	391

Notes: Each coefficient row reports the high-throughput IV estimate from the specification in Table 2. The dependent variable is flag-aware 2001 QCEW private employment in the indicated sector or sectors divided by the same private-service denominator. County counts are mapped to fixed 1850 boundaries before shares are formed; commuting-zone counts are summed before shares are formed. All specifications include state and manufacturing-peak-cohort fixed effects, the five 1850 controls, and complete 1850–1870 histories. Standard errors in parentheses use a 100-kilometer Conley cutoff. The three component coefficients add exactly to the full-numerator coefficient. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Excluding NAICS 55 reduces the county estimate from 0.187 to 0.147 and the commuting-zone estimate from 0.324 to 0.254: NAICS 51 and 54 retain approximately 79 percent of the effect at both geographies. The commuting-zone estimate remains significant conventionally ($p = 0.035$) and under Anderson–Rubin inference ($p = 0.012$). NAICS 54 is the largest component at both geographies and independently significant at commuting-zone level. Headquarters contribute about one-fifth of the estimate rather than driving the persistent HSPS-service response.

D Model Details and Policy Extension

This appendix documents historical examples of producer-service codification, the equilibrium and proposition proofs, calibration and sensitivity exercises, and the policy extension.

D.1 Historical examples of producer-service codification

The model distinguishes producer-service functions from how they are performed. Work organization, quality control, inventory management, accounting, procurement, and personnel management can rely

on local experience, tacit knowledge, and informal supervision, or on codified records, procedures, metrics, job categories, and decision rules. Repeated, measurable tasks make manufacturing and industrial procurement natural settings for codification. The resulting routines can then diffuse to department stores, hospitals, banks, retailers, offices, and universities.

Table D.1 summarizes the mapping from the model to several historical examples.

Table D.1: Examples of producer-service tasks moving from low-skill to high-skill mode

Producer-service task	Low-skill mode	Industrial codification	Later service sector use
Work organization	Foreman judgment, local rules of thumb, informal supervision	Motion study, workflow design, timing of tasks, written procedures, training systems	Department stores, offices, hospitals, and other service organizations use workflow design, scheduling, clerical organization, and personnel practices
Quality control	Inspection by eye, informal rejection of defective output	Testing protocols, defect records, statistical process control, control charts	Clinical laboratories and hospitals use control charts and statistical quality-control routines to monitor measurement processes
Accounting and managerial control	Bookkeeping and transaction recording	Cost accounting, budgeting, return-on-investment calculations, divisional reporting, performance metrics	Retail chains, department stores, banks, hospitals, universities, and other multi-unit service organizations use budgeting and management-control systems
Inventory, procurement, and logistics	Visual stock checks, ad hoc ordering, relationship-based purchasing	Stock-control systems, procurement planning, supplier coordination, delivery scheduling	Retail chains, wholesalers, hospitals, restaurants, and distribution firms use inventory and logistics routines
Personnel and training	Informal apprenticeship, personal hiring, on-the-job learning	Job analysis, hiring rules, training manuals, personnel departments, performance evaluation	Department stores, hotels, hospitals, offices, and other service organizations use formal hiring and training systems

Work organization: from factory motion study to department stores. In low-skill work organization, foremen and supervisors rely on experience and local knowledge. Repeated factory operations made worker motions, delays, output, and defects observable and measurable, enabling scientific management to codify work organization through motion study, workflow design, written procedures, and training.

Frank and Lillian Gilbreth developed motion-study methods initially oriented toward industrial work. Lillian Gilbreth later adapted motion study and industrial psychology to service and household settings, including department-store employees and consumers (Graham, 1992, 1999). Her work at Macy's in 1925–1928 applied scientific-management and personnel-management ideas to hiring, training, employee handling, clerical organization, and work organization (Graham, 2000). This illustrates the model's diffusion mechanism: routines codified in industrial work subsequently serve other sectors.

Quality control: from industrial process control to clinical laboratories. Informal quality control relies on direct inspection and experienced workers' judgment. Recording manufacturing defects, tolerances, downtime, and output variation enabled statistical process control. Walter Shewhart introduced the control chart at Bell Laboratories in the 1920s and formalized statistical monitoring of production variation in his later book (Shewhart, 1931; Best and Neuhauser, 2006).

Clinical laboratories subsequently adopted statistical quality control to monitor laboratory measurements. Levey and Jennings introduced control charts for clinical laboratories in 1950, applying process-control logic to repeated medical measurements (Levey and Jennings, 1950; Westgard et al., 1981). Informal inspection became a routine based on records, thresholds, and statistical decision rules. In the model, public organizational knowledge K_t similarly facilitates service-sector adoption of tasks first formalized in industry.

Accounting and managerial control: from industrial cost control to retail management. Accounting moves from transaction recording and bookkeeping to high-skill managerial control when managers must compare plants, products, inventories, costs, divisions, and capital allocations. The historical literature traces cost accounting and managerial control to large industrial and transportation enterprises, scientific management, and formal budgeting and performance measures at firms such as DuPont and General Motors (Johnson, 1975; Kaplan, 1984; Chandler, 1977).

These routines also became tools for service firms. James O. McKinsey professionalized budgeting and managerial accounting as consulting services through McKinsey (1922) and McKinsey (1924). After a 1935 consulting engagement for the Chicago department-store chain Marshall Field's, he became its chairman and chief executive (Flesher, 1985). The transferable object was a producer-service routine for measuring, budgeting, and managing a complex organization: the model's transition from low-skill bookkeeping to codified high-skill managerial control.

D.2 Equilibrium details

Producer-service supply. Formal and informal producer services are supplied competitively using skilled and unskilled labor, respectively. If $\mathcal{L}_t^H$ and $\mathcal{L}_t^L$ denote the corresponding employment levels, output is

$$Y_t^H = A_t^H h_s \mathcal{L}_t^H, \quad Y_t^L = A_t^L h_u \mathcal{L}_t^L. \quad (52)$$

Perfect competition gives $p_t^H = w_t^s/A_t^H$ and $p_t^L = w_t^u/A_t^L$, since wages are prices per efficiency unit. Formal services enter gross production, recurring coordination, and procedure creation and adaptation; informal services enter gross production and recurring coordination. Both producer-service productivities follow (29) with exponent ϕ_c .

Aggregate state and prices. The beginning-of-period aggregate state is

$$\mathcal{X}_t \equiv (L_t, S_t, K_t, \{A_{jt}, Z_{jt}, \mu_{jt}\}_{j \in \mathcal{J}}, A_t^H, A_t^L, \{e_{i,t}\}_{i \in \{a,m,s\}}). \quad (53)$$

Here $\mu_{jt}(a, s)$ is the measure of active sector- j establishments over permanent ability and inherited formalization frontiers. Establishment mass, employment, and the employment-weighted formal share are

$$M_{jt} = \int d\mu_{jt}(a, s), \quad N_{jt} = \int n_{jt}^*(a, s) d\mu_{jt}(a, s), \quad x_{jt} = \frac{1}{N_{jt}} \int n_{jt}^*(a, s) q(s) d\mu_{jt}(a, s). \quad (54)$$

The aggregate x_{jt} summarizes the establishment distribution; each establishment's productive and recurring producer-service demands depend on its own inherited share $q(s)$.

Let

$$\mathcal{P}_t \equiv (\{p_{jt}, p_{jt}^c\}_{j \in \mathcal{J}}, p_t^H, p_t^L, w_t^s, w_t^u), \quad p_{mt} \equiv 1, \quad (55)$$

where p_{jt} and p_{jt}^c are producer and consumer prices and manufacturing output is the numeraire. Let $\mathcal{Q}_t$ collect household allocations, establishment input and organizational policies, entry, and producer-service production.

Perfect-foresight competitive equilibrium. Given an initial state $\mathcal{X}_0$ and deterministic paths for exogenous fundamentals and government policies, a perfect-foresight competitive equilibrium is a sequence

$$\{\mathcal{P}_t, \mathcal{Q}_t, \mathcal{X}_{t+1}\}_{t=0}^{\infty} \quad (56)$$

such that, at every date t , the following conditions hold.

1. *Households.* Households choose sectoral consumption, fertility, and education optimally subject to their budget and time constraints. Children choose whether to acquire skill according to the random-utility rule in (27). Households take current prices and dividends as given and correctly anticipate future wages and skill premia.
2. *Final-good establishments.* For every sector j and state (a, s) , an incumbent chooses current labor, materials, and output-linked producer services to solve (14). It chooses its next-period frontier $u \geq s$ according to (21). The inherited share $q(s)$ governs current production and recurring coordination; the chosen frontier affects production from $t + 1$ onward.
3. *Producer-service suppliers.* Formal and informal suppliers maximize profits under (52). Their outputs equal the sums of output-linked demand, recurring organizational demand, and, for formal services, the inputs used to create and adapt procedures.
4. *Entry.* Potential entrants pay the setup cost before drawing $a \sim G$ and enter with $s = 0$. Entry and establishment counts satisfy (23).

5. *Market clearing and resource feasibility.* Agricultural, manufacturing, and consumer-service output is allocated to household consumption, intermediate use, and entry setup. Formal and informal producer-service markets clear separately. Demand for skilled and unskilled efficiency units equals the corresponding household supply. Establishment profits net of organizational investment, entry costs, and taxes are distributed to households, and the government budget balances.
6. *Aggregation and laws of motion.* Individual allocations aggregate according to (54). Let $u_{jt}^*(a, s)$ be the organizational policy implied by the Bellman equation. For every measurable set B of establishment states,

$$\begin{aligned} \mu_{j,t+1}(B) = & (1 - \delta_f) \int \mathbb{1}\{(a, u_{jt}^*(a, s)) \in B\} d\mu_{jt}(a, s) \\ & + I_{j,t+1} \int \mathbb{1}\{(a, 0) \in B\} dG(a). \end{aligned} \quad (57)$$

Public knowledge follows (22); population, skills, and productivities follow (28)–(29).

7. *Perfect foresight.* Forecasts of future aggregate prices and states coincide with their equilibrium realizations: for every $k \geq 1$,

$$\widehat{\mathcal{P}}_{t+k|t} = \mathcal{P}_{t+k}, \quad \widehat{\mathcal{X}}_{t+k|t} = \mathcal{X}_{t+k}. \quad (58)$$

Perfect foresight concerns aggregate variables. Establishments retain idiosyncratic exit risk and entrants draw heterogeneous ability; with a continuum of establishments, these risks aggregate according to their known distributions without aggregate uncertainty.

D.3 Establishment and household decisions and proposition proofs

D.3.1 Proof of Proposition 1

I first derive the reduced static establishment problem used in the proof. Conditional cost minimization in the CES labor nest gives

$$w_{jt}^v = \left[\alpha_j^{\sigma_L} (w_t^s)^{1-\sigma_L} + (1 - \alpha_j)^{\sigma_L} (w_t^u)^{1-\sigma_L} \right]^{\frac{1}{1-\sigma_L}}. \quad (59)$$

Because the labor aggregator is homogeneous of degree one, headcount is $n = \nu_{jt}\ell$, where

$$\nu_{jt} = \frac{\alpha_j^{\sigma_L}}{h_s} \left(\frac{w_{jt}^v}{w_t^s} \right)^{\sigma_L} + \frac{(1 - \alpha_j)^{\sigma_L}}{h_u} \left(\frac{w_{jt}^v}{w_t^u} \right)^{\sigma_L}. \quad (60)$$

Recurring organizational cost per unit of the labor composite is therefore

$$c_{jt}^o(q) = \frac{\nu_{jt}}{N} [qp_t^H + r(1 - q)p_t^L]. \quad (61)$$

Fix (j, t, a, s) and write $q = q(s)$. Before conditional inputs are eliminated, the establishment solves

$$g_{jt}^*(a, s) = \max_{\ell, m^a, m^m, H^p, L^p \geq 0} \left\{ p_{jt} Z_{jt} (A_{jt} a \ell^\eta)^{\theta_{jt}} (m^a)^{\gamma_{jt}^a} (m^m)^{\gamma_{jt}^m} [P^p(H^p, L^p; q)]^{\gamma_{jt}^P} - w_{jt}^v \ell - p_{at} m^a - p_{mt} m^m - p_t^H H^p - p_t^L L^p - c_{jt}^o(q) \ell \right\}. \quad (62)$$

For fixed P^p , CES duality implies expenditure $\omega_t^P(q) P^p$, where

$$\omega_t^P(q) = [q(p_t^H)^{1-\sigma_P} + (1-q)(p_t^L)^{1-\sigma_P}]^{\frac{1}{1-\sigma_P}}.$$

Next fix value added $v = A_{jt} a \ell^\eta$ and optimize over (m^a, m^m, P^p) . If $\mathcal{R}$ denotes gross revenue at the conditional optimum, the first-order conditions imply

$$p_{at} m^a = \gamma_{jt}^a \mathcal{R}, \quad p_{mt} m^m = \gamma_{jt}^m \mathcal{R}, \quad \omega_t^P(q) P^p = \gamma_{jt}^P \mathcal{R}. \quad (63)$$

Substitution into the gross-output technology gives

$$\mathcal{R} = \left[p_{jt} Z_{jt} \left(\frac{\gamma_{jt}^a}{p_{at}} \right)^{\gamma_{jt}^a} \left(\frac{\gamma_{jt}^m}{p_{mt}} \right)^{\gamma_{jt}^m} \left(\frac{\gamma_{jt}^P}{\omega_t^P(q)} \right)^{\gamma_{jt}^P} \right]^{\frac{1}{\theta_{jt}}} v. \quad (64)$$

Conditional expenditure on materials and productive producer services is $(1 - \theta_{jt}) \mathcal{R}$, so revenue net of these inputs is $\theta_{jt} \mathcal{R} = p_{jt}^v(q) v$. Hence the static problem reduces to

$$g_{jt}^*(a, q) = \max_{\ell \geq 0} \{ p_{jt}^v(q) A_{jt} a \ell^\eta - [w_{jt}^v + c_{jt}^o(q)] \ell \}. \quad (65)$$

The unique interior solution satisfies

$$\ell_{jt}^*(a, q) = \left[\frac{\eta p_{jt}^v(q) A_{jt} a}{w_{jt}^v + c_{jt}^o(q)} \right]^{\frac{1}{1-\eta}}, \quad n_{jt}^*(a, q) = \nu_{jt} \ell_{jt}^*(a, q). \quad (66)$$

Proof of Proposition 1. By the envelope theorem,

$$\begin{aligned} \frac{\partial g_{jt}^*(a, q)}{\partial q} &= v_{jt}^*(a, q) \frac{\partial p_{jt}^v(q)}{\partial q} - \ell_{jt}^*(a, q) \frac{\partial c_{jt}^o(q)}{\partial q} \\ &= v_{jt}^*(a, q) \frac{\partial p_{jt}^v(q)}{\partial q} + \frac{n_{jt}^*(a, q)}{\bar{N}} (r p_t^L - p_t^H), \end{aligned} \quad (67)$$

where the second equality uses

$$\frac{\partial c_{jt}^o(q)}{\partial q} = \frac{\nu_{jt}}{\bar{N}} (p_t^H - r p_t^L).$$

The first-order condition for ℓ implies

$$\eta p_{jt}^v(q) v_{jt}^*(a, q) = [w_{jt}^v + c_{jt}^o(q)] \frac{n_{jt}^*(a, q)}{\nu_{jt}}.$$

Substituting into (67) gives

$$\frac{\partial g_{jt}^*(a, q)}{\partial q} = n_{jt}^*(a, q) \left[\frac{w_{jt}^v + c_{jt}^o(q)}{\eta \nu_{jt}} \frac{\partial \log p_{jt}^v(q)}{\partial q} + \frac{rp_t^L - p_t^H}{\bar{N}} \right] = n_{jt}^*(a, q) \Psi_{jt}(q).$$

Thus, whenever $\Psi_{jt}(q) > 0$, the marginal operating gain from formalization is positive and proportional to employment.

It remains to establish the converse effect from formalization to desired scale. Differentiating (66) gives

$$\frac{\partial \log n_{jt}^*(a, q)}{\partial q} = \frac{1}{1 - \eta} \left[\frac{\partial \log p_{jt}^v(q)}{\partial q} + \frac{\nu_{jt}}{\bar{N}} \frac{rp_t^L - p_t^H}{w_{jt}^v + c_{jt}^o(q)} \right]. \quad (68)$$

If $\Psi_{jt}(q) > 0$, then

$$\frac{\partial \log p_{jt}^v(q)}{\partial q} > -\frac{\eta \nu_{jt}}{\bar{N}} \frac{rp_t^L - p_t^H}{w_{jt}^v + c_{jt}^o(q)}.$$

Hence, when $rp_t^L > p_t^H$, the bracket in (68) is strictly positive. Therefore

$$\frac{\partial \log n_{jt}^*(a, q)}{\partial q} > 0.$$

This proves both statements of the proposition. □

D.3.2 Proof of Proposition 2

I first derive the household decision rules used in the proof. Define

$$\zeta_t \equiv 1 + \frac{d_t}{\bar{w}_t}, \quad \chi_t \equiv \tau_e(S_t).$$

For a household of employment origin i , the time cost of one child receiving education $e_{i,t}$ is

$$T_{i,t}(e_{i,t}) = \tau_i + \chi_t e_{i,t}.$$

Conditional on the aggregate growth and skill environment, the first-order condition for fertility gives

$$n_{i,t} = \frac{\gamma \zeta_t}{\tau_i + \tau_e(S_t) e_{i,t}}. \quad (69)$$

For an interior education choice, the first-order condition is

$$\frac{g_t}{(e_{i,t} + \tau_h)(e_{i,t} + \tau_h + g_t)} = \frac{\tau_e(S_t)}{\tau_i + \tau_e(S_t) e_{i,t}},$$

which rearranges to

$$\tau_e(S_t)(e_{i,t} + \tau_h)^2 = g_t[\tau_i - \tau_e(S_t)\tau_h].$$

Hence

$$e_{i,t}^* = \left[\sqrt{g_t \left(\frac{\tau_i}{\tau_e(S_t)} - \tau_h \right)} - \tau_h \right]_+, \quad (70)$$

where $[x]_+ \equiv \max\{x, 0\}$. Aggregate fertility is

$$\bar{n}_t = \sum_{i \in \{a, m, s\}} \lambda_{i,t} n_{i,t}. \quad (71)$$

Proof of Proposition 2. Fix $S \in (0, 1)$ and define

$$t \equiv \tau_e(S), \quad a \equiv \gamma\zeta(S) - \tau_s, \quad b \equiv \tau_s - t\tau_h.$$

The maintained interiority conditions imply $a > 0$ and $b > 0$. Equations (70), (69), and (29) reduce in the terminal service economy to

$$g(L) = \left[1 + \lambda_A(SL)^{\phi_c}\right]^{1/\Delta} - 1, \quad (72)$$

$$e(L) = \left[\sqrt{\frac{b}{t}} g(L) - \tau_h\right]_+, \quad (73)$$

$$n(L) = \frac{\gamma\zeta(S)}{\tau_s + te(L)}. \quad (74)$$

Population therefore follows $L_{t+1} = T(L_t)$, where $T(L) = Ln(L)^{1/\Delta}$.

The function $g(L)$ is continuous and strictly increasing from zero to infinity. Hence $e(L)$ is continuous and nondecreasing, while $n(L)$ is continuous and nonincreasing. Moreover,

$$n(0) = \frac{\gamma\zeta(S)}{\tau_s} > 1, \quad \lim_{L \rightarrow \infty} n(L) = 0.$$

There is therefore a unique $L^\infty(S) > 0$ satisfying $n(L^\infty(S)) = 1$.

To establish global stability, first consider an interval in which education is interior. Let $\epsilon_g(L) = d \log g(L) / d \log L$. Concavity of $[1 + x]^{1/\Delta} - 1$ implies $0 < \epsilon_g(L) \leq \phi_c$. From (73),

$$\frac{de(L)}{d \log L} = \frac{1}{2} [e(L) + \tau_h] \epsilon_g(L).$$

Since $b > 0$ implies $t[e(L) + \tau_h] < \tau_s + te(L)$,

$$-\frac{d \log n(L)}{d \log L} = \frac{t}{\tau_s + te(L)} \frac{de(L)}{d \log L} < \frac{\epsilon_g(L)}{2} \leq \frac{\phi_c}{2}.$$

Therefore

$$\frac{d \log T(L)}{d \log L} = 1 + \frac{1}{\Delta} \frac{d \log n(L)}{d \log L} > 1 - \frac{\phi_c}{2\Delta} > 0.$$

Where education is at its corner, $n(L)$ is locally constant and $T(L)$ remains strictly increasing. Thus T is globally increasing. If $L < L^\infty(S)$, then $L < T(L) < L^\infty(S)$; if $L > L^\infty(S)$, then $L > T(L) > L^\infty(S)$. Hence the population sequence is monotone and bounded and converges to $L^\infty(S)$ from every $L_0 > 0$, while fertility converges to replacement.

At the fixed point, (74) implies

$$e^\infty(S) = \frac{\gamma\zeta(S) - \tau_s}{\tau_e(S)} = \frac{a}{t}.$$

Using the interior education rule then gives

$$\begin{aligned} g^\infty(S) &= \frac{[a + t\tau_h]^2}{tb} \\ &= \frac{[\gamma\zeta(S) - \tau_s + \tau_e(S)\tau_h]^2}{\tau_e(S)[\tau_s - \tau_e(S)\tau_h]} = \mathcal{G}(S), \end{aligned}$$

which proves (30). Inverting (72) at $g(L^\infty) = \mathcal{G}(S)$ gives (31).

Finally, since $t'(S) = -(1 - \tau_e^s) < 0$, logarithmic differentiation yields

$$\begin{aligned} \frac{\mathcal{G}'(S)}{\mathcal{G}(S)} &= \frac{2\gamma\zeta'(S)}{a + t\tau_h} \\ &\quad + t'(S) \left[\frac{2\tau_h}{a + t\tau_h} - \frac{1}{t} + \frac{\tau_h}{\tau_s - t\tau_h} \right]. \end{aligned} \tag{75}$$

The first term is nonnegative when $\zeta'(S) \geq 0$. Since $t'(S) < 0$, the second term is strictly positive whenever the bracketed expression is negative. Rearranging this condition gives

$$\tau_h = 0 < \frac{\tau_s[\gamma\zeta(S) - \tau_s]}{\tau_e(S)[2\gamma\zeta(S) - \tau_s]},$$

Hence $\mathcal{G}'(S) > 0$ under the conditions stated in the proposition. $\square$

D.3.3 Proof of Proposition 3

Proof of Proposition 3. Consider first the terminal service economy. For a stationary service establishment, the adaptation cost in (19) and the discounted recurring-coordination gain imply

$$q_f(K) = \min \left\{ K, \left[1 - \frac{1}{e\mathcal{A}_f} \right]_+ \right\}.$$

The employment-weighted formalized-task share,

$$x(K) = \frac{1}{N^s} \int n_f q_f(K) d\mu^s(f),$$

is therefore weakly increasing in public organizational knowledge. Together with the producer-service demands in (11), the skill-supply rule (27), and labor-market clearing, this implies

$$K_{A,3} > K_{B,3} \iff S_A^\infty > S_B^\infty.$$

Proposition 2 then gives

$$g_A^\infty > g_B^\infty \iff K_{A,3} > K_{B,3}. \tag{76}$$

It remains to determine when the manufacturing path generates the larger public stock. Under the period-2 restrictions used in the proposition, the creation cutoff and diffusion law imply

$$K_{A,3} > K_{B,3} \iff \bar{e}_m^o n_m > \bar{e}_c^o n_c.$$

Thus the comparison depends on education embodied in organizational labor and on the firm scale over which newly created procedures are used.

Taking $\tau_h \rightarrow 0$ in (70) gives

$$\frac{\bar{e}_m^o}{\bar{e}_c^o} = \left(\frac{g_m}{g_c} \frac{\tau_m}{\tau_s} \right)^{1/2}.$$

With $\Delta = 1$, (29) implies

$$\frac{g_m}{g_c} = (S_1 L_1)^{\phi_m - \phi_c}.$$

Under specialization and symmetry, establishment value is proportional to employment and entry is inversely related to average establishment size. Combining these relationships with (23), under the assumed equality of price and operating-surplus terms across the two paths, gives

$$\frac{n_m}{n_c} = \left(\frac{F_m}{F_c} \right)^{\frac{1}{1+\phi_E}}.$$

Consequently,

$$\frac{\bar{e}_m^o n_m}{\bar{e}_c^o n_c} = \left[\frac{\tau_m}{\tau_s} (S_1 L_1)^{\phi_m - \phi_c} \right]^{1/2} \left(\frac{F_m}{F_c} \right)^{\frac{1}{1+\phi_E}}.$$

The manufacturing path therefore generates more public organizational knowledge if and only if

$$\frac{\tau_m}{\tau_s} (S_1 L_1)^{\phi_m - \phi_c} \left(\frac{F_m}{F_c} \right)^{\frac{2}{1+\phi_E}} > 1,$$

which is exactly condition (32). Combining this result with (76) proves the proposition. $\square$

D.4 Calibration details

This subsection documents target construction, assigned parameters, numerical implementation, fit, local identification, and robustness. The main text reports the parameter estimates and summarizes the fit.

D.4.1 Calibration moments and measurement

The calibration uses 100 moment rows collected in 13 blocks. For moment k in block b , the normalized discrepancy is

$$r_{bk}(\Theta) = \frac{m_{bk}(\Theta) - \hat{m}_{bk}}{\max\{|\hat{m}_{bk}|, 0.01\}}. \quad (77)$$

The objective averages squared discrepancies within each block before applying the block weights:

$$\mathcal{L}(\Theta) = \frac{\sum_b \omega_b K_b^{-1} \sum_{k=1}^{K_b} r_{bk}(\Theta)^2}{\sum_b \omega_b}. \quad (78)$$

Averaging within blocks prevents longer paths from mechanically receiving greater weight. Table D.2 lists the moment rows. The peak-date target imposes zero loss for 1930–1960 and a quadratic penalty outside that interval.

Producer-service employment. Producer and consumer services are distinguished by use, not the supplier’s industry. I combine 2019 IPUMS USA microdata with the 2019 BEA Summary Use Table, separating occupations within agriculture, manufacturing, and consumer services into value-added work and organizational or technical producer-service work. Producer-service workers are classified as high skill if they have at least a bachelor’s degree. Employment in industries primarily supplying intermediate services is allocated to destination sectors in proportion to BEA intermediate-use expenditures; the household share is assigned to consumer-service value added. The replication files contain the IPUMS–BEA crosswalk and occupational classification.

This allocation assumes that each supplier industry’s skill composition and labor requirement per dollar of output are invariant across buyers, because input-output tables do not link transactions to workers’ characteristics. Applying the same rule to model and data produces the three HSPS shares and three within-value-added skill shares. The aggregate HSPS moment uses common sector weights and is an accounting check rather than an independent fourth source of variation.

Firm scale. The 2019 Census Statistics of U.S. Businesses targets are employees per employer *firm*: 7.435 in agriculture, 49.694 in manufacturing, and 20.454 in broad services. Accordingly, the model’s producing unit is interpreted as an employer firm. Including all service industries makes the service target conservative relative to a narrower employer-sector definition. Historical initial scale is anchored to the 1880 Census of Manufactures. Modern scale moments primarily discipline the three entry costs; entry-supply curvature and ability dispersion are assigned externally.

Productivity, skill, and demand. Sectoral productivity targets are mean annual value-added-per-worker growth over 1970–2010. GDP per capita is normalized to one in 1850, so its path identifies growth rather than a level unit. The terminal skill target is the college-educated employment share; the earnings premium is skilled relative to unskilled earnings. Employment shares at 16 dates over 1850–2010 and the manufacturing peak discipline nonhomothetic demand.

D.4.2 Assigned parameters and normalizations

Table D.3 separates unit choices, external anchors, and maintained benchmarks fixed outside the 19-dimensional calibration. Evidence on related objects anchors empirical magnitudes without supplying direct estimates. Generation length is 20 years (Lagerlöf, 2006), final-good substitution is 0.30 (Comin et al., 2021), and both skill-substitution elasticities are 1.20 (Krusell et al., 2000; Acemoglu and Autor, 2011). Input-output coefficients move gradually from value-added production in 1850 to modern BEA coefficients in 2025, allowing intermediate producer services to emerge endogenously over the transition.

With $\bar{N} = 25$, recurring coordination absorbs 5.42 percent of agricultural employment, 5.04 percent of manufacturing employment, and 5.19 percent of consumer-service employment in 2019. This is comparable in magnitude to the organization-capital evidence in Atkeson and Kehoe (2005), which anchors recurring organizational inputs rather than directly estimating $\bar{N}$.

Table D.2: Calibration moment blocks

Block	Rows	Measurement
Sectoral employment paths	48	Three sectors at 16 benchmark years
Manufacturing peak share	1	Maximum manufacturing employment share
Manufacturing peak timing	1	Interval target, 1930–1960
Fertility path	16	Aggregate fertility at benchmark years
Sectoral fertility ratio	1	Agriculture relative to services
GDP-per-capita path	18	U.S. series normalized to one in 1850
Sectoral VA productivity growth	3	Annual means, 1970–2010
Aggregate skilled share	1	Employment share, 2019
Sectoral HSPS shares	3	Means over 2015–2020
Aggregate HSPS share	1	Sector-weighted accounting aggregate
Within-VA skilled shares	3	Means over 2015–2020
Skilled earnings premium	1	Skilled/unskilled earnings, 2019
Sectoral firm size	3	Employees per employer firm, 2019
Total	100	

The diffusion parameter. Let E_t denote the employment-weighted exposure to privately created routines that have not yet entered the public stock. The maintained accumulation law is

$$K_{t+1} = K_t + \min\{1 - K_t, \delta_K E_t\}. \quad (79)$$

Thus δ_K converts current exposure into an annual addition to the stock of publicly usable organizational knowledge.

Diffusion evidence provides an approximate annual scale. Across 12 process innovations in four industries, [Mansfield \(1961\)](#) reports half-adoption times of 0.9–15 years, averaging 7.8 years. Under a constant annual closing rate, $d(T) = 1 - 2^{-1/T}$, so $d(7.8) = 0.085$ and $d(15) = 0.045$. [Greenstein \(1994\)](#) finds that average mainframe-computing practice advanced 14.4 percent annually over 1968–1983 and lagged best practice by at most six to seven years. Slow and fast annual benchmarks of 0.04 and 0.14 imply the midpoint $\delta_K = 0.09$, also close to Mansfield’s mean-based rate.

These studies measure adoption of capital-embodied process innovations and installed computer performance, respectively, whereas the model tracks publicly available organizational routines. They therefore anchor the annual scale of diffusion rather than directly estimating δ_K . Section D.4.6 varies δ_K over 0.01–0.15 and changes the accumulation law.

The remaining assigned parameters are held fixed before the joint search. The calibration chooses

$$\Theta = \{\tau, \tau_a^n, \phi_a, \phi_m, \phi_c, g_{GO}, c_{col}, \alpha_a, \alpha_m, \alpha_c, \epsilon_c, w_a, w_c, R, \xi_a, \xi_c, F_a, F_m, F_c\}. \quad (80)$$

Identification is joint rather than one parameter per moment. Fertility disciplines τ and τ_a^n ; GDP and productivity paths discipline ϕ_j and g_{GO} ; skill shares and the earnings premium discipline c_{col} and α_j ; structural change disciplines demand; HSPS shares discipline R , ξ_a , and ξ_c ; and firm size disciplines F_j . In particular, ξ_c is calibrated, not normalized.

Table D.3: Assigned parameters, external anchors, and normalizations

Parameter	Value	Interpretation or source
<i>A. Units and reference normalizations</i>		
$p_m; h_u; \beta_\pi$	1; 1; 1	Manufacturing-price, unskilled-efficiency, and skill-index units
$\log A_{j0}$	0	Initial value-added productivity normalization
τ_e^u	1	Education-time unit
$\epsilon_m; w_m$	1; 2.5	Manufacturing-demand reference values
$\kappa; \xi_m$	1; 1	Task-complexity scale and manufacturing-adaptability normalization
<i>B. Parameters disciplined by external evidence</i>		
β	0.96	Annual discount factor; Kydland and Prescott (1982)
δ_f	0.04	Annual establishment exit rate; Asturias et al. (2023)
η	0.85	Span-of-control curvature; Atkeson and Kehoe (2005)
$\bar{N}$	25	Organizational-scale benchmark; recurring coordination absorbs about 5 percent of employment; Atkeson and Kehoe (2005)
σ_a	0.16	Dispersion of firm ability, calculated from World Bank Enterprise Survey data
ϕ_E	1.552	Entry-supply curvature; Gutiérrez et al. (2021)
σ	0.30	Elasticity of substitution across final goods; Comin et al. (2021)
$\sigma_L; \sigma_P$	1.20; 1.20	Substitution across skill types; Krusell et al. (2000) ; Acemoglu and Autor (2011)
Δ	20	Generation length in years; Lagerlöf (2006)
λ_A	0.015	Productivity-growth scale; Fernald (2015)
h_s	1.60	Skilled-efficiency scale, benchmarked to the U.S. college earnings premium; Acemoglu and Autor (2011)
δ_K	0.09	Annual diffusion benchmark; Mansfield (1961) ; Greenstein (1994)
<i>C. Model-specific maintained benchmarks</i>		
σ_{skill}	0.10	Scale of the smooth education-choice margin; McFadden (1974)
τ_e^s	0.50	Skill-education cost complementarity ^a ; Cunha and Heckman (2007) .
$\tau_m^n; \tau_s^n$	1; 1	Common child-time cost outside agriculture ^b
ρ	0.40	Adaptation cost relative to creation cost; sensitivity reported below
γ	0.05	Child-preference weight, benchmarked to an education-expenditure share of approximately 5 percent
ϵ_a	0.01	Near-subsistence agricultural-demand benchmark; Comin et al. (2021)

Notes: Panel A contains unit choices and reference normalizations. Panel B contains parameters fixed before the joint calibration using external empirical evidence. Panel C contains model-specific benchmark values that are not directly estimated from the calibration moments. The citations in Panel C motivate the relevant functional form or empirical scale and are not interpreted as one-for-one estimates of the reported parameter.

The ability dispersion σ_a is calculated from the pooled, employment-weighted dispersion of firm employment in the World Bank Enterprise Surveys. The knowledge parameter δ_K governs the conversion of exposure to privately created routines into additions to the public stock of organizational knowledge; it is not a physical depreciation rate.

^a [Cunha and Heckman \(2007\)](#) motivate the education-environment complementarity. I set $\tau_e^s = 0.5$ as a conservative benchmark: at the calibrated skilled share of $S \simeq 0.34$, it implies that the time cost of education is only about 17 percent lower than in a fully unskilled environment.

^b Setting $\tau_m^n = \tau_s^n = 1$ shuts down systematic differences in the fixed time cost of children between manufacturing and service households. This is consistent with the absence of a persistent manufacturing-service fertility gap in the U.S. historical data and with the developing-country evidence discussed by [Peters et al. \(2026\)](#).

D.4.3 Numerical implementation

The model is solved annually over 1790–2289, extending beyond reported outcomes to limit terminal-condition effects on 2019 and policy transitions. Exact calibration uses nine productivity types and 101 organizational-task grid points, jointly solving entry, scale, education, formalization, and public knowledge.

Strictly positive parameters are searched in logs using multiple starts and damped local updates, followed by exact-grid evaluation. Candidates must converge on the exact grid and keep every 2019 sectoral firm-size error below three percent. The selected vector has $\mathcal{L} = 0.011878$ and maximum firm-size error 0.08 percent. The cold-start fixed point is accepted after 30 iterations; the last discrete-formalization update is 0.00646, below the three-grid-cell epsilon-equilibrium tolerance of 0.03, and the 2019 update is 0.00171. Education-budget and size-accounting residuals are zero to machine precision.

All 12 independent cold starts converge to the same exact-grid equilibrium within solver tolerance. Calibration and cross-country counterfactuals are also solved independently on the exact grid. Finite differences perturb the converged benchmark, using its solution as the common initialization for nearby parameters.

D.4.4 Detailed fit

Figure D.1 compares targeted HSPS shares, firm size, and value-added productivity growth. Agricultural, manufacturing, and service HSPS shares are 0.325, 0.461, and 0.399 in both model and data at reported precision; the maximum discrepancy is 0.04 percentage points. Model firm sizes of 7.44, 49.72, and 20.45 workers are within 0.08 percent of their targets. Annual productivity growth is 3.85, 2.98, and 1.31 percent, versus 3.71, 3.17, and 1.42 percent in the data.

Untargeted employment-weighted standard deviations of formalization are 0.176 in agriculture, 0.187 in manufacturing, and 0.049 in services, versus data-based proxies of 0.229, 0.205, and 0.211. Without sector-specific dispersion parameters, the model reproduces 76.8, 91.1, and 23.1 percent of the respective dispersions.

D.4.5 Local identification and uncertainty

Let $r(\theta)$ denote the vector of weighted residuals and let $\theta = \log \Theta$ denote the 19-dimensional vector of calibrated parameters. I compute the Jacobian

$$J = \frac{\partial r(\theta)}{\partial \theta'} \quad (81)$$

using exact-grid central differences around the frozen benchmark equilibrium. Every perturbation uses the same benchmark-equilibrium anchor. For parameter p and step h , the numerical derivative is

$$J_p(h) = \frac{r(\theta + he_p) - r(\theta - he_p)}{2h}. \quad (82)$$

I use $h = 0.025$ and $h = 0.05$. All 77 required exact simulations—one baseline and four perturbations for each of 19 parameters—converge.

The 100-by-19 objective Jacobian has full column rank at both steps. At the preferred five-percent step, its condition number is 952.1 and its effective rank is 17 at a one-percent relative singular-value cutoff and 19 at a 0.1-percent cutoff. Removing the locally flat peak-timing penalty, aggregate-HSPS identity, and one redundant sectoral employment share per date gives a smooth 82-by-19 Jacobian with full column rank, condition number 789.9, and the same effective ranks. Thus two local directions are weak, but no calibrated parameter is exactly redundant. Each column retains at least 14.0 percent of its norm after projection on the other 18. The identified subspace is stable across step sizes except in the weakest direction.

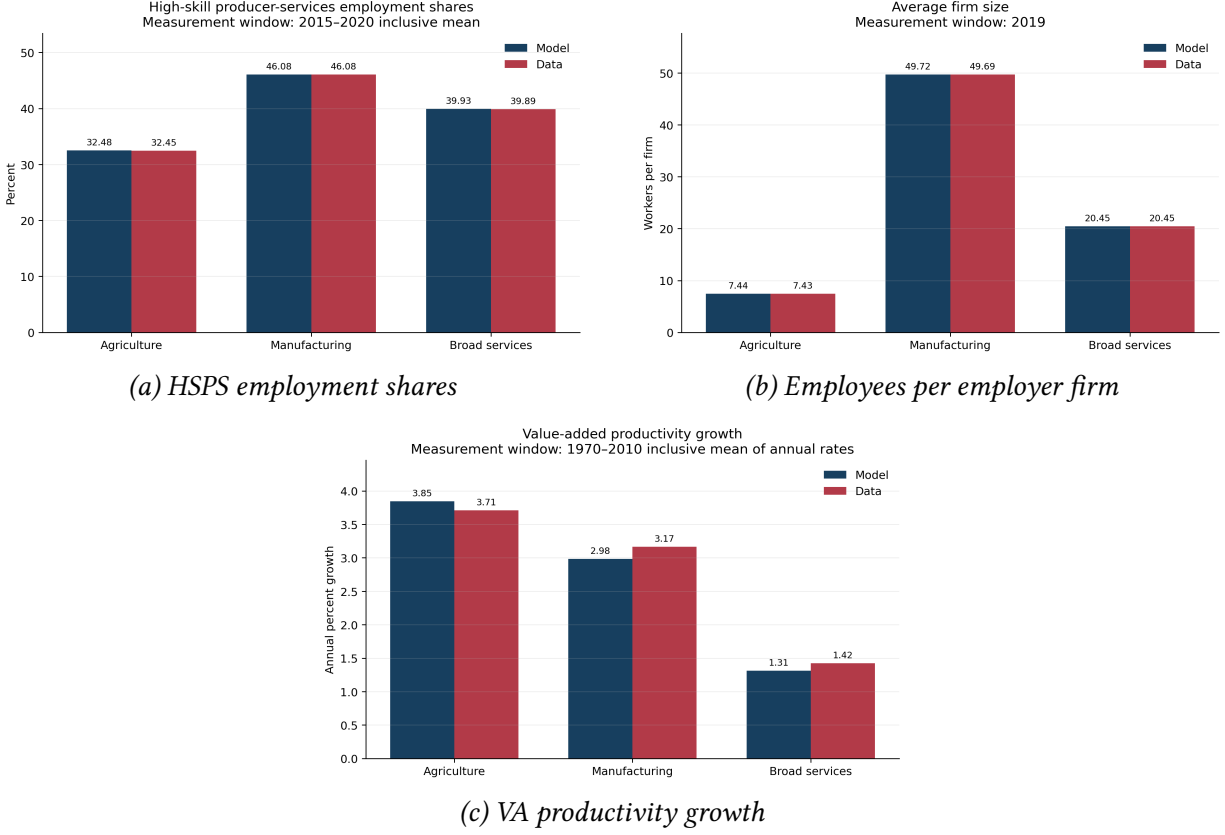


Figure D.1: Targeted calibration fit. Bars compare the calibrated model with the corresponding U.S. targets. HSPS shares are means over 2015–2020, firm size is measured in 2019, and productivity growth is the annual average over 1970–2010.

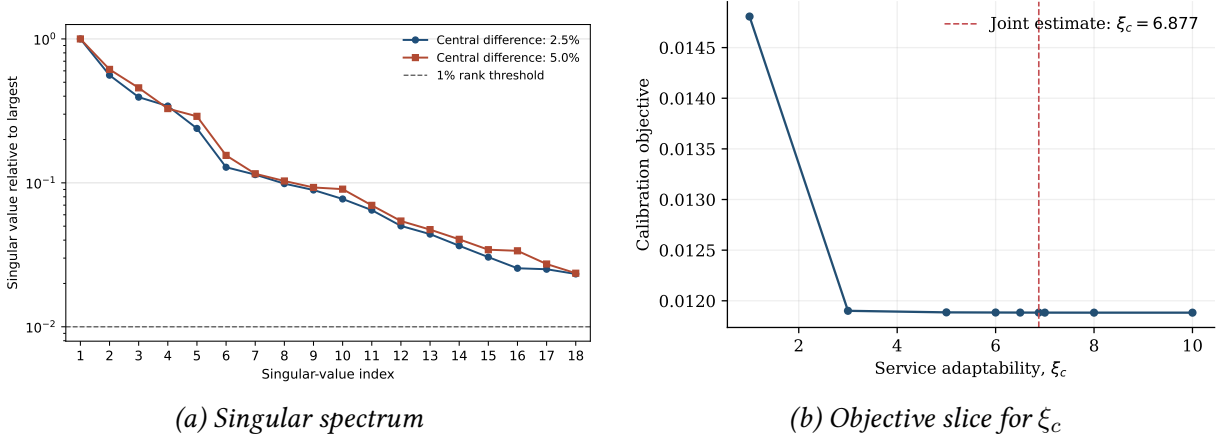


Figure D.2: Identification diagnostics. Panel (a) reports singular values of the exact weighted-moment Jacobian. Panel (b) evaluates the exact objective over ξ_c while holding the other 18 calibrated parameters at their reported values.

Service adaptability illustrates the difference between identification and precision. The exact objective falls from 0.014807 at $\xi_c = 1$ to 0.011901 at $\xi_c = 3$ and 0.0118833 at the calibrated $\xi_c = 6.877$. It is

nearly flat in the saturation region: 0.0118837 at $\xi_c = 6.5$, 0.0118832 at 7, and 0.0118829 at 8. Service-formalization moments reject low adaptability but do not precisely locate ξ_c within this region, explaining its large local standard error.

Table D.5 reports diagnostics from the smooth, nonredundant system. The unique share divides a Jacobian column's norm after projection on other columns by its original norm. HC1 residual-based minimum-distance standard errors are expressed as percentages of estimates. They summarize local curvature and residual fit conditional on the target vector, not sampling confidence intervals for the heterogeneous data sources.

For formal SMM inference, let Ω denote the sampling covariance matrix of the empirical moments and W the calibration weight matrix. The corresponding sandwich covariance is

$$\text{Avar}(\hat{\theta}) = (J'WJ)^{-1}J'W\Omega WJ(J'WJ)^{-1}. \quad (83)$$

Estimating Ω must preserve serial dependence and cross-moment covariance within each historical, administrative, or microdata source. The rank, objective-slice, and residual-based exercises establish local numerical identification of the 19 parameters without treating heterogeneous moment rows as a common random sample.

D.4.6 Robustness and Sensitivity

Two exercises assess the manufacturing–development relationship: 18 alternative specifications and a finer grid for knowledge diffusion δ_K . Parameters not varied remain calibrated, without recalibration. Slopes use the same five economies, differing only in manufacturing-demand weights. Every specification preserves the association of a higher manufacturing peak with higher GDP per capita, faster GDP-per-capita growth, greater tertiary-education and HSPS shares, and lower population growth. Magnitudes respond most to diffusion and firm scale, and less to adaptation costs, producer-service substitution, and the accumulation law.

Table D.6 reports slope changes relative to the benchmark. All 90 slopes across 18 specifications retain their signs. Median absolute changes are 2.9 percent for GDP levels, 8.5 percent for GDP growth, 10.2 percent for tertiary education, 2.4 percent for HSPS, and 4.3 percent for population growth. Excluding zero diffusion and equal sector weights, the largest absolute changes are 15.1, 17.5, 27.3, 16.4, and 13.7 percent, respectively.

Removing diffusion attenuates GDP-level and GDP-growth slopes by 30.4 and 33.8 percent, tertiary education by 77.9 percent, HSPS by 56.6 percent, and the magnitude of population growth by 34.3 percent. Equal sector weights reduce education and HSPS slopes by 38.5 and 24.0 percent and GDP levels and growth by 16.7 and 13.4 percent. Diffusion therefore matters particularly for education and organizational formalization, although preserved signs without diffusion show that spillovers are not the sole source of these relationships.

Firm-scale assumptions change magnitudes without reversing signs. Varying span of control changes GDP-level slopes by -10.4 to 15.1 percent and education slopes by -12.7 to 27.2 percent. Lower ability dispersion raises GDP-growth and education slopes by 17.5 and 27.3 percent. Reducing routine repetition by 25 percent raises the growth slope by 16.3 percent but lowers HSPS by 16.4 percent. Initial knowledge $K_0 = 0.10$ attenuates education by 24.4 percent, GDP levels by 5.0 percent, and HSPS by 3.1 percent. Education is thus more sensitive than GDP levels to several organizational assumptions.

In $\tau_e(S) = 1 - S(1 - \tau_e^s)$, lower τ_e^s strengthens the fall in education costs with skill. Relative to the benchmark 0.5, setting $\tau_e^s = 0.25$ lowers the education slope by 11.6 percent and the population-growth magnitude by 9.1 percent; setting it to 0.75 raises them by 8.8 and 8.3 percent. GDP-level, GDP-growth, and HSPS slopes change by at most 3.5 percent. This complementarity chiefly affects education and demographics over the tested range.

Changing adaptation costs by 25 percent moves every slope by less than 4 percent and GDP levels by less than 1 percent. Producer-service substitution of 0.90 or 1.50 changes slopes by at most 1.3 percent. Replacing (79) with an exponential-hazard law having the same first-order flow changes every slope by less than 0.7 percent. Results are therefore insensitive to the benchmark's hard cap and only modestly sensitive to adaptation costs over these ranges.

The finer diffusion exercise varies δ_K from 0.01 to 0.15 in steps of 0.01. Table D.7 reports slopes for a 10-percentage-point increase in peak manufacturing employment. Responses are nonmonotonic, with several slopes largest near the benchmark. Very slow diffusion chiefly attenuates skill and HSPS relationships.

Over the empirically motivated interval $[0.04, 0.14]$, slope changes relative to $\delta_K = 0.09$ range from -13.8 to 0.6 percent for GDP levels, -6.9 to 5.0 percent for GDP growth, -27.8 to 4.1 percent for tertiary education, -22.4 to 1.6 percent for HSPS, and -16.2 to 0.6 percent for the population-growth magnitude. Over $[0.01, 0.15]$, maximum attenuations are 28.2, 30.1, 73.7, 49.8, and 32.8 percent, respectively. Every sign is preserved; slow diffusion weakens education and organizational formalization more than the GDP relationship.

Table D.4: Parameter estimates and reported calibration fit

<i>Panel A: Jointly estimated parameters and principal disciplines</i>		
Parameter	Estimate	Principal empirical discipline
Child-rearing time cost, τ	0.03236	Aggregate fertility path
Agricultural fertility shifter, τ_a^n	0.44017	Agriculture-to-services fertility ratio
Agricultural growth elasticity, ϕ_a	0.99840	Agricultural VA productivity growth
Manufacturing growth elasticity, ϕ_m	0.91919	Manufacturing VA productivity growth
Service growth elasticity, ϕ_c	0.69046	Service VA productivity growth
Gross-output productivity growth, g_{GO}	0.00318	GDP-per-capita path and sectoral growth
Skill-acquisition cost, c_{col}	0.56821	Aggregate skilled-employment share
Agricultural skill weight, α_a	0.24338	Skilled share within agricultural VA
Manufacturing skill weight, α_m	0.33001	Skilled share within manufacturing VA
Service skill weight, α_c	0.40995	Skilled share within service VA
Service nonhomotheticity, ϵ_c	1.01048	Sectoral employment paths
Agricultural demand weight, w_a	0.65105	Sectoral employment paths
Service demand weight, w_c	3.82326	Sectoral employment paths
Discounted repetitions, R	22.78696	Sectoral HSPS employment shares
Agricultural adaptability, ξ_a	3.56402	Agricultural HSPS employment share
Service adaptability, ξ_c	6.87739	Service HSPS employment share
Agricultural setup-cost shifter, F_a	25.872	Agricultural firm size
Manufacturing setup-cost shifter, F_m	187.625	Manufacturing firm size
Service setup-cost shifter, F_c	85.117	Service firm size
<i>Panel B: Reported scalar targets and model outcomes</i>		
Moment	Data	Model
Agricultural HSPS share	0.325	0.325
Manufacturing HSPS share	0.461	0.461
Service HSPS share	0.399	0.399
Agricultural firm size	7.435	7.44
Manufacturing firm size	49.694	49.72
Service firm size	20.454	20.45
Agricultural VA productivity growth (%)	3.71	3.85
Manufacturing VA productivity growth (%)	3.17	2.98
Service VA productivity growth (%)	1.42	1.31
Manufacturing peak employment share (%)	28.28	24.93
Manufacturing peak year	1950	1928

Notes: Panel A reproduces the 19 estimates in Table 3 and their principal disciplines; the parameters are estimated jointly, so these are not one-parameter–one-moment mappings. Panel B collects the numerical fit values reported in the main text and Appendix D.4.4. HSPS shares are fractions and use 2015–2020 means; firm size is employees per employer firm in 2019; productivity growth is the annual mean over 1970–2010. Entries retain the source’s reported precision. The observed manufacturing peak is in 1950, while the estimation loss assigns zero timing penalty to dates from 1930 through 1960. Figures 5, 4, and D.1 report the corresponding paths and graphical fit. Other target blocks and their measurement are listed in Table D.2; no unreported fit values are imputed here.

Table D.5: Parameter-level local identification diagnostics

Parameter	Estimate	Unique share (%)	HC1 relative SE (%)
τ	0.03236	17.92	4.86
τ_a^n	0.44017	30.66	5.92
ϕ_a	0.99840	15.49	10.15
ϕ_m	0.91919	14.04	10.38
ϕ_c	0.69046	15.97	10.86
gGO	0.00318	61.33	24.31
c_{col}	0.56821	35.90	5.36
α_a	0.24338	81.42	2.81
α_m	0.33001	84.90	2.96
α_c	0.40995	82.84	2.62
ϵ_c	1.01048	24.87	30.15
w_a	0.65105	34.49	12.44
w_c	3.82326	25.57	39.53
R	22.78696	33.06	5.33
ξ_a	3.56402	64.17	88.56
ξ_c	6.87739	70.22	491.31
F_a	41.24479	80.00	48.09
F_m	277.23883	69.56	60.27
F_c	275.73211	91.00	22.85

Notes: The unique shares use the exact, smooth, nonredundant 82-by-19 Jacobian. HC1 standard errors are computed in log-parameter space and translated to levels by the delta method.

Table D.6: Sensitivity of cross-country slopes

Specification	GDP level	GDP growth	Tertiary share	HSPS share	Population growth
Adaptation cost: -25%	0.8	1.4	3.3	1.8	0.6
Adaptation cost: $+25\%$	0.1	2.5	3.6	0.4	0.5
Knowledge diffusion: -25%	2.4	7.7	11.3	4.6	2.3
Knowledge diffusion: $+25\%$	-4.4	-11.2	-20.8	-8.4	-4.6
Routine repetition: -25%	-7.0	16.3	4.2	-16.4	-8.2
Routine repetition: $+25\%$	1.8	-11.1	-9.0	5.5	3.0
Span of control: 0.80	15.1	13.0	27.2	0.8	13.7
Span of control: 0.90	-10.4	-9.8	-12.7	2.6	-12.0
Producer-service substitution: 0.90	0.1	0.2	0.0	0.2	0.0
Producer-service substitution: 1.50	-0.4	-0.9	-1.3	-0.8	-0.4
Firm-ability dispersion: -25%	5.0	17.5	27.3	6.9	4.0
Firm-ability dispersion: $+25\%$	-2.3	-5.6	-4.4	-2.1	-1.6
Education complementarity: $\tau_e^s = 0.25$	3.5	-2.5	-11.6	-2.0	-9.1
Education complementarity: $\tau_e^s = 0.75$	-2.0	-0.5	8.8	1.6	8.3
Initial knowledge: $K_0 = 0.10$	-5.0	-9.2	-24.4	-3.1	-6.0
Equal sector weights	-16.7	-13.4	-38.5	-24.0	-20.1
No knowledge spillovers	-30.4	-33.8	-77.9	-56.6	-34.3
Exponential-hazard accumulation	-0.1	0.5	0.6	0.0	-0.2

Notes: Entries are percentage changes in the absolute magnitude of the cross-country slope relative to the benchmark. Positive entries indicate a stronger relationship; negative entries indicate attenuation. All slopes retain their benchmark sign. Outcomes are log GDP per capita in 2019, annual GDP-per-capita growth over 2010–2019, average tertiary-education and HSPS shares over 2010–2019, and cumulative log population growth over 1850–2019. Each specification uses the same five manufacturing-demand economies. Parameters not listed in a row remain fixed. The benchmark and 18 alternatives comprise 95 economies, all of which satisfy the numerical convergence criterion. This broader exercise uses a coarser numerical approximation than Table D.7; all percentage changes are calculated relative to the benchmark computed with the same approximation.

Table D.7: Cross-country slopes by the rate of knowledge diffusion

δ_K	Log GDPpc	GDPpc growth	College share	HSPS share
0.01	0.1312	0.00112	0.0119	0.0333
0.02 [†]	0.1230	0.00105	0.0081	0.0291
0.03 [†]	0.1322	0.00128	0.0164	0.0364
0.04 [†]	0.1477	0.00139	0.0222	0.0450
0.05 [†]	0.1604	0.00151	0.0281	0.0520
0.06	0.1676	0.00157	0.0308	0.0558
0.07	0.1677	0.00155	0.0308	0.0564
0.08	0.1725	0.00156	0.0320	0.0588
0.09	0.1714	0.00150	0.0307	0.0579
0.10	0.1708	0.00148	0.0299	0.0574
0.11	0.1693	0.00147	0.0303	0.0569
0.12	0.1682	0.00145	0.0297	0.0564
0.13	0.1667	0.00143	0.0287	0.0558
0.14	0.1652	0.00140	0.0278	0.0553
0.15	0.1636	0.00137	0.0265	0.0541

Notes: Each entry is the OLS slope associated with a 10-percentage-point increase in the endogenous peak manufacturing employment share across five manufacturing-demand economies. All other parameters remain at their calibrated values. GDP per capita is measured in 2019; growth and education and HSPS shares refer to 2010–2019. Growth rates and shares are expressed as fractions. The population-growth slope, omitted from the table, is negative at every value of δ_K . [†]In one economy in each marked row, the discrete formalization decision continues to oscillate beyond the numerical tolerance, although the residuals for knowledge, employment, and establishment counts are below 8.5×10^{-4} . The reported values retain the same solution procedure across specifications. As an additional check, using a neighboring economy’s solution as the starting point at $\delta_K = 0.02$ changes the four reported slopes to (0.1188, 0.00105, 0.0082, 0.0291) and preserves all five signs. The precise magnitudes in the marked rows should therefore be interpreted with this numerical qualification.

D.4.7 Interpreting the manufacturing demand counterfactual

Varying manufacturing demand w_m changes manufacturing's relative pull during the transition and its endogenous employment peak, holding technology, fertility and education preferences, and production primitives fixed.

This counterfactual has an equivalent price-wedge representation. Final demand over agriculture, manufacturing, and consumer services implies relative demand terms of the form

$$K_a = \left(\frac{p_a}{p_m} \right)^{1-\sigma} Y^{(1-\sigma)(k_a-1)} \frac{w_a}{w_m^{k_a}}, \quad K_c = \left(\frac{p_c}{p_m} \right)^{1-\sigma} Y^{(1-\sigma)(k_c-1)} \frac{w_c}{w_m^{k_c}},$$

where $k_j \equiv \varepsilon_j / \varepsilon_m$. Changing the manufacturing demand weight from w_m to $f_m w_m$ therefore changes the relative demand terms to

$$K_a(f_m) = K_a(1) f_m^{-k_a}, \quad K_c(f_m) = K_c(1) f_m^{-k_c}.$$

Now consider an economy with the original w_m , but in which households face effective final-good prices

$$\tilde{p}_{a,t} = (1 + \tau_a) p_{a,t}, \quad \tilde{p}_{m,t} = p_{m,t}, \quad \tilde{p}_{c,t} = (1 + \tau_c) p_{c,t}.$$

The wedge economy generates the same relative demand schedules as the $f_m w_m$ economy if

$$(1 + \tau_a)^{1-\sigma} = f_m^{-k_a}, \quad (1 + \tau_c)^{1-\sigma} = f_m^{-k_c}.$$

Equivalently,

$$\boxed{1 + \tau_a = f_m^{-k_a/(1-\sigma)}, \quad 1 + \tau_c = f_m^{-k_c/(1-\sigma)}}. \quad (84)$$

The w_m counterfactual can therefore be represented as either a manufacturing-demand shifter or effective final-good prices. This equivalence concerns household relative-demand schedules conditional on income and producer prices; it does not require a primitive preference interpretation. Table D.8 maps the selected economies in Figure 7 into price wedges.

Table D.8: Manufacturing-demand counterfactuals and equivalent final-consumption wedges

w_m	f_m	Peak manufacturing employment (%)	Agriculture wedge τ_a (%)	Consumer-services wedge τ_c (%)
0.60 [†]	0.240	11.19	2.06	684.67
1.16	0.464	15.92	1.10	202.97
1.80	0.720	20.62	0.47	60.67
2.50 (benchmark)	1.000	24.93	0.00	0.00
2.56	1.024	25.27	-0.03	-3.37
3.49	1.396	29.88	-0.48	-38.22
4.69	1.876	34.56	-0.89	-59.67
6.10	2.440	38.80	-1.27	-72.41

Notes: The calibrated benchmark is $w_m^* = 2.5$, and $f_m \equiv w_m / w_m^*$. The wedges are computed from $1 + \tau_j = f_m^{-k_j/(1-\sigma)}$, using $\sigma = 0.30$, $k_a = 0.0100$, and $k_c = 1.0105$. The normalization holds the effective manufacturing price fixed. Positive values raise the effective final-consumption price relative to manufacturing; negative values lower it. Peak manufacturing employment is endogenous. The wedges are demand-equivalent shadow prices, not statutory tax rates.

The asymmetry between the two wedges follows directly from the calibrated demand system. For agriculture, the relevant exponent is

$$-\frac{k_a}{1-\sigma} = -0.0143,$$

so even large changes in w_m imply small movements in the effective agricultural price. For consumer services, the corresponding exponent is

$$-\frac{k_c}{1-\sigma} = -1.4435.$$

The service-price mapping is therefore substantially more responsive and becomes nonlinear as f_m falls. For example, the $w_m = 1.16$ counterfactual implies $1 + \tau_c = 3.03$: under the normalization in the table, the effective consumer-service price is approximately three times its benchmark level. At $w_m = 1.80$, the corresponding multiplier is 1.61.

These shadow prices summarize combinations of market access, distribution costs, markups, regulation, product availability, relative quality, and other pecuniary or nonpecuniary forces producing the same relative-demand schedules. The counterfactuals introduce no government revenue or tax rebates, and the wedges play no independent calibration role. Endogenous manufacturing peaks discipline the variation; the wedge mapping interprets the reduced-form demand differences.

D.5 Policy extension

This subsection documents the LMIC calibration and the implementation of the policy experiments in Section 7. Parameters not listed below are inherited from the benchmark model.

D.5.1 Distortions and LMIC calibration

Before drawing permanent ability $a \sim G$, an entrant chooses setup intensity $h \in [\underline{h}, 1]$, which determines effective productivity $x = ha$. In manufacturing and consumer services, the productivity-correlated distortion is

$$1 - \tau_{i,t} = \kappa_{j,t} x_{i,t}^{-d_j}, \quad x_{i,t} = h_{i,t} a_i, \quad j \in \{m, c\}, \quad (85)$$

where $\kappa_{j,t}$ sets the revenue-weighted mean of $\tau_{i,t}$ to zero. Setup intensity requires

$$e(h, d_j) = \exp\{-\lambda_E d_j (1 - h)\} \quad (86)$$

units of the benchmark setup input. We set $d_m = d_c = 0.41$, following the excess productivity-correlated distortion measured by [Bento and Restuccia \(2017, 2021\)](#), and $\lambda_E = 12$. The permanent-reform experiment reduces d_m and d_c linearly during 2026–2035 and sets them to zero thereafter.

The joint LMIC parameter vector contains the organizational-scale normalization, three sectoral entry-cost shifters, three sectoral productivity-growth parameters, the manufacturing-demand shifter, and the skill-acquisition cost. The selected vector has $\tilde{N} = 3$ and retains the externally disciplined LMIC productivity dispersion of 0.16. The historical establishment-count initialization from the benchmark transition is retained. Table D.9 reports the moments used in the calibration and their model counterparts.

Table D.9: Calibration of the LMIC policy economy

Block	Target moment	Data	Model
Productivity	Agricultural VA productivity growth, 1990–2018 (%)	2.620	2.716
	Manufacturing VA productivity growth, 1990–2018 (%)	2.013	2.084
	Services VA productivity growth, 1990–2018 (%)	1.111	1.145
Firm scale	Agricultural average establishment size, 2019	2.220	2.123
	Manufacturing average establishment size, 2019	6.238	5.864
	Services average establishment size, 2019	1.883	1.714
Education	Tertiary-skilled employment share, 2019 (%)	12.000	11.980
Demand	Peak manufacturing employment, 1990–2018 (%)	11.713	10.442
Organization	High-skill producer-service employment, 2019 (%)	1.326	1.352

Notes: Productivity growth and peak manufacturing employment use the GGDC/UNU–WIDER Economic Transformation Database. The firm-size targets use the LMIC target construction based on [Bento and Restuccia \(2021\)](#); the agricultural entry is a low-scale proxy because harmonized agricultural establishment employment is unavailable for the same country sample. The high-skill producer-service target is the unweighted median employment share of university-educated workers in ISCO groups 1–4 and the maintained producer-service industries across the ten available LMIC censuses in the IPUMS sample. The denominator is total employment. The model counterpart is total high-skill producer-service employment.

D.5.2 Policy implementation

Four programs operate during 2026–2045. The college scholarship reimburses part of the private cost of acquiring skill. Firm-formalization support reimburses part of the one-time cost of creating an organizational procedure. The mixed package applies both reimbursements at equal statutory rates. The manufacturing wage subsidy reimburses part of the skilled and unskilled manufacturing wage bill. Table D.10 reports the rates used in each distortion environment.

The scholarship operates through the existing education-choice margin. Let c_t be the calibrated cost of acquiring skill, τ_t^e the scholarship rate, and p_t^e the skilled share of the entering cohort. The calibrated cost is treated as pecuniary tuition, so

$$p_t^e = \Lambda \left(\frac{\beta_\pi \log(w_t^s/w_t^u) - (1 - \tau_t^e)c_t}{\sigma_e} \right). \quad (87)$$

To convert the choice-index cost into monetary resources, one unit of c_t is assigned $\kappa_{\text{edu}} = 1.696$ annual unskilled-wage units. The conversion uses the tuition coefficient in [Cordeiro and Cox \(2023\)](#), education-choice dispersion $\sigma_e = 0.10$, eight semester payments, and mean earnings for workers with completed secondary education ([Instituto Brasileiro de Geografia e Estatística, 2020](#)):

$$\kappa_{\text{edu}} = \frac{1000 \sum_{j=0}^7 0.96^{j/2}}{0.10 \times 2.049 \times (12 \times 1788)} = 1.696. \quad (88)$$

With cohort mass L_t/g and $g = 20$, government scholarship spending is

$$G_t^e = \tau_t^e \kappa_{\text{edu}} c_t w_t^u \left(\frac{L_t}{g} \right) p_t^e. \quad (89)$$

All eligible skilled entrants receive the reimbursement, including inframarginal entrants. At zero subsidy, tuition is already included in calibrated household final expenditure. Under the policy, the government pays the subsidized share and finances it with a contemporaneous lump-sum household tax. The welfare calculation therefore does not count private tuition twice.

Table D.10: Temporary-policy rates

Policy	Distortions remain	Full reform
College scholarship	32.0%	29.0%
Firm-formalization support	90.0%	90.0%
Mixed package: scholarship/formalization	36.0%/36.0%	27.5%/27.5%
Manufacturing wage subsidy	6.702%	6.702%

Notes: The mixed-policy rates apply separately to tuition and procedure-creation costs; equal rates do not imply equal realized spending. Rates are chosen using equilibrium take-up to keep present-value expenditure near 0.6 percent of discounted policy-economy GDP. Formalization support is evaluated at the highest stable reimbursement rate used in the exercise.

Firm-formalization support reimburses the stated share of the one-time procedure-creation cost; firms continue to pay for the producer-service labor used to operate installed procedures. The manufacturing subsidy applies to the contemporaneous skilled and unskilled manufacturing wage bill. All government outlays are financed by contemporaneous lump-sum household taxes. Present-value fiscal cost is discounted government spending divided by discounted GDP during 2026–2045, and consumption-equivalent variation is evaluated over 2026–2089.

D.5.3 Withdrawal and outcome construction

Withdrawal is unexpected. Agents first solve the 2026–2045 equilibrium believing that the policy will continue. At the start of 2046, the policy rate is set to zero and the continuation begins from the inherited distribution of firms, setup intensities, installed procedures, private localized knowledge, and public organizational knowledge. The permanent-policy and withdrawal solutions are identical through 2045. Each policy path is compared with a no-temporary-policy path under the same permanent-distortion environment.